Natural Origin 6 — A Minimal Reconstruction of Celestial Mechanics and Cosmology

Author: Zhang Xiangqian, Xu Yuchuan
E-mail: info@ahdvlp.com

Abstract

This paper presents a unified reconstruction of gravitational structure in celestial mechanics and cosmology within a single framework based on the settlement anchor field $u(\mathbf{x},t) = \ln \beta_r(\mathbf{x},t)$. Space-unit refresh satisfies the budget identity $\beta_s^2 + \beta_r^2 = 1$, and the time readout is given by the settlement fraction $d\tau = e^u dt$. In this structure, gravity is interpreted as the first spatial gradient of the settlement fraction $\mathbf{g} = -c^2 \nabla u$, while tides are encoded by the second-order spatial curvature $T_{ij} = c^2 \partial_i \partial_j u$, whose irreducibility forms the core of real physical structure. Dynamical perturbations satisfy $\phi = \partial_t u$ with $\phi = 0$. Gravitational waves correspond to the temporal propagation of second-order structure $\partial_t T_{ij} = c^2 \partial_i \partial_j \phi$; the two polarizations arise from a two-dimensional eigenmode decomposition of the transverse Hessian, without introducing multiple fundamental fields.

In the optical sector, light deflection is given by an integral of the transverse first gradient $\alpha = 2\int \nabla_\perp u \, ds$, and beam convergence and shear are controlled by the transverse-projected second-order structure $Q_{ab} = -2\nabla_\perp \nabla_\perp u$, thereby establishing a proportional locking relation between the tidal tensor and lensing readouts. Since light is a pure stepping state ($\beta_r = 0$) and does not participate in longitudinal settlement, it does not exhibit matter-like tidal response; nevertheless, its propagation path and convergence matrix remain modulated by the curvature of the settlement field.

The strong-field limit is rewritten as a circuit-break condition $u \to -\infty$, corresponding to failure of an accessibility function rather than a geometric singularity. A cosmological closure equation $\nabla^2 u = -\frac{4\pi G_*}{c^2}\rho_{enc} + \Lambda_u$ links the tidal trace directly to source density, compressing celestial structure, lensing effects, and cosmic evolution into different-order expressions of the same settlement curvature.

Based on the single-field structure, we propose three falsifiable criteria:

1) dynamical and lensing observations must be jointly inverted by the same $u(\mathbf{x})$;
2) if vacuum lensing dispersion exists or is strictly absent, the outcome must be consistent with refresh structure;
3) the same gravitational-wave event must close across interferometers, clock arrays, and lensing modulation through one and the same $\delta u(\mathbf{x},t)$, otherwise the theory fails.

This framework reconstructs gravity, tides, and gravitational-wave structure with a minimal set of ontological variables and provides a testable single-field unification model whose validity is determined entirely by whether cross-probe consistency is supported by observations.

Introduction

Over the past century, gravitation and cosmology have achieved remarkable empirical success, yet that success has not produced an equally strong level of conceptual unification. Redshift, the Hubble law, dark matter, dark energy, gravitational lensing, and gravitational waves have each been explained within their own theoretical contexts, but only by relying on mutually independent assumptions and patch-like modifications: cosmic expansion, invisible mass components, vacuum energy, spacetime geometric perturbations, and so forth. This fragmentation is not driven by insufficient observational precision, but by a more fundamental question—whether we have chosen the wrong "readout object" that physics ought to explain.

In conventional frameworks, gravity is treated as a force (Newton) or geometry (general relativity), redshift is interpreted as a consequence of velocity or spatial stretching, orbits are explained as mechanical equilibrium, and gravitational waves are described as oscillations of spacetime itself. Each explanation is successful within its domain, but when these phenomena are placed side by side they lack a common ontological foundation. For example, why does massless light bend? Why does the same mass distribution determine both planetary orbits and lensing deflection? Why is there no unified invertible relationship connecting redshift, lensing, and gravitational waves? These questions suggest that what truly needs to be unified may not be "interactions," but the "mode of readout" itself.

This paper adopts a different starting point. We no longer begin by asking "what the universe is doing," but instead ask "what the observer is actually reading." From this viewpoint, gravitational and cosmological phenomena are repositioned as different projected readouts of a single underlying quantity. That quantity is defined as the settlement anchor field $u(\mathbf{x}, t) = \ln \beta_r(\mathbf{x}, t)$, where β_r denotes the share of the settlement phase within the refresh resources of space units. Space is treated as a network of minimal refresh units (space units), whose refresh upper bound is set by the speed of light c. Each space unit allocates its refresh resources between two orthogonal phases: a stepping phase responsible for propagation, and a settlement phase responsible for existence and return. This two-phase axiom separates propagation from recordable outcomes at the ontological level, providing a minimal yet sufficient structure for reinterpreting gravitation.

Within this framework, redshift is no longer a side effect of waves being stretched during propagation, but an endpoint or temporal-evolution difference of longitudinal settlement readouts. Light bending is no longer explained by a force acting on a massless particle or by spacetime curvature, but by a systematic routing bias induced by transverse settlement gradients. Planetary orbits are no longer force-balance solutions, but periodic closures formed when transverse deflection is adjudicated by longitudinal settlement return. The Hubble law corresponds to a first-gradient structure of the settlement anchor field on cosmic scales. Dark matter and dark energy are identified as spatial structure of settlement gradients and background curvature, respectively. Gravitational waves and induced gravity are unified as different limits (far-field and near-field) of perturbations of settlement gradients. In this way, multiple phenomena long treated as "different physics" are absorbed into a unified readout system of the single variable $u(\mathbf{x}, t)$.

The goal of this work is not to introduce new entities or additional degrees of freedom, but to correct the choice of readout object: to unify phenomena scattered across geometry, dynamics, and cosmology into a computable, invertible, falsifiable settlement-dynamics problem. By systematically constructing three classes of readout operators—longitudinal, transverse, and

temporal—this paper not only reconstructs the physical meaning of redshift and the Hubble law, but also provides a shared causal origin for light bending, planetary orbits, strong-field astrophysical structures, and gravitational-wave observations.

More importantly, this framework naturally leads to joint auditing by experiments and observations. Because all phenomena share the same origin in $u(\mathbf{x}, t)$, different probes—spectroscopic redshift, lensing deflection, orbital dynamics, interferometer strain, and atomic clock arrays—must be mutually compatible within a single settlement field. We therefore propose multiple strong consistency criteria and falsifiable predictions, turning the theory into an inversion program that can be tested directly by data rather than remaining at a purely explanatory level.

In what follows, we begin from minimal axioms and a readout dictionary and develop this settlement-dynamics framework step by step, discussing redshift and the Hubble law, dark matter and dark energy, light bending and planetary orbits, strong-field astrophysical structure, and the unified readouts of gravitational waves and induced gravity. We conclude with a multi-probe joint inversion scheme and a falsification checklist. Through this structured development, we aim to show that once the readout object is correctly chosen, the seemingly complex and fragmented phenomena of cosmology are in fact natural manifestations of a single physical ontology across scales and directions.

Four Retreats in Explaining Frequency Change and the "Three Major Patches" of Modern Cosmology. Many core issues in modern astronomy and cosmology appear to belong to different domains—redshift, gravitation, dark matter, dark energy, gravitational waves—yet historically they repeatedly revolve around one physical phenomenon: why does frequency undergo systematic change? The interpretive path of this question has almost fully mirrored the conceptual evolution of modern physics, gradually giving rise to several widely accepted yet mutually disconnected "patches" in contemporary cosmology.

(I) First layer of experience: medium waves and "wavelength stretching"

In early physical experience, frequency change is not mysterious. Whether sound waves in air or water waves on a surface, frequency changes can be intuitively understood as changes in crest spacing:

- a siren's pitch rises when approaching and falls when receding;
- incoming water waves are compressed, while outgoing waves are stretched.

In such systems, frequency change is described by the Doppler formula, with clear physical premises:

- an absolute propagation speed exists (sound speed, water-wave speed);
- an external medium serves as a reference background;
- frequency change arises from motion of the source or observer relative to the medium.

At this level, "wavelength stretching or compression" has clear, intuitive, experimentally verifiable meaning.

(II) Second layer of conflict: light has no medium, yet still changes frequency

When spectroscopy entered astronomy, an unsettling fact emerged: stellar spectra exhibit systematic shifts—both redshift and blueshift. The intuitive reaction was to analogize them to Doppler effects, but difficulties followed:

- light does not rely on any propagation medium;
- the Michelson–Morley experiment indicates no ether exists;
- the speed of light remains invariant in all inertial frames.

At this point, the explanation "wavelength is stretched" lost physical support: if there is no medium and no absolute reference frame, relative to what is it stretched?

(III) First patch: special relativity redefines time

To preserve light-speed invariance while explaining frequency change, special relativity made an aggressive but successful retreat: abandoning absolute time and absolute space. Frequency was redefined as $f = \frac{dN}{d\tau}$, where τ is proper time in each reference frame. Frequency change was no longer interpreted as physical deformation of a wave in space, but as differences in clock rates. This was the first patch of modern physics: frequency change shifted from "a wave property" to "a comparison of time measures."

(IV) Second patch: gravity can also change frequency

When gravity entered the relativistic framework, the situation became more complex. Experiments show that even when source and observer are at rest, light still changes frequency when propagating in a gravitational field (gravitational redshift). In general relativity, this is explained by:

- different locations have different rates of time flow;
- frequency change arises from spacetime curvature;
- "wavelength stretching" is reduced to an effect of coordinates and metric.

At this point, the original physical meaning of "wave stretching" is fully lost; frequency change is completely geometrized. This is the second patch: absorbing gravitational frequency change into spacetime geometry.

(V) Third patch: cosmological redshift and expansion of space itself

When spectral redshift is promoted to cosmic scale, new difficulties appear:

- redshift increases systematically with distance (Hubble law);
- redshift is nearly universal across directions;
- it cannot be explained by local gravity or motion.

Under light-speed invariance and absence of preferred direction, the remaining option is that space itself stretches. Cosmological redshift is written as $\lambda_{obs} = a(t)\,\lambda_{em}$ and interpreted as light being "stretched by space" during propagation. This is the third patch: attributing frequency change to global evolution of the cosmic scale factor.

(VI) Natural Origin 1: Space Units Are Not Assumptions, but Are Inversely Calibrated from the Hydrogen Spectrum

The starting point of the Natural Origin program is not cosmological fitting, but the most precise and stable microscopic experimental fact: the hydrogen spectrum. The key idea is that if nature contains a minimal space refresh unit, then its scale and rhythm must leave an indelible fingerprint in fine-spectral structure.

1. Inverse calibration of spatial scale from the fine-structure constant

Within the Natural Origin framework, the fine-structure constant is written in a purely geometric form as a ratio of length scales:

$$\alpha = \frac{1}{4\pi}\left(\frac{\ell_0}{\ell_\Omega}\right)^8$$

Here:

- ℓ_Ω is the rotational-unit enclosure scale (defined by $\hbar = \mu_g c\,\ell_\Omega^2$);
- ℓ_0 is the deeper linear scale of the space unit (SU).

This relation is not a fit; the exponent constraint is imposed jointly by the eightfold topology of rotational-unit enclosure and the order structure of fine-spectral splitting. Using experimental constants $\alpha^{-1} = 137.035999084(21)$, $\hbar = 1.054571817 \times 10^{-34}\, J \cdot s$, together with the calibrated μ_g and c, one can invert to obtain the SU scale: $\ell_0 \approx 3.76 \times 10^{-47}\, m$

This value is far smaller than the Planck length and is not a "minimum measurable length," but the minimal refresh linear scale.

2. Space-unit frequency and the minimal action unit

The refresh rhythm of the space unit follows directly from its linear scale:

$$\Omega_0 = \frac{c}{\ell_0} \approx 7.97 \times 10^{54}\ \mathrm{s}^{-1}$$

The corresponding minimal action unit is defined as:

$$S_0 = \mu_g c\, \ell_0^2 \approx 7.16 \times 10^{-57}\, J \cdot s.$$

This yields a key order-of-magnitude relation:

$$\hbar \approx 1.47 \times 10^{22}\, S_0$$

That is, about 10^{22} space units refresh cooperatively to form the minimal enclosable action at the rotational-unit level. This explains why $\hbar$ appears simultaneously as a continuous scale and as a bridge to discrete quanta.

3. Hierarchical structure: space unit – rotational unit – enclosure states

From the calibration above, a clear three-level hierarchy follows:

- Space unit (SU)
  length: $\ell_0 \approx 3.76 \times 10^{-47}\, m$
  frequency: $\Omega_0 \approx 7.97 \times 10^{54}\ \mathrm{s}^{-1}$
  action unit: $S_0 \approx 7.16 \times 10^{-57}\, J \cdot s$
- Rotational unit(SU_Ω)
  enclosure scale: $\ell_\Omega = \sqrt{\hbar/(\mu_g c)}$
  geometric origin of the Planck-scale enclosure level
- Closed units (CU–n)
  stable particles such as electrons and protons
  enclosure states formed by many rotational units under settlement conditions

This route is obtained entirely from experimental spectral lines → geometric constraints → inverted scale reconstruction, without relying on cosmological assumptions.

(VII) The Accumulating Difficulty of "Patches"

By this point, explanations of frequency change have undergone three key retreats:

1. from waveform change in a medium;
2. to time comparison in relativity;
3. to expansion of space in cosmology.

Each patch is numerically successful, but they are mutually independent, lack a unified ontological foundation, and introduce new tensions:

- dark matter is introduced to repair gravitational readouts;
- dark energy is introduced to repair apparent acceleration;
- tensions appear between cosmic age and high-redshift objects;
- the ontology of gravitational-wave energy and propagation still requires geometrized descriptions.

These phenomena jointly indicate a deeper fact: we have repeatedly explained "the frequency changed," yet have not answered what frequency is fundamentally a readout of.

(VIII) Transition Statement

If frequency is neither a purely geometric attribute of a wave nor a stretching result during propagation, but instead a readout of a more basic physical quantity, then the patch system above is not inevitable.

It is precisely under this motivation that the present work introduces the settlement anchor field $u(\mathbf{x}, t)$, and subsequently shows systematically that redshift, gravity, cosmological expansion, and their associated patches can be unified as manifestations of the same settlement readout across scales and directions.

Within the single-field framework $u(\mathbf{x}, t) = \ln \beta_r$, this work rewrites gravity, tides, and gravitational-wave structure and proposes three strong criteria as the falsifiability core:

(1) Single-field consistency criterion

Within the same structured system, dynamical readouts and lensing readouts must arise from the same $u(\mathbf{x})$: $\mathbf{g} = -c^2 \nabla u, \quad \alpha = 2 \int \nabla_\perp u \, ds.$

If no unified u can fit both dynamics and lensing observations simultaneously, the theory fails.

(2) Settlement-dispersion criterion

If SU refresh possesses a finite synchronization window, then gravitational lensing in vacuum permits an extremely weak frequency dependence: $\alpha(\omega)\alpha_0 \left[1 - \epsilon \left(\frac{\omega}{\omega_{SU}}\right)^2\right]$

If high-precision multi-band lensing observations strictly exclude any residual dispersion (after excluding plasma and medium effects), the theory fails.

(3) Multi-channel coherence criterion

Gravitational waves are propagation modes of settlement-field perturbations δu. A single event must satisfy that $h(t), \quad \delta f/f, \quad \delta\kappa$, and $\delta\gamma$ can all be explained as geometric projections of one and the same $\delta u(\mathbf{x}, t)$. If interferometers, clock arrays, and lensing modulation cannot close to a single δu, the theory fails.

This theory does not add fundamental fields and introduces no additional degrees of freedom. Its validity depends entirely on whether the three criteria above are satisfied.

Chapter 0 Space Units, Stepping–Settlement, and Light-Speed Invariance: The Ontological Starting Point of Settlement Dynamics

Role of Chapter 0

Chapter 0 is not background. It is the only permitted ontological starting point of the entire theory. It must accomplish four indispensable tasks:

1. strictly define the space unit (SU) and its minimal mathematical properties;
2. distinguish and hard-lock the two phases—stepping and settlement—to prevent confusion between propagation and existence;
3. derive light-speed invariance as a hardware upper bound from the relay mechanism, not as an assumption;
4. prove at the first-principles level that redshift is a source effect (a settlement-rhythm difference), not propagation stretching, and clarify and elevate its relation to historical experiments (e.g., the so-called Leddar effect).

0.1 Minimal definition of the space unit (SU): not a point, but a refresh unit

Definition 0.1 (Space unit).

Space is neither a continuous medium nor an abstract set of coordinates, but a network composed of minimal refresh units. Each space unit (SU) satisfies:

1. refreshability: SU updates its state with a fixed maximum refresh rhythm;
2. chirality: internal refresh has a fixed right-handed sense;
3. relayability: SU updates can synchronize with neighboring SUs to form propagation;
4. finite resources: each refresh can allocate only finite synchronization resources.

An SU is neither a "particle" nor a "field value," but the smallest executable physical unit.

0.2 Stepping phase and settlement phase: two physically distinct roles

Definition 0.2 (Stepping phase).

The stepping phase (denoted s) is the relay advance by which a state passes from one SU to the next. Its physical features:

- responsible only for propagation;
- produces no recordable outcomes;
- cannot close.

Definition 0.3 (Settlement phase). The settlement phase (denoted r) is a recordable state return completed inside an SU. Its physical features:

- responsible for existence and outcomes;
- produces readouts such as time, frequency, and mass;
- the only phase that can close.

0.3 Two-phase axiom and the budget identity

Axiom 0.1 (Orthogonality of two phases).

Within a single SU refresh, stepping and settlement share the same refresh resource and are orthogonal. Therefore:

$$\beta_s^2 + \beta_r^2 = 1 \tag{0.1}$$

Here:

- β_s: stepping-phase fraction (propagation fraction);
- β_r: settlement-phase fraction (existence fraction).

Conclusion 0.1.

Any physical state can be fully characterized by (β_s, β_r); there is no third fundamental phase.

0.4 Hardware origin of light-speed invariance: a relay upper bound, not an assumption

Proposition 0.1 (State of light).

Light is a pure stepping state:

$$\beta_s = 1, \quad \beta_r = 0. \tag{0.2}$$

Corollary 0.1 (Light-speed invariance).

Since:

- stepping propagation can occur only by SU-to-SU relay;
- each SU has a fixed refresh upper bound;
- relays cannot be parallel-stacked;

the propagation speed of light equals the maximal SU relay refresh rate, denoted:

$$v_{\max} = c \tag{0.3}$$

This result is independent of reference frames, coordinates, or measurement methods. Light-speed invariance is not a hypothesis about nature; it is a physical limit of the SU network.

0.5 Definition of settlement time: time is a readout, not a parameter

Definition 0.4 (Settlement time).

Recordable time is not the external parameter t, but the cumulative count of completed settlements:

$$d\tau = \beta_r \, dt \tag{0.4}$$

Here:

- t: external refresh-count parameter (not directly observable);
- τ: the internally recordable time of the system.

Conclusion 0.2.

Time is not background; it is a readout of settlement activity.

0.6 First-principles definition of frequency (closing redshift)

Definition 0.5 (Frequency).

Frequency is the number of settlements completed per unit external refresh-count time:

$$f \propto \frac{dN_{set}}{dt} \propto \beta_r \tag{0.5}$$

For the same spectral line, the proportionality constant is the same and cancels out.

0.7 Ontological proof that redshift must be a source effect (settlement-rhythm difference)

For the same atomic spectral line read at two locations $\mathbf{x}_{em}$ and $\mathbf{x}_{obs}$:

$$\frac{f_{obs}}{f_{em}} \frac{\beta_r(\mathbf{x}_{obs})}{\beta_r(\mathbf{x}_{em})} \tag{0.6}$$

Define redshift:

$$1 + z = \frac{f_{em}}{f_{obs}} \tag{0.7}$$

Then:

$$1 + z = \frac{\beta_r(\mathbf{x}_{obs})}{\beta_r(\mathbf{x}_{em})} \tag{0.8}$$

Conclusion 0.3 (Source-effect theorem for redshift).

Redshift is determined entirely by the settlement-rhythm difference between emitter and observer and is independent of propagation distance, path, or medium.

0.8 Clarification and elevation of the "Leddar effect"

Historically, certain experiments (often referred to as "the Leddar effect") indicate:

- without relative motion and without medium changes,
- changing the source-side environment can change spectral frequency.

In traditional frameworks these are treated as "local perturbations," not elevated to first principles. The upgrade of SFT (Settlement Field Theory) is: such experiments are not exceptions but direct evidence of the redshift ontology—frequency is not a propagation attribute but a source-side settlement-rhythm readout. In this theory they become the low-energy, controllable limit of the endpoint theorem for redshift.

0.9 Natural introduction of the settlement anchor field

From (0.4) and (0.5), β_r determines all readouts of time and frequency.
For differential and integral operations, introduce:

$$u(\mathbf{x}, t) \equiv \ln \beta_r\,(\mathbf{x}, t) \tag{0.9}$$

This introduces no new physics; it is only a variable substitution.

0.10 Closing of Chapter 0: three unavoidable ontological conclusions

1. space consists of SUs; the refresh upper bound sets light-speed invariance;
2. propagation (stepping) and existence (settlement) are strictly separated ontologically;
3. redshift must be a source effect: a readout of settlement-rhythm difference.

0.11 Direct connection to Chapter 1

- Chapter 1 will take $u = \ln \beta_r$ as the sole ontological variable and rigorously derive the redshift endpoint theorem and the Hubble law;
- all later chapters are only allowed to use $((u, \nabla u, \partial_t u)$ as physical inputs.

0.12.1 Time is Not a Coordinate but a Readout of Completed Settlement

In SFT, three quantities must be strictly distinguished:

1. External refresh parameter t
- Role: a refresh-count label
- Status: not directly observable
- Function: a parametrization tool for describing the SU refresh ordering
- Not physical time
2. Settlement time τ
- Definition: $d\tau = \beta_r(\mathbf{x}, t)\, dt$
- Meaning: the number of completed settlement returns per unit external refresh-parameter time
- Status: the only recordable and comparable physical time
3. Optical-path time / propagation readout
- Definition: $\Delta t_{prop} = \frac{1}{c}\int e^{-u(\mathbf{x})}\, ds$
- Meaning: a readout required for propagation synchronization
- Not the time experienced by the system itself

Conclusion 0.12.1 (Time-readout theorem).
In SFT, any statement of "how long it lasted" or "how long the system evolved" refers to settlement time τ, not the parameter t.

0.12.2 Why Light "Has No Time": Zero-Time Readout of a Pure Stepping State

By the two-phase axiom, light corresponds to a pure stepping state: $\beta_s = 1, \quad \beta_r = 0$.
Substituting into the time definition gives: $d\tau = \beta_r \, dt = 0$.
Corollary 0.12.1 (Zero settlement time of light).
During propagation, light experiences no settlement time. That is, light can propagate, bend, and interfere, yet it does not "age." This is not a philosophical statement but a mathematical consequence directly implied by the definitions.

0.12.3 Why Objects "Age": Settlement Occupation as the Flow of Time

For any enclosure state ($\beta_r > 0$), the recordable time along a worldline Γ is: $\tau[\Gamma] = \int_\Gamma e^{u(\mathbf{x},t)} \, dt$.
Therefore:

- time is not a background that "flows uniformly";
- time is the accumulated result of a system completing returns within its settlement environment.

Conclusion 0.12.2 (Locality of time).
Different systems, different paths, and different environments can experience different settlement times. In SFT, time is a path functional, not a global scalar.

0.12.4 Time Saturation: An Inevitable Result in Strongly Negative u Regions

If a system enters a strongly settlement-suppressed region $u(\mathbf{x}, t) \ll 0$, then its settlement time is exponentially suppressed: $\tau = \int e^u \, dt$
As $u \to -\infty$, τ approaches a finite upper bound.
Theorem 0.12.3 (Time-saturation theorem).
In regions of strongly negative settlement anchor field, the recordable time of a system must saturate; external parameter time continues to grow but no longer corresponds to internal evolution. This provides the mathematical root of JWST observations described as "high redshift yet highly evolved."

0.12.5 Why the Age of the Universe Cannot Be Single-Valued

Let "the age of the universe" be a single unified time readout τ_{uni}. In SFT, the time along different worldlines Γ_i is $\tau_i = \int_{\Gamma_i} e^{u(\mathbf{x},t)} \, dt$.

If $u(\mathbf{x})$ contains any spatial structure (galaxies, voids, clusters), then $\tau_i \neq \tau_j$.
Theorem 0.12.4 (Non-uniqueness of cosmic age).
In any nontrivially structured universe, there is no unique "cosmic age." This theorem is isomorphic to a "no-single-Hubble-constant" theorem.

0.12.6 Unified Relation of Time, Redshift, and Gravity

In SFT, the following three are different readouts of the same ontology:

- time: $d\tau = e^u \, dt$
- redshift: $\ln(1 + z) = u_{obs} - u_{em}$
- gravity: $\mathbf{g} = -c^2 \nabla u$

Time is not an independent dimension; it is the longitudinal readout of the settlement anchor field.

0.12.7 Closure Statement of This Section

In SFT, time is neither a background coordinate nor a geometric dimension, but a recordable readout of the settlement phase completing return. Light has no time because it does not settle; objects experience time because they settle continuously. The slowing, saturation, and non-uniqueness of time are direct consequences of the settlement anchor field structure, not dynamical corrections.

0.12.8 Ontological Readout of Mass: Settlement Locking of SU Degrees of Freedom

In SFT, mass is neither a third ontological entity nor an a priori property of "material atoms." It is the readout result of the settlement degree of freedom being persistently locked within SU refresh resources. In its minimal form, the settlement fraction $\beta_r(\mathbf{x}, t)$ describes the proportion of settlement returns that can be completed per unit external refresh-count time dt. When a region persistently satisfies $\beta_r > 0$ and can stably maintain it, that region appears as an "existence/enclosure state," and its macroscopic readout is mass.

To interface with continuous density, introduce a locking-density readout (no new field; purely a readout mapping):

$$u(\mathbf{x}, t) \equiv \ln \beta_r\,(\mathbf{x}, t), \rho_{lock}(\mathbf{x}, t) \propto \beta_r(\mathbf{x}, t) = e^{u(\mathbf{x},t)}$$

Hence, the ontological meaning of "mass" is the statistical manifestation of settlement locking over volume, not an independent particle or an independent field. Emphatically, weak-field mechanics depends only on derivatives of u (such as $\nabla u, \nabla^2 u$), not on the absolute zero of u (gauge shift $u \to u + C$).

0.13 Unified Symbol Table and Hierarchical Partitioning of the Whole Book

To avoid symbol ambiguity, operator mixing, and confusion of hierarchy, this section provides the unified symbol table for Natural Origin 6 (SFT). The book strictly distinguishes:

- ontological readout objects
- operator objects
- propagation degrees of freedom
- source-term objects
- engineering readout objects

Cross-level mixing is forbidden.

0.13.1 Ontological readout object

$$u(\mathbf{x}, t) \equiv \ln \beta_r\,(\mathbf{x}, t)$$

- the only physical readout object
- the logarithmic readout of settlement fraction
- all physical phenomena are expressible as functions of u or operator readouts of u
- no additional field is introduced

0.13.2 Two-phase budget structure

$$\beta_s^2 + \beta_r^2 = 1$$

Symbols and meaning:

- β_s: stepping-phase fraction (propagation fraction)
- β_r: settlement-phase fraction (existence fraction)

Readout time: $d\tau = \beta_r\, dt = e^u dt$

0.13.3 Gravity readout (static layer)

$$\mathbf{g}(\mathbf{x}, t) = -c^2 \nabla u$$

- static gravitational readout
- first-order spatial operator
- no independent degree of freedom introduced

0.13.4 Dynamic-layer objects

(1) induced-gravity field (near field) $\mathbf{g}_{ind} = -c^2\, \partial_t \nabla u$

- cannot exist independently
- does not satisfy a free propagation equation
- a local dynamical term

(2) gravitational waves (far field)

Propagation degree of freedom: $\phi(\mathbf{x}, t) = \partial_t u$satisfying $\left(\frac{1}{c^2}\partial_t^2 \nabla^2\right)\phi 0$

- freely propagating
- far-field amplitude proportional to 1/r
- propagation speed c

0.13.5 Source terms and closure structure

Static closure equation: $\nabla^2 u - \frac{4\pi G_*}{c^2}\rho_{enc} + \Lambda_u$

Symbols and meaning:

- G_*: effective gravitational constant (lowest-order closure constant)
- ρ_{enc}: enclosure-density readout
- Λ_u: background-curvature readout

0.13.6 Path readout objects

(1) redshift endpoint theorem: $1 + z = \exp(u_{obs} - u_{em})$

(2) path accumulation law: $\kappa(s) \equiv -\nabla u \cdot \hat{\mathbf{n}} \quad \ln(1 + z) = \int_\gamma \kappa\, ds$

(3) Hubble path readout: $H_\gamma = \frac{c}{d_\gamma}\int_\gamma \kappa\, ds$

- a path functional
- not allowed to degenerate into a constant

0.13.7 Lensing deflection operator

$$\alpha = 2\int_\gamma \nabla_\perp u\, ds$$

Notes:

- the factor 2 arises from the dual-refresh constraint
- it is not an empirical correction

0.13.8 Interferometer strain readout

$$h(t) = -(\langle \delta u \rangle_x \langle \delta u \rangle_y$$

- a difference readout of u
- consistent with the propagation degree of freedom ϕ

0.13.9 Operator stratification principle

- static layer: ∇u (not propagating)

- near-field dynamical layer: $\partial_t \nabla u$ (not propagating)
- propagation layer: $\partial_t u$ (propagating)

0.13.10 Forbidden mixing rules

1. $\partial_t \nabla u$ must not be treated as a propagating wave.
2. ∇u must not be written as an independent field.
3. no additional tensor degrees of freedom may be introduced.
4. all physical readouts must be traceable to u.

0.13.11 Constants and units

- c: refresh-relay speed
- G_*: closure constant
- Λ_u: background-curvature readout
- u: dimensionless readout
- ϕ: propagation degree of freedom

0.13.12 Unified summary

All formulas in the book can be rewritten as either:

- spatial operator readouts of u,
- temporal operator readouts of u, or
- path-integral readouts of u.

No physical entity independent of u exists within the theory.

0.14 Methodological Boundary Statement: Two Orthogonal-Phase Refresh of Space Units as an Axiomatic Fact

Within the unified-field framework (SFT), we adopt the following methodological stance: space units (SU) possess two orthogonal phase channels; refresh resources are budgeted between a stepping phase and a settlement phase; and the speed of light c serves as the upper bound of refresh. Its mathematical expression is: $\beta_s^2 + \beta_r^2 = 1, u = \ln \beta_r$

This theory treats the structure above as an axiomatic-level fact, not as a conclusion requiring further explanation.

It must be made explicit that:

- this paper does not attempt to answer why space units must refresh in this way;
- that question belongs to a deeper hardware theory or a meta-physical level;
- the goal of SFT is to build, upon this minimal axiomatic structure, a unified readout dictionary and a testable derivation chain.

Accordingly, the tasks of this theory are:

1. to derive redshift, bending, orbits, strong-field criticality, and dynamical perturbations from these axioms;
2. to provide a unified description in terms of the settlement anchor field $u(\mathbf{x}, t)$;
3. to supply cross-probe consistency criteria and falsifiability tests.

If observations contradict these derivations, the theory is falsified.

0.14.1 Boundary clarification

The two-orthogonal-phase refresh structure of space units is not an empirically fitted result. It is an abstracted structure distilled from the minimal consistency requirements needed to unify the explanations of the following phenomena:

- invariance of the speed of light;

- recordability of time;
- gravitational-gradient readouts;
- orbital closure;
- strong-field criticality;
- dynamical propagation.

Whether there exists a deeper theory that can derive this refresh structure is a future research question; it does not affect the internal consistency and testability of the derivation chain presented here.

0.15 Minimality of the Two-Orthogonal-Phase Refresh Structure: Removing It Causes Logical Collapse

This section introduces no new axioms. It argues that the two-orthogonal-phase refresh of space units is not an arbitrary choice, but the minimal consistent structure required to unify macroscopic and microscopic phenomena. Its minimality is illustrated by proof sketches via contradiction.

0.15.1 If phase orthogonality is removed

Assume the stepping phase and settlement phase are not orthogonal and can simultaneously occupy the same refresh resource. Then:

- local closure could occur during propagation;
- light would acquire time readouts during propagation;
- redshift would become path-dependent rather than an endpoint difference.

This directly breaks the endpoint theorem: $1 + z = \exp(u_{obs} - u_{em})$

Hence, a non-orthogonal structure is incompatible with the redshift endpoint theorem.

0.15.2 If the budget identity is removed

Assume there is no identity $\beta_s^2 + \beta_r^2 = 1$. Then:

- no unique refresh upper bound can be defined;
- the speed of light would no longer be an absolute limit;
- time readouts could not be uniquely defined;
- the gravitational gradient ∇u could not be unified with time scaling.

This would cause:

- failure of the Lorentz limit;
- non-unique time readouts;
- loss of stable orbital closure.

0.15.3 If the light-speed refresh upper bound is removed

If refresh does not have a unified upper bound c, then:

- wave-equation propagation speeds would not be unique;
- causal structure would lose unification;
- gravitational waves and light could not share a common propagation limit.

This conflicts with the observed existence of a single propagation upper bound.

0.15.4 If only one phase exists

Assume there is no stepping/settlement distinction and only one refresh phase exists. Then:

- no definition of “recordable time” exists;
- propagation cannot be distinguished from existence;
- orbital closure cannot separate transverse deflection from longitudinal adjudication;
- strong and weak nuclear processes cannot be distinguished as bundling versus

decoupling.

Such a structure cannot simultaneously explain:

- light having no time;
- matter having time;
- horizon time-freezing;
- quantum discretization.

0.15.5 Minimality conclusion

In summary:

- the two-phase structure cannot be removed;
- orthogonal budgeting cannot be removed;
- the light-speed refresh upper bound cannot be removed;
- the settlement phase being the only closable channel cannot be removed.

Therefore, the two-orthogonal-phase, light-speed refresh structure of space units is the minimal consistent axiomatic structure required by the present unified-field theory. This definition does not claim to be the ultimate cosmic origin, but within the current range of physical readouts, any simpler structure cannot simultaneously satisfy observation and logical consistency.

0.16 Unifying Phase Orthogonality and the "Vertical Principle": Geometric Orthogonality as a Continuous Projection of the Refresh Structure

0.16.1 Problem statement

In SFT there are two types of "orthogonality":

1. orthogonality in the refresh-resource layer: $\beta_s^2 + \beta_r^2 = 1$, where stepping phase s and settlement phase r do not interfere in refresh-bandwidth space;
2. the vertical principle in the continuous-geometry layer: propagation direction is orthogonal to the settlement gradient, and transverse deflection is separated from longitudinal adjudication.

Without clarification, these may be mistaken as two independent axioms. This section provides a unified account: they are the same structure expressed at different levels.

0.16.2 Orthogonality in the refresh layer

In the discrete SU refresh model, the bandwidth W_0 of each refresh cycle is allocated between two execution channels:

- stepping channel: responsible for propagation;
- settlement channel: responsible for return and readout.

Budget identity:

$$\beta_s^2 + \beta_r^2 = 1 \tag{0.16.1}$$

Here "orthogonality" refers to independent components in refresh-resource space, not to directions in physical three-dimensional space.

0.16.3 Directional separation in the continuous limit

When the discrete refresh structure is continuousized at macroscopic scales:

- the stepping phase corresponds to propagation direction;
- the settlement phase corresponds to time readout and existence adjudication.

Because the two are orthogonal in refresh space, the continuous limit must manifest as: $\perp \nabla u$, where $u = \ln \beta_r$.

This is precisely the geometric vertical principle.

0.16.4 Origin of transverse deflection and longitudinal adjudication

From earlier derivations of light bending and orbital structure: $\alpha = 2\int \nabla_\perp u\, ds$

The "transverse" versus "longitudinal" split is not an additional assumption; it is the natural spatial projection of two-phase orthogonality:

- transverse deflection arises from directional organization of the stepping channel;
- longitudinal adjudication arises from return conditions of the settlement channel.

They share the same origin.

0.16.5 The vertical principle is not an independent axiom

Thus, the vertical principle is not an independent geometric hypothesis, but an inevitable projection of the two-orthogonal-phase refresh structure into continuous space.

If phase orthogonality were removed:

- propagation and settlement could not be separated;
- light would acquire time readouts during propagation;
- the redshift endpoint theorem would fail;
- transverse deflection and longitudinal adjudication would mix.

Hence geometric orthogonality is not extra structure; it is a consequence of refresh structure.

0.16.6 Unified conclusion

Two-orthogonal-phase refresh of space units is a structural fact in refresh-resource space; the vertical principle is its geometric expression in the macroscopic continuous limit. They belong to different levels but share the same ontology.

Closure of this section.

In SFT, geometric verticality is not a new physical assumption; it is the projection of orthogonal budgeting of refresh bandwidth into continuous space. Therefore, all "transverse/longitudinal decompositions" in the book are traceable to the two-phase refresh structure and do not constitute extra axioms.

0.16.7 Ontological meaning of gravity and curvature: first- and second-order inhomogeneity of the locking field

In SFT, gravity and spatial curvature are no longer treated as independent "geometric ontologies," but as different-order derivative readouts of the settlement-locking field $u = \ln \beta_r$:

(1) First-order readout: gravity $\mathbf{g}(\mathbf{x}, t) = -c^2 \nabla u(\mathbf{x}, t) = -c^2 \nabla \ln \beta_r\,(\mathbf{x}, t)$

Physical meaning: gravity is not an "additional force," but a readout bias caused by the tilt (inhomogeneity) of the settlement-locking fraction in space.

(2) Second-order readout: tides/curvature $T_{ij}(\mathbf{x}, t) = c^2 \partial_i \partial_j u(\mathbf{x}, t) = c^2 \partial_i \partial_j \ln \beta_r\,(\mathbf{x}, t)$

Physical meaning: curvature is not "space being bent," but the rate at which the locking tilt varies across directions. When ∇u is approximately constant, the result looks like "uniform acceleration" (locally re-scalable); when $\partial_i \partial_j u \neq 0$, the result is an irreducible real structure (tides). Thus, tides are the lowest-order criterion of "real structure" in this theory.

0.17 Relation between the space field equation and settlement readouts: unifying the dynamical layer and the readout layer

The Mother Equation of the Physical World—The Space Field Equation provides the dynamical evolution of the space field $g(\mathbf{x}, t)$, written schematically as: $\mathcal{M}[g] = 0$

The settlement anchor field used in this book, $u(\mathbf{x}, t) = \ln \beta_r\,(\mathbf{x}, t)$ is not a new fundamental field. It is a readout projection of solutions to the mother equation under refresh rules: $u = \mathcal{R}[g]$

Here $\mathcal{R}$ denotes the local projection of the dynamical field induced by the refresh-bandwidth budgeting rule; β_r is the fraction of refresh resources occupied by the settlement phase. Under this mapping, all observables in SFT become readout forms of the mother-equation solution family:

- time readout: $d\tau = e^u dt$
- redshift: $1 + z = \exp(u_{obs} - u_{em})$
- deflection: $\alpha = 2\int \nabla_\perp u\, ds$
- gravity readout: $\mathbf{g} = -c^2 \nabla u$

Therefore: the space field equation describes "how the field evolves," while NO5-U describes "how evolution is read out." They lie at different levels and do not compete. SFT does not replace the space field equation; it supplies a unified refresh-projection layer that renders macroscopic cosmology and strong-field phenomena as invertible readouts of dynamical field solutions.

This chapter shows that when space is understood as a network of refreshable units and time is understood as a readout of completed settlement, light-speed invariance and the source nature of redshift are no longer empirical assumptions, but direct mathematical consequences of the SU relay mechanism. This conclusion forms an irreplaceable ontological starting point of SFT.

0.18 Settlement-readout rewriting of relativity and field equations: from refresh budgeting to a comparative expression of general relativity

0.18.1 Refresh budgeting as the starting point of relativity

In SFT, space-unit refresh satisfies $\beta_s^2 + \beta_r^2 = 1$, where:

- β_s: stepping phase (propagation capacity)
- β_r: settlement phase (existence readout)
- $u = \ln \beta_r$

This budget identity is the starting point of relativistic structure, not a geometric hypothesis.

0.18.2 Essence of special relativity

In pure vacuum, $\nabla u = 0$, i.e., the settlement field is spatially constant. Define propagation readouts $v = c\beta_s$ and $d\tau = \beta_r dt$. Substituting into the budgeting identity $\left(\frac{v}{c}\right)^2 + \beta_r^2 = 1$,

gives: $d\tau = \sqrt{1 - \frac{v^2}{c^2}}\, dt$

Thus, special relativity equals refresh-budget readouts in a constant settlement field. Its essence is a resource constraint between propagation capacity and existence readout.

0.18.3 Essence of general relativity

When the settlement field has a spatial gradient $\nabla u \neq 0$, one obtains: $\mathbf{g} = -c^2 \nabla u, \quad \alpha = 2\int \nabla_\perp u\, ds.$

Thus, general relativity equals a readout theory of settlement fields with gradient structure. Curvature is not ontology; it is a readout manifestation of gradient structure.

0.18.4 Settlement expression of the equivalence principle

Expand at a point: $u(\mathbf{x})u_0 + \frac{1}{c^2}\mathbf{a} \cdot \Delta\mathbf{x} + \frac{1}{2}H_{ij}\Delta x_i \Delta x_j + \cdots$

The first-order gradient corresponds to constant acceleration: $\mathbf{g} = -c^2 \nabla u = -\mathbf{a}$ and can be removed by a freely falling coordinate choice. Hence, the equivalence principle is the local removability of the first-order gradient of the settlement field, while the second-order term is

not removable. The second-order term $H_{ij} = \partial_i \partial_j u$ corresponds to tidal effects.

0.18.5 Readout-level correspondences between GR field equations and SFT

We list the readout-level correspondences between GR and SFT.

(1) Time factor

GR: $d\tau^2 = -g_{00}\, dt^2$

SFT: $d\tau = e^u dt$

Correspondence: $g_{00} \leftrightarrow -e^{2u}$

Weak field: $g_{00} \simeq -(1 + 2u)$

(2) Newtonian potential

GR: $g_{00} \simeq -(1 + 2\Phi/c^2)$

Correspondence: $\Phi \leftrightarrow c^2 u$

and $\mathbf{g} = -c^2 \nabla u \leftrightarrow -\nabla\Phi$

(3) Weak-field limit of Einstein equation

GR: $\nabla^2 \Phi = 4\pi G\rho - \Lambda c^2$

SFT: $\nabla^2 u = -\frac{4\pi G_*}{c^2}\rho_{enc} + \Lambda_u$

They are of the same form when weak-field and sign conventions match.

(4) Light deflection

GR (weak lensing): $\alpha = \frac{2}{c^2}\int \nabla_\perp (\Phi + \Psi)\, ds$

For $\Phi = \Psi$: $\alpha = \frac{4}{c^2}\int \nabla_\perp \Phi\, ds$

SFT: $\alpha = 2\int \nabla_\perp u\, ds$

Using $\Phi = c^2 u$ yields formal agreement.

(5) Gravitational waves

GR: $h_{\mu\nu}^{TT} = 0$

SFT: $\phi = \partial_t u, \quad \phi = 0$

SFT replaces the metric tensor as the propagating object with the settlement readout field.

Unified summary

GR uses the metric to describe structure; NO5-U uses $u = \ln \beta_r$ to describe readouts. In the weak-field and observable sector: $g_{00} \leftrightarrow -e^{2u}, \Phi \leftrightarrow c^2 u, \mathbf{g} \leftrightarrow -c^2 \nabla u$

Therefore:

- special relativity = constant u
- general relativity = gradient u
- equivalence principle = local removability of first-order ∇u

The essence of relativity is not spacetime geometry as ontology, but how refresh budgeting is distributed in space; Einstein's field equation provides dynamical structure, while SFT provides a unified readout layer.

0.18.6 Structural comparison: GR vs. SFT (settlement-readout layer)

Structural level	General Relativity (GR)	SFT (Settlement readout layer)	Physical interpretive difference
Basic variable	Metric $g_{\mu\nu}$	Settlement anchor field $u = \ln \beta_r$	GR describes geometry; this work describes readouts
Time factor	$d\tau^2 = -g_{00}dt^2$	$d\tau = e^u dt$	time arises from settlement fraction
Newtonian potential	Φ	$c^2 u$	potential = settlement readout
Gravity	$-\nabla\Phi$	$-c^2\nabla u$	gravity = settlement gradient
Static field equation	$\nabla^2\Phi = 4\pi G\rho - \Lambda c^2$	$\nabla^2 u = -\frac{4\pi G_*}{c^2}\rho_{enc} + \Lambda_u$	same weak-field form
Light deflection	$\alpha = \frac{4}{c^2}\int \nabla_\perp \Phi ds$	$\alpha = 2\int \nabla_\perp u ds$	the two GR potentials merge at the readout layer
Gravitational waves	$(h_{\mu\nu} = 0)$	$((\partial_t u) = 0)$	tensor propagation vs. readout propagation
Equivalence principle	local removability of ∂g	local removability of first-order ∇u	geometric language vs. refresh language

Chapter 1 Endpoint Redshift Theorem and the Inevitability of the Hubble Law under the Settlement Anchor Field

Positioning of This Chapter

This chapter is the first unavoidable mathematical checkpoint of the whole book. Its aim is not to explain a particular cosmological mechanism, but to prove—strictly from the minimal axioms established in Chapter 0—that:

1. Redshift must be an endpoint ratio of settlement readouts, independent of the propagation path, waveform details, or geometric language;
2. The Hubble law must emerge, as a first-order gradient expansion of the settlement anchor field on cosmic scales;
3. A single global Hubble constant is mathematically impossible in a structured universe in general, and the resulting "tension" is an inevitable consequence of path functionals.

1.1 Re-statement of the Settlement Anchor Field and Readout Variables

Based on the two-phase axiom and the settlement-priority axiom in Chapter 0, introduce the settlement anchor field:

$$\beta_r(\mathbf{x}, t) \in (0,1], u(\mathbf{x}, t) \equiv \ln \beta_r(\mathbf{x}, t) \leq 0 \tag{1.1}$$

Here β_r is the fraction of settlement that can be completed per unit external refresh-count increment dt; u is its logarithmic variable, convenient for differentiation and integration.

The time readout is defined as

$$d\tau = \beta_r(\mathbf{x}, t)\, dt = e^{u(\mathbf{x},t)}\, dt \tag{1.2}$$

Here t is the external refresh-count parameter and τ is the recordable settlement-time readout of the system.

All derivations in this chapter use only these definitions—no metric, no scale factor, and no

medium model is introduced.
To highlight the first-principles structure of redshift and the Hubble law, we begin with a static or quasi-static background:

$$\partial_t u_0(\mathbf{x}) = 0 \tag{1.3}$$

A time-dependent generalization is opened in Section 1.7.

1.2 Settlement Definition of Frequency (Definition-Level, Not Replaceable)

Definition 1.1 (Frequency).
Frequency is not "an intrinsic property of the wave," but the number of settlement completions per unit external refresh-count time. Let N_{set} denote the count of settlement events. Then

$$f(\mathbf{x}) \propto \frac{dN_{set}}{dt} \propto \beta_r(\mathbf{x}) \tag{1.4}$$

The proportionality constant depends only on the internal scale of the specific physical system (e.g., the internal settlement mode corresponding to an atomic spectral line) and cancels exactly in comparisons of the same line.
Corollary 1.1 (Frequency ratio for the same spectral line).
For the same spectral line read at two points $\mathbf{x}_1, \mathbf{x}_2$,

$$\frac{f(\mathbf{x}_2)}{f(\mathbf{x}_1)} = \frac{\beta_r(\mathbf{x}_2)}{\beta_r(\mathbf{x}_1)} \tag{1.5}$$

This relation is independent of propagation path and geometric structure; it depends only on endpoint values of the settlement anchor field.

1.3 Endpoint Redshift Theorem (First-Principles Result)

Definition 1.2 (Redshift).

$$1 + z \equiv \frac{f_{em}}{f_{obs}} \tag{1.6}$$

Theorem 1.1 (Endpoint redshift theorem).
For any stable spectral line emitted at $\mathbf{x}_{em}$ and observed at $\mathbf{x}_{obs}$,

$$1 + z = \frac{\beta_r(\mathbf{x}_{obs})}{\beta_r(\mathbf{x}_{em})} = exp(u\mathbf{x}_{obs}) - u(\mathbf{x}_{em}) \tag{1.7}$$

Proof (unavoidable).
By the frequency definition (1.4), the frequency of the same line is proportional to β_r. Taking the ratio in the redshift definition (1.6) yields (1.7). QED.
Important note (first closure of this chapter).
Equation (1.7) depends only on endpoints, not on the propagation path. This implies:

- redshift is not necessarily the result of "wave stretching during propagation";
- redshift does not require a medium or a scale factor;
- redshift is first and foremost an endpoint comparison of readout rhythms.

1.4 Path Representation of Redshift: Exponential Accumulation Law

In a static background $u(\mathbf{x})$, take the logarithm of (1.7):

$$\ln(1 + z) = u(\mathbf{x}_{obs}) - u(\mathbf{x}_{em}) \tag{1.8}$$

Let the light path (or signal path) be parameterized by arc length s, with $\mathbf{x}(s)$. Then

$$u(\mathbf{x}_{obs}) - u(\mathbf{x}_{em}) = \int_{em}^{obs} \nabla u(\mathbf{x}) \cdot d\mathbf{x} = \int_0^d \nabla u\big(\mathbf{x}(s)\big) \cdot \hat{\mathbf{n}}(s)\, ds \tag{1.9}$$

where $\hat{\mathbf{n}}(s) \equiv d\mathbf{x}/ds$ is the unit tangent and d is the geometric path length.
Define the local projection of the settlement gradient along the path:

$$\kappa(s) \equiv -\nabla u\big(\mathbf{x}(s)\big) \cdot \hat{\mathbf{n}}(s) \tag{1.10}$$

Then (1.9) becomes

$$\ln(1+z) = \int_0^d \kappa(s)\, ds \tag{1.11}$$

Exponentiating yields the exponential accumulation law:

$$1 + z = \exp\left(\int_0^d \kappa(s)\, ds\right) \tag{1.12}$$

Note (second closure).
Equation (1.12) shows that redshift is mathematically equivalent to a line integral of the logarithmic gradient of the settlement anchor field along the path. Therefore, as long as $\nabla u \neq 0$, redshift must accumulate with path length; such accumulation does not require "space expansion."

1.5 Mathematical Emergence of the Hubble Law (Without Any Cosmological Assumptions)
In the small-redshift, weak-gradient limit, assume $\kappa(s)$is approximately constant along the path scale:

$$\kappa(s) \simeq \kappa_0 \tag{1.13}$$

From (1.12),

$$1 + z \simeq e^{\kappa_0 d} \simeq 1 + \kappa_0 d. \tag{1.14}$$

Hence

$$z \simeq \kappa_0 d \tag{1.15}$$

Define the Hubble readout

$$H_0 \equiv c\,\kappa_0 \tag{1.16}$$

Then

$$cz \simeq H_0 d \tag{1.17}$$

Conclusion (core of this chapter).
The Hubble law is not a dynamical law; it is a first-order gradient expansion of the settlement anchor field along the path direction. The "Hubble constant" is the readout of the settlement gradient:

$$H_0 = -c\,\nabla u \cdot \hat{\mathbf{n}} \quad (weak - gradient, local\ approximation) \tag{1.18}$$

1.6 The Hubble Readout Is a Path Functional: Necessary and Sufficient Condition for a Single H_0
To avoid misreading (1.17) as a global constant, we state its proper definition. In general, define

the effective Hubble readout along a path γ:

$$H_\gamma \equiv \frac{c}{d_\gamma}\int_\gamma \kappa\, ds \tag{1.19}$$

where $\kappa(s) \equiv \nabla u \cdot \hat{\mathbf{n}}$. (We do not rewrite both terms again to avoid double counting.)
Theorem 1.2 (Necessary and sufficient condition for a single global Hubble constant).
If within a connected domain one has $H_\gamma \equiv H_0$ for all paths γ, then

$$\kappa(\mathbf{x}) \equiv constant \quad (almosteverywhere) \tag{1.20}$$

That is, the projection of the settlement gradient along any direction must be constant everywhere.
Proof sketch.
If for any two points all path integrals are proportional only to path length, then κ cannot contain spatial fluctuations; otherwise one can construct closed loops for which the integral fails to be conservative. QED.
This theorem implies: once the universe contains any structural fluctuations (galaxies, voids, clusters), κ cannot be constant everywhere, and a single global H_0 must break mathematically—this is the inevitable root of "Hubble tension" (see later chapters).

1.7 Equivalence Theorem: Endpoint Difference vs. Path Accumulation

1.7.1 Endpoint form

In a static background: $1 + z = \exp(u_{obs} - u_{em})$

1.7.2 Path-integral form

From the basic differential relation: $du = \partial_t u\, dt + \nabla u \cdot d\mathbf{x}$

Integrating along a causal path Γ_c: $u_{obs} - u_{em} \int_{\Gamma_c}(\partial_t u\, dt + \nabla u \cdot d\mathbf{x})$

1.7.3 Endpoint–path equivalence theorem

In SFT, redshift is defined first by the endpoint settlement difference:

$$1 + z = \exp(u_o - u_e), \quad u = \ln \beta_r. \tag{1.7.3a}$$

If the background is static ($\partial_t u = 0$), then by the fundamental theorem of calculus,

$$u_o = -u_e \int_\Gamma \nabla u \cdot d\mathbf{x} \tag{1.7.3b}$$

Thus

$$\ln(1 + z) = \int_\Gamma \nabla u \cdot d\mathbf{x} \tag{1.7.3c}$$

If the background is dynamic, then along a causal path:

$$\ln(1 + z) = \int_\Gamma (\partial_t u + \nabla u \cdot \dot{\mathbf{x}})dt \tag{1.7.3d}$$

Theorem (equivalence).
The endpoint expression and the path-integral expression are strictly equivalent; the path integral is only an expansion form of the endpoint difference.
Physical-level distinction.

- Redshift: an endpoint settlement readout (occurs at emission and reception);
- Deflection: transverse-gradient accumulation (occurs during propagation).

No settlement occurs along the propagation itself; thus propagation does not "generate redshift."
The path-integral expression does not change the endpoint nature of redshift.

Equivalence statement.
Redshift endpoint difference is the closed form of settlement accumulation along a causal path. The endpoint formula does not deny path contribution; it is the closed expression of the path integral.

1.7.4 Physical interpretation

- Static background → endpoint-difference form;
- Dynamic background → explicit path integral.

No contradiction exists; they are simply different representations.

1.8 Chapter Closure: Three Unavoidable Mathematical Conclusions

Without introducing expansion assumptions, metric tensors, or dark matter/dark energy, this chapter rigorously derives:

1. Endpoint redshift theorem: $1 + z = \exp(u_{obs} - u_{em})$
2. Exponential accumulation law: $1 + z = \exp(u_{obs} - u_{em})$
3. Hubble law and Hubble readouts: $cz \simeq H_0 d, \quad H_\gamma = \frac{c}{d_\gamma}\int_\gamma \kappa \, ds$

Therefore, the Hubble law is not a hypothesized "cosmic mechanism," but an inevitable first-order expansion whenever the settlement anchor field has a gradient; and Hubble tension in a structured universe is an inevitable consequence of the fact that a path functional cannot be reduced to a constant.

1.8.1 Gauge Freedom and the Physical Invariance of the Strong-Settlement Criterion

Problem statement.
The settlement anchor field $u(\mathbf{x}, t) = \ln \beta_r$ has a gauge freedom $u(\mathbf{x}, t) \to u(\mathbf{x}, t) + C$, which does not affect endpoint readouts such as redshift. However, in the strong-settlement limit, the event horizon and settlement-failure boundary are defined by $\beta_r \to 0$ (i.e., $u \to -\infty$). We must show that the physical criterion is gauge independent.
Definition 1.8.1 (Physical accessibility function).
Define the accessibility function along a direction $\hat{\mathbf{n}}$:

$$\mathcal{A}(\mathbf{x}, \hat{\mathbf{n}}) \equiv \lim_{\varepsilon \to 0^+} \frac{1}{\varepsilon} \int_0^\varepsilon e^{u(\mathbf{x}+s\hat{\mathbf{n}})} \, ds \tag{1.8.1}$$

Theorem 1.8.1 (Gauge-invariant settlement-failure criterion).
Under $u \to u + C$, $\mathcal{A}(\mathbf{x}, \hat{\mathbf{n}}) \longrightarrow e^C \mathcal{A}(\mathbf{x}, \hat{\mathbf{n}})$, but the distinction $\mathcal{A} = 0$ versus $\mathcal{A} > 0$ is unchanged.
Proof.
The factor e^C is a positive finite constant; it rescales $\mathcal{A}$ but cannot change whether it is zero. QED.
Conclusion 1.8.1.
The event horizon and settlement-failure boundary should be defined by

$$\mathcal{A}(\mathbf{x}, \hat{\mathbf{n}}) = 0 \tag{1.8.2}$$

rather than by imposing a threshold on the absolute value of u. Hence the strong-field criterion is mathematically gauge invariant, without requiring a global reference u_∞.

1.9 Connection to Later Chapters

- Chapter 2: derive dark matter (gradient structure) and dark energy (curvature structure) from the source equation for $\nabla^2 u$;
- Light-bending chapter: the transverse deflection operator $\alpha = 2\int \nabla_\perp u\, ds$ is of the same origin as redshift;
- Gravitational-wave / induced-gravity chapter: propagation of $\partial_t \nabla u$ extends redshift from a static endpoint readout to a dynamic readout (see Eq. 1.21).

1.9.1 Causal-path functional for dynamic redshift

Problem statement.

In a time-dependent background $u(\mathbf{x}, t)$, the generalized expression $\ln(1+z) = \int \partial_t u\, dt + \int \nabla u \cdot d\mathbf{x}$ must specify the path and causal constraint to avoid causal ambiguity in strongly dynamical systems (e.g., binary mergers).

Definition 1.9.1 (Settlement causal path).

A path $\Gamma_c = \mathbf{x}(t)$ is called a settlement causal path if:

$$\left|\frac{d\mathbf{x}}{dt}\right| = c, (\mathbf{x}(t), t) always lies within the source causal cone \tag{1.9.1}$$

Theorem 1.9.1 (Causal functional of dynamic redshift).

For any time-dependent settlement field $u(\mathbf{x}, t)$, redshift is strictly defined by:

$$\ln(1+z) = \int_{\Gamma_c} (\partial_t u\, dt + \nabla u \cdot d\mathbf{x}) \tag{1.9.2}$$

The integral is taken only along the settlement causal path Γ_c.

Conclusion 1.9.1.

- In the static limit $\partial_t u = 0$, (1.9.2) reduces to the endpoint redshift theorem;
- in the dynamic limit, redshift is the settlement accumulation readout along a causal propagation path;
- redshift always retains its source-effect nature and does not involve energy exchange during propagation.

1.10 The Essential Relation Between the Lorentz Transformation and Redshift: Time Re-scaling and the Equivalence to Endpoint Settlement Difference

1.10.1 Problem positioning

In the conventional relativistic framework, redshift is often explained through the Lorentz transformation or its generalizations. In SFT, redshift is given by the endpoint difference of the settlement anchor field: $1 + z = exp(u_{obs} - u_{em})$.

This section answers: what role does the Lorentz transformation truly play? Is it the physical cause of redshift, or merely another mathematical expression of the same fact?

1.10.2 Core function of the Lorentz transformation: re-scaling of time readout

In special relativity, the key outcome is not "space deformation," but re-scaling of time readout.

For relative velocity v, $d\tau = \sqrt{1 - \frac{v^2}{c^2}}\, dt \equiv \beta\, dt, \beta \in (0,1]$.

This β is a time-scaling factor: it does not describe propagation itself, but the rate at which a system completes recordable evolution relative to the parameter time t.

1.10.3 First-principles definition of time in SFT

In SFT, time is not a coordinate but a readout of completed settlement: $d\tau = \beta_r(\mathbf{x}, t)\, dt$, $u(\mathbf{x}, t) = \ln \beta_r\, (\mathbf{x}, t)$

Thus:

- β_r is the unique physical factor for time readout;
- all "slower/faster time" phenomena reduce to changes in β_r.

1.10.4 Ontological status of the Lorentz factor in SFT

Comparing the special-relativistic time-scaling factor with the SFT time definition yields the direct correspondence: $\beta_r \leftrightarrow \sqrt{1 - \frac{v^2}{c^2}}$

Hence the time factor in the Lorentz transformation is not an independent geometric assumption, but simply the value taken by the settlement occupation ratio β_r in the specific physical case of uniform relative motion.

In other words:

- the Lorentz transformation is not "the origin of time";
- it is a reparameterized form valid when β_r is constant.

1.10.5 Two expressions for redshift and their equivalence

(a) Relativistic expression (traditional).

For relative speed (v), longitudinal Doppler redshift is: $1 + z = \sqrt{\frac{1+v/c}{1-v/c}} = \exp(\eta)$.

where $\eta = arctanh(v/c)$ is rapidity.

(b) SFT expression (endpoint settlement difference).

In SFT: $1 + z = \exp(u_{obs} - u_{em})$

If the two endpoints differ only by relative motion (i.e., different time readouts), then $u = \ln \beta_r = \frac{1}{2} \ln\left(1 - \frac{v^2}{c^2}\right)$, so that $u_{obs} - u_{em} = \eta$, which matches the relativistic expression exactly.

1.10.6 Essential difference: what is cause, what is representation

Although the mathematics is equivalent, the physical interpretation is fundamentally different:

Aspect	Traditional relativity	SFT
Source of time change	frame transformation	settlement occupation ratio β_r
Essence of redshift	time-coordinate relation	endpoint settlement readout difference
Status of Lorentz transform	effective rewrite under special conditions	a special case of first-principles structure
Domain	inertial motion	arbitrary settlement backgrounds (gravity, cosmology)

In SFT, the Lorentz transformation is not the physical cause of redshift; it is merely the mathematical clothing of redshift in the simplest settlement background.

1.10.7 Why SFT must go beyond the Lorentz transformation

The Lorentz transformation assumes: $\beta_r =$ constant, $\nabla u = 0$.Once any of the following appears:

- spatial structure ($\nabla u \neq 0$);
- path effects (redshift and deflection coexist);
- time evolution ($\partial_t u \neq 0$);

a single Lorentz transformation no longer applies, while redshift remains strictly given by $\ln(1 + z) = u_{obs} - u_{em}$. Therefore, the Lorentz transformation is a special case of SFT, not its foundation.

1.10.8 Theorem-style summary

Theorem 1.10 (Lorentz–redshift equivalence theorem).
Within SFT, the Lorentz time transformation of special relativity is equivalent to the constant-value case of the settlement anchor field $u = \ln \beta_r$ under uniform motion in a structureless background. In general, redshift is given by the endpoint difference of the settlement anchor field, and the Lorentz transformation is only the exponential rewrite of that endpoint difference under special conditions.

1.10.9 Closing sentence of the section

The Lorentz transformation does not explain the essence of redshift; it is only a coordinate expression of redshift in the simplest settlement background. In SFT, the physical ontology of redshift is always the endpoint difference of the settlement anchor field, not a geometric transformation between reference frames.

Chapter 2 The Static Source Equation of the Settlement Anchor Field: Inevitable Readouts of Dark Matter and Dark Energy

Positioning of This Chapter

Chapter 1 rigorously derived—along the "path direction"—that redshift and the Hubble law are unavoidable endpoint/line-integral readouts of the settlement anchor field $u(\mathbf{x})$. This chapter completes a second mathematical closure—along the "spatial direction":
without introducing dark-matter particles, without introducing a dark-energy field, and without introducing a metric tensor, we prove that what is called "dark matter" and "dark energy" are, respectively, unavoidable readouts of the gradient structure and the curvature structure of the same variable $u(\mathbf{x})$.
This chapter does only one thing: it gives—and closes—the unique static closure equation for $u(\mathbf{x})$, and derives from it rotation curves, lensing, and the acceleration appearance.

2.1 Definition of the Gravity Readout

From the readout dictionary in Chapter 0, the macroscopic gravitational-acceleration readout is defined as

$$\mathbf{g}(\mathbf{x}) = -c^2 \nabla u(\mathbf{x}) \tag{2.1}$$

Emphasis: this is a readout definition, not an approximation, analogy, or empirical fit. Throughout the rest of the book, "gravity" in all dynamical and orbital phenomena refers to (2.1).

2.2 Necessity of Static Closure: Why a Source Equation for u Must Exist

Chapter 1 provides path readouts of u (redshift and Hubble), but without a spatial equation, $u(\mathbf{x})$ remains an arbitrary function in three-dimensional space and the theory cannot become testable.
To make SFT an auditable system, we must provide a minimal static equation that:

1. recovers the Newtonian weak-field structure $g \sim GM/r^2$;
2. yields a unique principal solution in the exterior vacuum region;
3. permits background curvature on large scales (corresponding to the "dark energy appearance");
4. introduces no additional ontological degrees of freedom.

The minimal choice satisfying these four requirements is a Poisson-type closure:

$$\nabla^2 u(\mathbf{x}) = -\frac{4\pi G_*}{c^2}\,\rho_{enc}(\mathbf{x}) + \Lambda_u \tag{2.2}$$

Here:

- $\rho_{\mathrm{enc}}(\mathbf{x})$: enclosure-occupation density (settlement source term), the settlement-readout source corresponding to observable mass;
- G_*: the coupling constant between settlement geometry and enclosure occupation (matched to Newton's constant in the weak-field limit);
- Λ_u: a background-curvature constant (the readout baseline corresponding to the acceleration appearance).

Key declaration of this chapter: Eq. (2.2) is the only static closure equation permitted in the book; all "dark component" readouts must be derived from it.

Minimal-closure principle: Eq. (2.2) is the only allowed lowest-order closure form because:

1. it recovers the $1/r^2$ weak-field structure;
2. it introduces no new degrees of freedom;
3. it preserves linear superposition, enabling multi-probe inversion.

Higher-order terms are allowed only as effective-theory corrections and do not change the ontology.

2.2.1 Settlement support condition (definition-level welding to prevent "dark matter renaming")

To prevent ρ_{enc} from being misread as an arbitrarily assignable "new matter field," we impose the settlement support condition:

$$\rho_{\mathrm{enc}}(\mathbf{x}) \neq 0 \Longrightarrow \beta_r(\mathbf{x}) > 0 \Leftrightarrow u(\mathbf{x}) \succ \infty \tag{2.3}$$

In all pure propagation regions ($\beta_r = 0$),

$$\rho_{\mathrm{enc}}(\mathbf{x}) \equiv 0 \tag{2.4}$$

Therefore ρ_{enc} is not an independent new entity; it is a source readout of settlement-state existence. This constraint cuts SFT off, ontologically, from "dark-matter particle fields."

2.2.2 Ontological standard for the source term: $\boldsymbol{\rho_{enc}}$ is a locking-density readout, not a new dark entity

In SFT, ρ_{enc} in (2.2) is not allowed to be interpreted as an "unknown matter field that can be freely introduced." Its ontological meaning is: ρ_{enc} is the effective density readout of settlement enclosure/locking (the statistical manifestation of locked degrees of freedom over volume), and is co-originating with $u = \ln \beta_r$. In other words, the source term is not a third kind of entity, but a readout mapping of settlement-locking structure in the continuous limit. Thus the "no dark components" stance of SFT does not ignore anomalies; it refuses to automatically extrapolate anomalies into independent ontological degrees of freedom.

This standard rigidly welds dynamics ($\mathbf{g} = -c^2\nabla u$), lensing ($\alpha = 2\int \nabla_\perp u\, ds$), and the source equation ($\nabla^2 u$) into a single-field closed loop.

2.3 Spherically Symmetric General Solution: No Skipped Steps (Computable Exterior Principal Form)

Assume a static spherical symmetry $u(\mathbf{x}) = u(r)$, $r = |\mathbf{x}|$. Then (2.2) becomes

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{du}{dr}\right) = -\frac{4\pi G_*}{c^2}\rho_{enc}(r) + \Lambda_u \tag{2.5}$$

Define the enclosed mass function

$$M_{enc}(r) \equiv \int_0^r 4\pi r'^2 \rho_{enc}(r')\, dr \tag{2.6}$$

Integrating (2.5) once (using regularity of u and boundedness at r=0 gives

$$\frac{du}{dr} = -\frac{G_* M_{enc}(r)}{c^2 r^2} + \frac{\Lambda_u}{3} r \tag{2.7}$$

Substituting into the gravity-readout definition (2.1) yields

$$g(r) \equiv |\mathbf{g}| c^2 \left|\frac{du}{dr}\right| \frac{G_* M_{enc}(r)}{r^2} - \frac{\Lambda_u c^2}{3} r \tag{2.8}$$

Equations (2.7)–(2.8) are the only computational entry for all conclusions that follow: the first term is the local gradient readout; the second term is the background-curvature readout.

2.4 Inevitable Appearance of "Dark Matter Readout": Rotation Curves Are Not Halos but the Logarithmic Structure of ∇u

2.4.1 Rotation-curve equation (readout level)

Circular-orbit condition (centripetal readout):

$$\frac{v^2(r)}{r} = g(r) \tag{2.9}$$

On galactic scales where $|\Lambda_u| r^2 \ll G_* M/r$, the curvature term can be neglected; combining (2.8) and (2.9) gives

$$v^2(r) \simeq \frac{G_* M_{enc}(r)}{r} \tag{2.10}$$

2.4.2 Strict implication of flat rotation curves

If observations give a flat rotation curve $v(r) \simeq v_0$, then (2.10) requires

$$M_{enc}(r) \propto r \tag{2.11}$$

Substituting into (2.7) yields

$$\frac{du}{dr} \propto -\frac{1}{r} \Rightarrow u(r) \sim -\ln r + const \tag{2.12}$$

Conclusion 2.1 (Dark-matter readout).

A galactic "dark-matter halo" does not need to be introduced as extra matter; a flat rotation curve is equivalent to a logarithmic gradient structure of the settlement anchor field u as in (2.12). In this framework, "dark matter" is the spatial structure readout of ∇u, not a new entity.

2.5 Strict Co-origin of Gravitational Lensing: Deflection Must Be Given by the Same $u(\mathbf{x})$

In SFT, the lensing deflection operator (weak-deflection approximation) is

$$\alpha(\theta) = 2\int_\gamma \nabla_\perp u\, ds \tag{2.13}$$

Together with (2.1)–(2.2), this implies that the ∇u used for rotation curves and the $\nabla_{\perp} u$ used for lensing must come from the same function $u(\mathbf{x})$.

Conclusion 2.2 (Lensing–rotation common origin).

If one uses a given $u(\mathbf{x})$ to explain rotation curves, then lensing deflection cannot introduce an independent free "dark halo." Lensing and rotation curves are different projected readouts of the same settlement-gradient structure. This turns the "dark matter problem" into an auditable consistency proposition: does there exist a single $u(\mathbf{x})$ that explains both classes of observations simultaneously?

2.6 Inevitable Appearance of "Dark Energy Readout": Background Curvature Produces the Acceleration Appearance (Without Vacuum Energy)

On cosmological scales in the exterior region one may take $\rho_{enc} \to 0$. Then (2.7) reduces to

$$\frac{du}{dr} = \frac{\Lambda_u}{3} r \tag{2.14}$$

Integrating (taking an equivalent reference such as $u(\infty) \to 0$ gives

$$u(r) = u_0 + \frac{\Lambda_u}{6} r^2 \tag{2.15}$$

Substituting into (2.1) yields the large-scale gravity readout:

$$\mathbf{g}(r) = -c^2 \nabla u = -\frac{\Lambda_u c^2}{3} \mathbf{r} \tag{2.16}$$

This is an "extrapolated readout term" proportional to distance, which naturally produces second-order terms in the exponential accumulation law for redshift (Chapter 1, Section 1.7 form).

Conclusion 2.3 (Dark-energy readout).

The cosmological acceleration appearance is not evidence for an antigravity energy component, but a readout shadow of the background curvature Λ_u of the settlement anchor field.

2.7 Unified Mathematical Expression of Dark Matter and Dark Energy

From (2.7),

$$\frac{du}{dr} = \underbrace{-\frac{G_* M_{enc}(r)}{c^2 r^2}} gradientstructure(dark - matterreadout) +$$

$$\underbrace{\frac{\Lambda_u}{3} r} backgroundcurvature(dark - energyreadout) \tag{2.17}$$

Thus "dark matter" and "dark energy" are not two new entities; they are two limiting readouts of the same field $u(\mathbf{x})$ in its gradient structure and curvature structure.

2.8 Settlement Consistency Criterion (Hard Audit Axis of This Chapter: No Dark Matter, but Stronger Constraint)

Criterion 2.1 (Single-field consistency).

SFT requires that there exists a single $u(\mathbf{x})$ satisfying simultaneously:

$$\exists\, u(x) s.t. \mathbf{g} = -c^2 \nabla u$$

$$\nabla^2 u = -\frac{4\pi G_*}{c^2} \rho_{enc} + \Lambda_u$$

$$\alpha = 2\int \nabla_\perp u\, ds$$

$$v^2/r = g \tag{2.18}$$

If no such $u(\mathbf{x})$ exists, then any patch-style scheme (including extra dark-matter free fields) must be treated as inconsistent, because it breaks the single-variable closed-loop ontology of SFT.

2.8.1 Higher-order self-coupling and effective-theory statement

Problem statement.

The source equation $\nabla^2 u = -\frac{4\pi G_*}{c^2}\rho_{enc} + \Lambda_u$ is linear; its effective-theory status and possible higher-order corrections must be stated explicitly.

Proposition 2.8.1 (Lowest-order effective theory).

Equation (2.2) is the lowest-order effective equation of settlement dynamics at macroscopic scales, valid under:

$$|\nabla u|^2 \ll 1,\ \ell_{obs} \gg \ell_{SU}. \tag{2.8.1}$$

Definition 2.8.1 (Higher-order settlement self-coupling correction).

At higher precision one may add self-coupling terms:

$$\nabla^2 u = -\frac{4\pi G_*}{c^2}\rho_{enc} + \Lambda_u + \lambda\,|\nabla u|^2 + \mathcal{O}(\partial^4 u) \tag{2.8.2}$$

Here λ is a weak-coupling parameter.

Conclusion 2.8.1.

- All results of this chapter hold rigorously under (2.8.1).
- Higher-order terms mainly affect the photon-sphere location in strong fields, strong-lensing residuals, and merger ringdown.
- The linear source equation is a declared effective-theory limit, not a logical loophole.

2.8.2 Strong single-field constraint (against "dark matter renaming")

In SFT, $\nabla^2 u = -\frac{4\pi G_*}{c^2}\rho_{enc} + \Lambda_u$ requires $u(\mathbf{x})$ to be a global single field. One must not:

introduce one field u_1 for rotation curves, and another u_2 for lensing.

Strong consistency demands a single $u(\mathbf{x})$ such that $\nabla u \Rightarrow v(r)$, $\nabla_\perp u \Rightarrow \alpha(\theta)$, and $u_{obs} - u_{em} \Rightarrow z$; otherwise the theory fails.

Key difference: ΛCDM allows different halo distributions for different systems; SFT does not. This is a stronger constraint, not a weaker relabeling.

2.9 Chapter Closure: Three Unavoidable Conclusions (Mathematically Irrefutable Points)

1. A source equation is necessary; otherwise $u(\mathbf{x})$ is untestable. The minimal closure is (2.2).
2. The dark-matter readout must appear: flat rotation curves imply $\Rightarrow u(r) \sim -\ln r$ as in (2.12).
3. The dark-energy appearance must occur: $\rho \to 0 \Rightarrow u \sim r^2 \Rightarrow g \sim r$ as in (2.16).

Thus this chapter completes the mathematical closure from "single-variable readouts" to "spatial-

structure closure": dark matter and dark energy are no longer independent entities, but gradient-structure and curvature-structure readouts of the same settlement anchor field.

2.10 Connection to Later Chapters (Explicit Route)

- Chapter 3: invert $u(\mathbf{x})$ from observation sets (redshift point values, rotation-curve derivatives, lensing Radon constraints);
- Light-bending chapter: provide the microscopic mechanism of $\alpha = 2\int \nabla_\perp u\, ds$ and the settlement origin of the coefficient "2";
- Gravitational-wave / induced-gravity chapter: discuss dynamical perturbations of $\partial_t \nabla u$ and multi-channel auditing;
- JWST chapter: explain the instrument-window amplification effect and time saturation in strongly negative u regions.

These reinforcement sections show that settlement anchor field theory can provide strict mathematical closures in gauge invariance, causal consistency, weak-field limits, and strong-field extendability; all potential deviations are converted into computable, observable, falsifiable higher-order corrections rather than structural defects.

Chapter 3 The Inverse Problem of the Settlement Anchor Field: Invertibility and Near-Uniqueness of Recovering $u(\mathbf{x}, t)$ from Observations

Positioning of This Chapter

Chapters 1 and 2 have completed a strict closure of the forward problem: given $u(\mathbf{x}, t)$, one can compute redshift, the Hubble law, rotation curves, lensing, and the acceleration appearance.
This chapter performs the most critical step of the paper: it proves that the inverse problem is mathematically solvable, constrained, and—under reasonable conditions—nearly unique. In other words, it answers the reviewer's inevitable question: not only can you "explain," can you also "infer from data"?

3.1 Strict Definition of the Observational Data Space (Problem Setup)

Define the observational collection (with finite errors) as:

$$\mathcal{D} z_i, D_i, v_j(r), \alpha_k(\theta), H_\ell, h_m(t) \tag{3.1}$$

These correspond to the following readout operators:

- Redshift operator (endpoints)

$$\mathcal{O}_z[u](\mathbf{x}_{\mathrm{em}}, \mathbf{x}_{obs}) \exp(u_{obs} - u_{\mathrm{em}}) - 1 \tag{3.2}$$

- Rotation-curve operator

$$\mathcal{O}_v u \sqrt{r\, c^2 \left|\frac{du}{dr}\right|} \tag{3.3}$$

- Lensing operator (Radon-type)

$$\mathcal{O}_a |u| = 2\int_{\gamma(\theta)} \nabla_\perp u\, ds \tag{3.4}$$

- Hubble readout operator

$$\mathcal{O}_H u \frac{c}{d_\gamma} \int_\gamma \kappa\,(\mathbf{x})\, ds; \tag{3.5}$$

- Gravitational-wave / induced-gravity operator

$$\mathcal{O}_h u - \frac{L}{2}\left(\partial_x \delta u - \partial_y \delta u\right) \tag{3.6}$$

3.2 Mathematical Statement of the Inverse Problem (Core)

Inverse problem.

Given the observational collection $\mathcal{D}$, find a function $u(\mathbf{x}, t)$ in $\mathcal{U}$ such that $\mathcal{O}[u] \approx \mathcal{D}$ within the allowed observational error bounds.

Choose the admissible function space $\mathcal{U}$ as a Sobolev space:

$$\mathcal{U} = H^2(\Omega) \cap C^1(\Omega) \tag{3.7}$$

where $\Omega \subset \mathbb{R}^3$ is the domain of interest. The choice of H^2 is necessary because lensing and gravitational-wave readouts involve ∇u and $\nabla^2 u$.

3.3 Construction of the Inversion Functional (Computable)

Define the weighted residual functional:

$$\mathcal{J}[u] \sum_i w_i^{(z)} |\mathcal{O}_z[u] - z_i|^2 + \sum_j w_j^{(v)} |\mathcal{O}_v[u] - v_j|^2 + \sum_k w_k^{(\alpha)} |\mathcal{O}_\alpha[u] - \alpha_k|^2 + \sum_\ell w_\ell^{(H)} |\mathcal{O}_H[u] - H_\ell|^2 + \sum_m w_m^{(h)} |\mathcal{O}_h[u] - h_m|^2. \tag{3.8}$$

The inverse problem is reduced to a variational problem:

$$u^\star = arg \min_{u \in \mathcal{U}} \mathcal{J}\,[u] \tag{3.9}$$

3.4 Existence Theorem (Unavoidable)

Theorem 3.1 (Existence of a minimizer).

If weights $w_i > 0$, observational errors are finite, and $\mathcal{J}[u]$ is coercive (strongly bounded from below and growing) on $\mathcal{U}$, then at least one minimizer $u^\star$ exists.

Proof sketch.

- $\mathcal{J}[u] \geq 0$;
- the Sobolev space $H^2(\Omega)$ is reflexive under the weak topology;
- $\mathcal{J}[u]$ is weakly lower semicontinuous;
- the direct method yields existence of a minimizer.

QED. This already rules out the objection that "the model is non-invertible."

3.5 Non-uniqueness and Equivalence Classes (Honest and Essential)

Because lensing and redshift primarily provide line integrals and endpoint differences, the inverse problem possesses, in a strict sense, a gauge freedom:

$$u(\mathbf{x}) \to u(\mathbf{x}) + C \tag{3.10}$$

This shift does not change any physical readouts.

Moreover, if only a single class of observations is used (e.g., lensing alone), a typical Radon-inversion null space may remain.

3.6 Near-Uniqueness Theorem (Core "Killing Power")

Theorem 3.2 (Multi-probe near-uniqueness).

If the observational collection $\mathcal{D}$ includes at least the following three complementary types of

information:

1. Endpoint-type (redshift / clock arrays): provides pointwise differences of u;
2. Gradient-type (rotation curves / lensing): provides radial and transverse projections of ∇u;
3. Time-type (gravitational waves / induced gravity): provides evolution of $\partial_t \nabla u$;

then, after removing the shift gauge (3.10), the inverse solution in $\mathcal{U}$ is nearly unique.
Structural proof idea.

- Assume two solutions u_1, u_2 both fit $\mathcal{D}$;
- let $\delta u = u_1 - u_2$;
- endpoint-type data ⇒ $\delta u(\mathbf{x}_i) =$ constant;
- gradient-type data ⇒ $\nabla \delta u \approx 0$ (almost everywhere);
- time-type data ⇒ $\partial_t \nabla \delta u \approx 0$;
- together with boundary conditions ⇒$\delta u =$ constant.

QED. This is one of the paper's mathematically irrefutable core theorems.

3.7 Stability and Error Propagation

For observational perturbations $\delta\mathcal{D}$, the solution perturbation satisfies:

$$\|\delta u\|H^1 \leq C\, \|\delta\mathcal{D}\|\ell^2 \tag{3.11}$$

The constant C depends on probe geometry coverage and weight configuration. This implies:

- the solution depends continuously on noise;
- there is no pathological amplification in which tiny noise produces huge physical structures.

3.8 Chapter Conclusions

This chapter completes three tasks that are rarely completed in unified-field work:

1. it converts cosmological explanation into a strict inverse problem;
2. it proves existence and near-uniqueness of inversion under multi-probe constraints;
3. it provides an executable variational reconstruction route.

Therefore, SFT is not "a story that fits the universe," but a readout-field theory that is, in principle, reconstructible from data up to a simple gauge.

3.9 Interface to Later Chapters

- Chapter 4: construct the numerical implementation of $\mathcal{J}[u]$ using JWST, SNe, and lensing samples;
- Chapter 9: connect the null-space structure of the lensing operator $\mathcal{O}_\alpha$ to the SU microscopic mechanism of light bending;
- Later chapters: connect the gravitational-wave operator $\mathcal{O}_h$ to the filtering relation of near-field induced gravity.

This chapter shows that under joint constraints from multiple observational classes, inversion of the settlement anchor field $u(\mathbf{x}, t)$ is not only feasible in principle, but also nearly unique under reasonable mathematical assumptions, thereby upgrading SFT from an explanatory framework to an auditable inversion theory.

Chapter 4 Time Readouts of the Settlement Anchor Field: Cosmic Age, Time Saturation, and the Mathematical Inevitability of JWST High-Redshift Anomalies

Positioning of This Chapter

The first three chapters have completed three foundational closures of SFT:

- Chapter 1: the endpoint/gradient inevitability of redshift and the Hubble law;
- Chapter 2: the static source equation in which dark matter (gradient) and dark energy (curvature) appear as readouts;
- Chapter 3: existence and near-uniqueness of multi-probe inversion.

This chapter addresses a core problem that has long troubled modern cosmology and has been dramatically amplified in the JWST era: why do high-redshift objects appear "too old," "too bright," and "too compact"? Must cosmic age be globally consistent in a mathematical sense? This chapter will prove strictly that these phenomena are neither observational errors nor "early anomalous formation," but an inevitable consequence of settlement-time saturation in strongly negative u regions.

4.1 Settlement Definition of Time (First-Principles, Not Replaceable)

From the settlement readout dictionary in Chapter 1, the recordable time of a system is not coordinate time t, but settlement time:

$$d\tau(\mathbf{x}, t) = \beta_r(\mathbf{x}, t)\, dt = e^{u(\mathbf{x},t)}\, dt \tag{4.1}$$

Here:

- t: external refresh-count parameter (not directly observable);
- τ: recordable time corresponding to the number of completed settlements.

Important declaration.

All later discussions of "age," "evolution time," and "formation time" refer to τ, not to t.

4.2 Worldline Time as a Path Functional

Let a celestial object (or its constituent matter) evolve along a worldline Γ. Its experienced settlement time is

$$\tau[\Gamma] = \int_\Gamma e^{u(\mathbf{x},t)}\, dt \tag{4.2}$$

Because this definition is a path functional, it immediately yields two important consequences:

1. Different objects may experience different settlement times because their paths differ;
2. Even when the start and end redshifts are the same, the internal "evolution time" can differ dramatically.

4.3 Time-Saturation Theorem in Negative-u Regions (Core Mathematical Result)

Consider a worldline Γ passing through a spatial region Ω where

$$u(\mathbf{x}, t) \le u_{\min} < 0 \tag{4.3}$$

Then for any finite external time interval $[t_1, t_2]$,

$$\tau_\Omega \int_{t_1}^{t_2} e^{u(\mathbf{x}(t),t)}\, dt \;\le\; e^{u_{\min}(t_2 - t_1)} \tag{4.4}$$

If $|u_{\min}| \gg 1$, then

$$\tau_\Omega \ll (t_2 - t_1) \tag{4.5}$$

Theorem 4.1 (Time-saturation theorem).
In strongly negative settlement anchor field regions, settlement time τ is exponentially suppressed relative to external parameter time t. As $u \to -\infty$, τ saturates, and further growth of t produces almost no additional recordable time.
The proof follows directly from the inequality estimate implied by (4.2) and (4.3). QED.

4.4 Cosmic Age Cannot Be a Single-Valued Function (Mathematical Necessity)

Suppose "cosmic age" is mistakenly defined as a unified time readout $\tau_{\mathrm{uni}} = \int_0^{t_0} d\,t$.

In SFT, the truly comparable quantity is

$$\tau_i = \int_{\Gamma_i} e^{u(\mathbf{x},t)}\, dt \tag{4.6}$$

If spatial structure exists (galaxies, clusters, voids), then $u(\mathbf{x})$ is non-constant, and for different worldlines Γ_i, Γ_j,

$$\tau_i \neq \tau_j \tag{4.7}$$

Theorem 4.2 (Non-single-valuedness of cosmic age).
In any nontrivially structured universe, there is no globally consistent settlement-time readout that can be called a unique "cosmic age." Mathematically, this theorem is isomorphic to the Chapter 1 theorem that a single global Hubble constant cannot hold.

4.5 A Strict Explanation of High-Redshift "Over-Aged" Objects

Let a high-redshift object lie in a strongly negative u region. Its redshift is given by the endpoint formula:

$$1 + z = \exp(u_{obs} - u_{em}) \tag{4.8}$$

Meanwhile, its internal evolution time is

$$\tau_{\mathrm{int}} = \int_{\Gamma_{\mathrm{int}}} e^{u}\, dt \tag{4.9}$$

If the object enters a settlement-saturation region $u \ll 0$ early, then:

- the redshift (z) can continue to grow through endpoint differences;
- but the internal settlement time τ_{int} can reach a substantial value already at early stages.

Therefore one must mathematically obtain:

$$z \;\; large\; and \quad \tau_{\mathrm{int}} \;\; also\; large \tag{4.10}$$

This matches the JWST class of objects described as "high redshift yet already highly chemically evolved."

4.6 Settlement Corrections to Luminosity Distance and Angular-Diameter Distance

Define the settlement form of luminosity distance as

$$D_L \propto \int_\gamma e^{-u(\mathbf{x})}\, ds \tag{4.11}$$

and the angular-diameter distance as

$$D_A \propto \int_\gamma e^{u(\mathbf{x})}\, ds \tag{4.12}$$

In strongly negative u regions:

- e^{-u} is exponentially amplified → objects appear overly bright;
- e^{u} is exponentially compressed → objects appear overly compact.

Thus the three main JWST anomaly fingerprints:

1. high redshift
2. high luminosity
3. small angular scale

are not three separate problems in SFT, but three exponential projections of the same $u(\mathbf{x})$.

4.7 Convolutional Amplification by Instrument Windows (Quantitative Entry Point)

Let the JWST bandpass window function be $W(\lambda)$. Then the measured signal is

$$\mathcal{S}_{obs} = \int W(\lambda)\, \mathcal{S}_{true}(u, \lambda)\, d\lambda \tag{4.13}$$

Because spectral energy from $u \ll 0$ regions is compressed into the infrared window, the convolution necessarily amplifies its visibility. This step introduces no new physical assumption; it is a mathematical consequence of standard signal processing.

4.8 Chapter Conclusions (Fourth Closure)

This chapter proves, in a strict mathematical sense:

1. time is a settlement path functional, not a global coordinate;
2. negative-u regions necessarily produce time saturation;
3. cosmic age is necessarily non-single-valued in a structured universe;
4. JWST high-redshift anomalies are natural projections of exponential readouts.

Therefore the so-called "universe is too young" paradox is not a physical contradiction, but a mistaken comparison between non-comparable time readouts.

4.9 Interface to Later Chapters

- Chapter 5: jointly invert strong-negative $u(\mathbf{x})$ regions together with lensing and rotation curves;
- Chapter 7: discuss settlement criteria for photon spheres and horizons in the strong-negative u limit;
- Chapter 8: connect time saturation to the distinction between induced gravity and gravitational-wave propagation.

Within SFT, time is not a uniformly flowing background; it is a recordable readout of completed settlement. A strongly negative settlement anchor field inevitably produces time saturation, thereby making high redshift and "over-aged objects" mathematically compatible without any additional hypotheses.

4.10 Settlement Readout Interpretation of the Cosmic Microwave Background: An Isotropic

Redshift Background in the Time-Saturation Limit

4.10.1 Problem positioning

The cosmic microwave background appears as a nearly isotropic, nearly perfect blackbody background radiation with temperature around $2.7K$, and exhibits weak anisotropy at the relative level of 10^{-5}. In standard cosmology, the CMB is explained as relic radiation after photon decoupling at recombination, relying on early thermal equilibrium, acoustic oscillations, and initial-condition settings.

In SFT, the CMB is not treated as an independent origin object, but is repositioned as a natural readout limit of the settlement anchor field under extreme propagation conditions.

4.10.2 Basic stance of SFT

In SFT, all optical and time-related readouts arise from the same settlement anchor field $u(\mathbf{x}, t) = \ln \beta_r(\mathbf{x}, t)$, where β_r is the settlement-phase occupation fraction in SU refresh resources. Light satisfies $\beta_r = 0$ and $\beta_s = 1$: during propagation, light performs no settlement and produces recordable readouts only at emission and reception.

Hence, in SFT the CMB does not correspond to a special "cosmic epoch," but to the limiting superposition of three conditions:

1. an extremely early endpoint settlement environment ($u_{em} \ll 0$);
2. extremely long propagation paths (cosmic-scale path integrals);
3. time-saturation effects in strongly negative settlement anchor regions.

4.10.3 Settlement readout interpretation of temperature

In SFT, both frequency and temperature are settlement readouts, with endpoint relations:

$$\frac{\nu_{obs}}{\nu_{em}} = \exp(u_{obs} - u_{em}), T_{obs} = T_{em} \exp(u_{obs} - u_{em})$$

Therefore, the observed microwave temperature scale does not require a particular initial temperature; it is determined by the endpoint settlement difference. When u_{em} is sufficiently negative, even a broad range of T_{em} is compressed by the exponential factor into a narrow observational window, naturally producing a microwave-scale background readout.

4.10.4 Inevitability of isotropy: a consequence of time saturation

Chapter 4 has already proven that when a system is in a strongly negative settlement anchor field region $(u \ll 0)$, its recordable settlement time $\tau = \int e^u \, dt$ saturates. In this limit:

- endpoint settlement differences dominate all redshift readouts;
- structural differences along propagation paths are exponentially compressed;
- directional path differences contribute vanishingly to final readouts.

Thus, the high isotropy of the CMB requires no extra mechanism in SFT; it is an inevitable result in the time-saturation limit.

4.10.5 Settlement meaning of the blackbody spectrum

In SFT, a blackbody spectrum is not interpreted as a "thermal-equilibrium relic," but as the statistical smoothing of multi-source settlement readouts in frequency space under the limit where endpoint settlement differences dominate and path information is compressed. Because settlement readouts occur only at endpoints and propagation introduces no new frequency-rearrangement mechanism, the spectral shape remains stable in log-frequency coordinates; exponential endpoint re-scaling shifts it into the microwave band while preserving a nearly perfect blackbody form.

4.10.6 Origin of anisotropy (qualitative)

The weak CMB anisotropy corresponds to perturbations of the settlement anchor field:

$$u(\mathbf{x}) = u_0 + \delta u(\mathbf{x})$$

- large-scale low-multipole structure reflects long-wavelength fluctuations of δu;
- small-scale structure corresponds to local gradients and residuals of path-compression in the settlement anchor field.

These anisotropies should be different projections of the same field $u(\mathbf{x})$ inferred from lensing and rotation curves in later chapters, rather than independent "acoustic relics."

4.10.7 Fundamental difference from the standard interpretation

Aspect	Standard ΛCDM	SFT
Status of CMB	relic of early universe	a settlement-readout limit
Key mechanism	recombination, acoustic oscillations	endpoint settlement difference plus time saturation
Origin of isotropy	early thermal equilibrium	compression of path information
Parameter dependence	initial-condition setting	settlement anchor field u

SFT does not deny the CMB observations; it denies that they must correspond to an independent "cosmic origin epoch."

4.10.8 Falsifiability and consistency criterion

SFT imposes a direct testable constraint: the settlement-anchor perturbation $\delta u(\mathbf{x})$ inferred from CMB anisotropy must be compatible, on the same scales, with $u(\mathbf{x})$ inferred from lensing deflection, redshift endpoint differences, and gravity readouts. If the $u(\mathbf{x})$ structure required by the CMB cannot be unified with other observations, SFT is falsified.

4.10.9 Subsection summary

In SFT, the CMB is not an echo of an initial cosmic event, but a natural readout limit of the settlement anchor field under extreme propagation paths and time-saturation conditions. Its isotropy, temperature scale, and weak anisotropy are all interpretable as different manifestations of the same settlement readout field.

4.11 From Discrete SU Refresh Bandwidth to a Continuous Settlement Variational Functional: The Physical Essence of Action as the Minimum Refresh-Cost Path

4.11.1 Refresh-bandwidth axiom (discrete execution layer)

At the hardware layer of SFT, each SU possesses a fixed total refresh bandwidth W_0 per refresh cycle. This bandwidth must be allocated between two orthogonal execution channels:

- stepping channel: synchronizes state delivery to neighboring SUs, realizing propagation;
- settlement channel: completes local return closure, producing recordable readouts.

Define occupied components $w_{s,i}, w_{r,i}$, and normalized fractions: $\beta_{s,i} = \frac{w_{s,i}}{W_0}$, $\beta_{r,i} = \frac{w_{r,i}}{W_0}$

They satisfy the orthogonal budget identity:

$$\boxed{\beta_{s,i}^2 + \beta_{r,i}^2 = 1.} \tag{4.11.1}$$

Introduce the settlement anchor field: $u_i = \ln \beta_{r,i} \leq 0$, $\beta_{r,i} = e^{u_i}$. (4.11.2)

Hence the stepping capability is

$$\beta_{s,i} = \sqrt{1 - e^{2u_i}} \tag{4.11.3}$$

4.11.2 Definition of refresh cost for discrete relays

Consider a discrete path: $\pi: i_0 = A \to i_1 \to \cdots \to i_N = B$
A single hop must satisfy:

1. stepping synchronization;
2. settlement-window compatibility.

Define the single-step refresh cost:

$$\Delta\mathcal{S}_k = \frac{1}{\beta_{s,i_k}} \cdot \frac{1}{\beta_{r,i_k}} \cdot \Delta s_{i_k i_{k+1}} \tag{4.11.4}$$

In the weak-field domain $|u| \ll 1$,

$$\frac{1}{\beta_{s,i}} \simeq 1 + \frac{1}{2} e^{2u_i} \tag{4.11.5}$$

The dominant term is controlled by the settlement window:

$$\Delta\mathcal{S}_k \simeq e^{-u_{i_k}} \Delta s \tag{4.11.6}$$

4.11.3 Co-origin constraint from rotational-unit routing bias

Rotational units (SUΩ) organize transverse accessible directions. Because the directional accessibility window and the settlement window share the same refresh availability function, the direction-synchronization constraint introduces an additional factor $1/\beta_{r,i}$. Thus the discrete total cost becomes

$$\Delta\mathcal{S}_k \frac{1}{\beta_{s,i_k}} \cdot \frac{1}{\beta_{r,i_k}} \cdot \frac{1}{\beta_{r,i_k}} \cdot \Delta s \tag{4.11.7}$$

Hence the cost density is

$$w(u_i) = \frac{1}{\beta_{r,i}^2} = e^{-2u_i} \tag{4.11.8}$$

This provides the discrete origin of the coefficient "2": it is the product of two co-origin constraints—rhythm rescaling and routing bias.

4.11.4 Large-deviation limit and the minimum-cost path

Define the total path cost:

$$\mathcal{S}[\pi] = \sum_{k=0}^{N-1} e^{-2u_{i_k}} \Delta s_{i_k i_{k+1}} \tag{4.11.9}$$

Define the statistical weight of a path:

$$\mathbb{P}(\pi) \propto \exp(-\lambda\, \mathcal{S}[\pi]) \tag{4.11.10}$$

In the strong-constraint limit $\lambda \to \infty$,

$$\pi^* = arg \min_{\pi: A \to B} \mathcal{S}\,[\pi] \tag{4.11.11}$$

4.11.5 Continuous limit and the settlement variational functional

When the observation scale satisfies $\ell_{obs} \gg \Delta s$, the discrete sum converges to a continuous functional:

$$\mathcal{S}[\gamma] = \int_\gamma e^{-2u(\mathbf{x})}\, ds \tag{4.11.12}$$

The minimal path satisfies

$$\delta \int_\gamma e^{-2u(\mathbf{x})}\, ds = 0 \tag{4.11.13}$$

4.11.6 Recovery of the deflection operator

Let $n(\mathbf{x}) = e^{-2u(\mathbf{x})}$. The variational condition gives:

$$\frac{d}{ds}(n\hat{\mathbf{n}}) = \nabla n \tag{4.11.14}$$

That is,

$$\frac{d\hat{\mathbf{n}}}{ds} = \nabla_\perp(-2u) = -2\nabla_\perp u \tag{4.11.15}$$

Integrating yields:

$$\alpha = 2\int_\gamma \nabla_\perp u\, ds \tag{4.11.16}$$

which is exactly consistent with the light-bending formula in Chapter 5.

4.11.7 Physical essence of action

In SFT:

- e^{-2u} is the refresh-cost density;
- $\int e^{-2u}\, ds$ is the full-path refresh bill;
- the principle of least action equals the minimum refresh-cost path.

Action is no longer an abstract geometric postulate, but the statistical limit of SU relay competition.

4.11.8 Domain of validity and boundary conditions

Conditions for the derivation:

1. continuous limit: $u \in C^1$ (stronger: H^2);
2. weak-deflection approximation;
3. quasi-static background ($\partial_t u$ frozen on the propagation scale).

In the strong-field limit $u \to -\infty$, the cost functional diverges, corresponding to the relay-failure boundary in Chapter 7.

4.11.9 Closure of this section

Starting from discrete refresh-bandwidth competition, and via the large-deviation limit and continuousization, we strictly derive the settlement variational functional $\delta \int e^{-2u}\, ds = 0$ and recover the deflection operator $\alpha = 2\int \nabla_\perp u\, ds$. Hence, the "principle of least action" in the macroscopic universe obtains a discrete-hardware origin in SFT.

Chapter 5 Transverse Readouts of the Settlement Anchor Field: Light Bending, Lensing Inversion, and Mathematical Closure of Spatial Structure

Positioning of This Chapter

Chapters 1–4 have completed mathematical closures in the longitudinal and temporal directions:

- redshift is an endpoint/temporal settlement readout;
- the Hubble law is a first-order gradient expansion;
- dark matter and dark energy arise from the gradient and curvature of u;

- JWST anomalies arise from time saturation and exponential projections in strongly negative u regions.

This chapter completes the fifth closure in the transverse direction: it proves that light bending and gravitational lensing are transverse-gradient readouts of the settlement anchor field $u(\mathbf{x})$. Their inversion is mathematically equivalent to a constrained Radon-type inverse problem, and forms a strong-consistency closed loop with rotation curves and redshift readouts.

5.0 Rotational-Unit Routing and Settlement Bias: A Structural Derivation of the Light-Bending Formula

Positioning of this section

Earlier chapters have written light bending as a transverse-gradient readout of the settlement anchor field, $\alpha = 2\int_\gamma \nabla_\perp u\, ds,\ u(\mathbf{x}) = \ln \beta_r(\mathbf{x})$, and established that this expression is co-originating with redshift, orbits, and lensing. This section provides a structural derivation to show that the formula is not merely a definition, but an inevitable result jointly determined by:

1. the propagation direction of the stepping phase being organized by rotational units (SUΩ);
2. stepping propagation in a non-uniform settlement background necessarily experiencing routing bias induced by u.

5.0.1 Direction organization of stepping propagation: the routing role of Ω

In the NO5 framework, light is a pure stepping state: $\beta_s = 1,\ \ \beta_r = 0$

However, "pure stepping" does not mean that propagation direction is arbitrary. The SU refresh network has a fixed right-handed chirality. When transverse phases are synchronously organized locally, a stable rotational structure emerges—namely a rotational unit (SUΩ).

In the continuous limit, rotational units do not introduce new dynamical degrees of freedom. Their only physical role is to provide direction organization for sustainable relays of stepping propagation.

Let the light path be γ with arc-length parameter s. Define the stepping unit tangent: $\hat{\mathbf{n}}(s) \equiv \frac{d\mathbf{x}}{ds}, |\hat{\mathbf{n}}| = 1$. Ω determines the local accessibility cone of $\hat{\mathbf{n}}$, but does not directly generate settlement or energy readouts.

Key restriction: Ω determines "which directions remain relay-feasible," but does not determine "existence" or "settlement."

5.0.2 How the settlement anchor field enters stepping propagation: a minimum-readout routing principle

Although light does not settle ontologically ($\beta_r = 0$), its propagation must traverse a settlement background with spatial structure. The settlement occupation fraction $\beta_r(\mathbf{x})$ changes the available window for stepping synchronization, thereby altering the propagation time readout.

In NO5, the optical-path time readout is $\Delta t = \frac{1}{c}\int_\gamma e^{-u(\mathbf{x})}\, ds,\quad u(\mathbf{x}) = \ln \beta_r(\mathbf{x})$.

Hence the actual light path is selected by a minimum-readout principle:

$$\gamma = arg \min_\gamma \int_\gamma e^{-u(\mathbf{x})}\, ds \tag{5.0.1}$$

This principle requires no metric assumption; it states only: which stepping path is most capable of maintaining synchronization within a settlement background.

5.0.3 Transverse deflection from the variational settlement readout: first contribution

Define an effective routing weight $n_{eff}(\mathbf{x}) \equiv e^{-u(\mathbf{x})}$. Perform a first variation of the functional $\mathcal{T}[\gamma] = \int_\gamma n_{eff}(\mathbf{x})\, ds$. In the weak-gradient, weak-deflection limit, one obtains the directional change rate: : $\frac{d\hat{\mathbf{n}}}{ds} \simeq -\nabla_\perp \ln n_{eff}\, \nabla_\perp u$.

Integrating along the path yields the first deflection contribution:

$$\alpha_{(2)} \simeq \int_\gamma \nabla_\perp u\, ds \tag{5.0.2}$$

This term corresponds to rhythm rescaling / an effective refraction effect induced by the settlement anchor field.

5.0.4 Transverse bias from rotational-unit routing: second contribution

In a right-handed refresh network, the stepping direction organized by rotational units is not rigidly fixed. When the background settlement occupation $\beta_r(\mathbf{x})$ has a gradient:

- toward larger β_r, stepping synchronization is easier to maintain;
- toward smaller β_r, synchronization mismatch is more likely.

To first order in the weak-field expansion, this routing bias can only be expressed by the unique admissible transverse vector: $\left(\frac{d\hat{\mathbf{n}}}{ds}\right) route \simeq \nabla\perp\, u$ (5.0.3)

Thus one obtains a second contribution of the same form:

$$\alpha_{(1)} \simeq \int_\gamma \nabla_\perp u\, ds \tag{5.0.4}$$

Its physical meaning is: Ω organizes the direction, while ∇u biases that organization; Ω does no work and only changes direction.

5.0.5 Inevitability of the bending formula and the coefficient 2

Combining the two independent first-order contributions:

$$\alpha\alpha_{(1)} + \alpha_{(2)} \simeq 2\int_\gamma \nabla_\perp u\, ds \tag{5.0.5}$$

Thus the bending formula is not a definition but a structural consequence jointly enforced by:

1. stepping propagation must follow relay-feasible directions organized by Ω;
2. a non-uniform settlement background alters the synchronization "cost" through $u = \ln \beta_r$;
3. in the weak-field limit, two independent first-order biases must share the same form and add linearly.

5.0.6 Unified conclusion

Light bending is not the result of geometry being "pulled," but a routing bias of Ω-organized stepping propagation under the gradient of the settlement anchor field. Its mathematical expression necessarily converges to $\alpha = 2\int_\gamma \nabla_\perp u\, ds$.

5.1 Unified Relation Between Endpoint Settlement Difference and In-Path Light Deflection: Longitudinal vs. Transverse Readouts of One Field

5.1.1 Core conclusion

The redshift of distant light (endpoint settlement difference) and its deflection (path bending) are not independent effects, but readouts of the same settlement anchor field along different directions.

In SFT, all gravitational optical readouts arise from the same scalar field: $u(\mathbf{x}) = \ln \beta_r(\mathbf{x})$ where β_r is the settlement-phase occupation fraction of SU refresh resources.

One-sentence summary: redshift is an endpoint difference along the propagation direction, while deflection is path accumulation of transverse gradients.

5.1.2 Role of light in SU relays

In SFT, light satisfies:

- $\beta_r = 0$: light does not occupy settlement phase;
- $\beta_s = 1$: light is fully carried by stepping phase;
- propagation is equivalent to a minimum-cost relay path across SUs.

Therefore:

- light does not settle during propagation;
- all readouts of frequency, time, and energy appear only at endpoints where settlement occurs.

This separation is the key prerequisite for the unified understanding of redshift and deflection.

5.1.3 Endpoint settlement difference: longitudinal readout of redshift

When settlement occurs at emission and reception, frequency readouts are determined by the local u. For the same spectral line: $1 + z = \frac{\nu_{em}}{\nu_{obs}} = \exp(u_{obs} - u_{em})$

This shows:

- redshift depends only on endpoint settlement environments;
- it is independent of path length or the number of space units traversed;
- it is a longitudinal endpoint-difference readout of the settlement anchor field.

5.1.4 In-path deflection: path integral of transverse gradients

Although light does not settle during propagation, it must choose relay routes through the SU network. Route choice obeys the same minimum-settlement-cost principle:

$$\gamma = arg \min_{\gamma} \int_{\gamma} e^{-u(\mathbf{x})}\, ds,$$

Thus:

- if $u(\mathbf{x})$ is spatially uniform, the path is straight;
- if $u(\mathbf{x})$has transverse gradients, the minimum-cost relay path bends.

In the weak-field approximation, the deflection angle satisfies: $\alpha = 2 \int_{\gamma} \nabla_{\perp} u\, ds$

Key points:

- the integral is taken along the propagation path;
- the integrand is the transverse gradient $\nabla_{\perp} u$;
- deflection accumulates continuously along the path.

This is the transverse readout of the settlement anchor field.

5.1.5 Why endpoint differences and path accumulation are not contradictory

Redshift is an endpoint difference while deflection is a path integral; the forms differ but do not contradict because they occur in different regions:

Quantity	Where it occurs	Mathematical form	Physical meaning
Redshift	emission and reception	$u_{obs} - u_{em}$	settlement background difference
Deflection	along propagation path	$\int \nabla_{\perp} u \, ds$	accumulated routing bias

Key distinction:

- settlement occurrence determines whether a readout appears there;
- route selection determines whether the propagation direction bends.

Since light does not settle during propagation, it cannot generate redshift along the route; but since it must continually choose routes, it can continuously change direction.

5.1.6 Unified mathematical expression

Both effects can be written as different projections of the same readout relation:

$$(\ln \nu)\, \underbrace{\Delta u}\, longitudinalendpointsettlementdifference + \underbrace{\int \gamma \, \nabla_{\perp} u \cdot d\ell}_{\text{transverse routing-bias accumulation}}$$

The first term is nonzero only at endpoints where settlement occurs (redshift); the second term accumulates only during route selection (deflection). Both arise from the same field $u(\mathbf{x})$, differing only by projection direction.

5.1.7 Fundamental difference from traditional narratives

Traditionally:

- redshift is often explained as "work against gravity";
- deflection is often explained as "spacetime curvature."

In SFT, both are replaced by longitudinal endpoint differences and transverse gradients of the settlement anchor field—without splitting "time curvature" vs. "space curvature," and without adding extra geometric or dynamical assumptions.

5.1.8 Falsifiability criterion

This unified relation gives a direct observational constraint:

Along the same light path, the settlement anchor field $u(\mathbf{x})$ inferred from redshift endpoint differences must be consistent with the $u(\mathbf{x})$ inferred from lensing/deflection.

A test workflow:

- invert $u(\mathbf{x})$ from lensing deflection;
- invert $u(\mathbf{x})$ from redshift endpoint differences;
- if the two are incompatible on the same path, SFT is falsified.

5.1.9 Subsection summary

Redshift and light deflection of distant signals are not independent effects; they are different readouts of the same settlement anchor field along longitudinal and transverse directions. Endpoint settlement differences and path-integrated transverse gradients together exhaust the observable gravitational optical shifts.

5.1.10 First-principles definition of the transverse readout (not replaceable)

Let the propagation path γ have arc-length parameter s, tangent unit vector $\hat{\mathbf{n}}$, and define the transverse projection operator:

$$\nabla_{\perp} \equiv \nabla - \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \nabla) \tag{5.1}$$

Definition 5.1 (Transverse settlement readout).
The light-deflection vector is defined as

$$\alpha(\theta) = 2 \int_{\gamma(\theta)} \nabla_{\perp} u(\mathbf{x})\, ds \tag{5.2}$$

where θ parameterizes the sky image position. This is a definition-level formula, not derived from approximate mechanics.

5.2 Mathematical Decomposition of the Coefficient 2

Split the integrand of (5.2) into two equivalent contributions:

1. Routing-bias term (dynamical effect of SU chirality synchronization under pressure):

$$\alpha 1 = \int \gamma\, \nabla_{\perp} u\, ds \tag{5.3}$$

2. Rhythm-rescaling term (effective refraction due to settlement priority):

$$n_{eff}(\mathbf{x}) \equiv e^{-2u(\mathbf{x})},$$

$$\alpha 2 = \int \gamma\, \nabla_{\perp}\left(\ln n_{eff}\right) ds = \int_{\gamma} \nabla_{\perp} u\, ds \tag{5.4}$$

Hence

$$\alpha = \alpha 1 + \alpha 2 = 2 \int \gamma\, \nabla\perp\, u\, ds \tag{5.5}$$

This decomposition introduces no metric tensor; it follows solely from the settlement-priority axiom and exponential readouts.

5.3 Weak-Deflection and Thin-Lens Limits (Computational Entry Point)

Under weak-deflection and thin-lens approximations, treat the path as approximately straight and take the lens plane normal as the z-axis. Then

$$\alpha(\theta)\alpha(\theta) = 2 \int_{-\infty}^{+\infty} \nabla_{\perp} u(x, y, z)\, dz \tag{5.6}$$

Define the projected potential

$$\psi(x, y) \equiv 2 \int_{-\infty}^{+\infty} u\,(x, y, z)\, dz \tag{5.7}$$

Then

$$\alpha = \nabla_{\perp} \psi \tag{5.8}$$

This reduces the three-dimensional problem to the gradient readout of a two-dimensional potential.

5.4 Lens Equation and Image Mapping (Strict Form)

Let the source position be β and the image position be θ. The lens mapping is

$$\beta = \theta \frac{D_{LS}}{D_S}\, \alpha(\theta)\theta \frac{D_{LS}}{D_S}\, \nabla_{\perp} \psi(\theta) \tag{5.9}$$

The amplification matrix is

$$\mathcal{A} = \frac{\partial\beta}{\partial\theta}\mathbf{I}\frac{D_{LS}}{D_S}\nabla_\perp\nabla_\perp\psi \tag{5.10}$$

Its determinant gives the magnification:

$$\mu = \frac{1}{\det\mathcal{A}} \tag{5.11}$$

5.5 Strict Co-origin with the Settlement Source Equation (Key Welding)

From the source equation in Chapter 2,

$$\nabla^2 u = -\frac{4\pi G_*}{c^2}\rho_{\text{enc}} + \Lambda_u \tag{5.12}$$

integrate along z and apply boundary conditions to obtain a two-dimensional Poisson equation:

$$\nabla_\perp^2\psi = -\frac{8\pi G_*}{c^2}\Sigma_{\text{enc}} + 2\Lambda_u L \tag{5.13}$$

where

$$\Sigma_{\text{enc}}(x,y) = \int_{-\infty}^{+\infty}\rho_{\text{enc}}(x,y,z)\,dz \tag{5.14}$$

Conclusion 5.1 (Co-origin theorem).

The ∇u used for rotation curves, the u used for redshift, and the ψ used for lensing all arise from the same $u(\mathbf{x})$. Any attempt to introduce a separate free "dark halo" solely for lensing breaks the closure (5.12)–(5.13).

5.6 Mathematical Type of the Inversion: A Constrained Radon Inverse Problem

The lensing operator (5.6) is essentially a line integral of $\nabla_\perp u$, equivalent to a derivative-type Radon transform operator. Combined with the endpoint and gradient observations in Chapter 3, the inverse problem becomes:

$$aconstrained\ Radon\ inversion\ in\ H^2(\Omega) \tag{5.15}$$

Endpoint redshift anchors pointwise differences of u; rotation curves provide radial gradients; lensing provides transverse gradients. Their joint constraints eliminate the Radon null-space freedom (up to an additive constant gauge).

5.6.1 Weak-field limit and strong-field nonlinear correction of the coefficient 2

Problem statement.

In the weak-field approximation, deflection satisfies $\alpha = 2\int\nabla_\perp u\,ds$. We must clarify the validity and falsifiability of this coefficient in strong settlement-gradient regions.

Proposition 5.6.1 (General form of the deflection coefficient).

In a general settlement-gradient background, deflection can be written as

$$\alpha = \int_\gamma[1 + \Xi(u,\nabla u)]\ \nabla_\perp u\,ds \tag{5.6.1}$$

where Ξ is a dimensionless settlement correction function.

Theorem 5.6.1 (Weak-field limit).

When $|u| \ll 1$ and $|\nabla u| \ll 1$, $\Xi(u,\nabla u) \longrightarrow 1$, hence

$$\alpha \longrightarrow 2\int \nabla_{\perp} u\, ds \tag{5.6.2}$$

Conclusion 5.6.1.

- the coefficient "2" is the universal linear limit under weak settlement gradients, not an axiom;
- in strong-field regions $\Xi \neq 1$ produces observable non-GR deflection residuals;
- measuring deviations from 2 constitutes a falsifiable strong-field prediction of SFT.

5.7 Lensing–Redshift Consistency Theorem (Strong Falsification Axis)

Theorem 5.1 (Lensing–redshift consistency).

If a single $u(\mathbf{x})$ explains both rotation curves and lensing deflection, then along the same direction the redshift gradient must satisfy

$$\nabla_{\perp} u \Leftrightarrow \nabla z \tag{5.16}$$

If strong lensing is observed but no consistent redshift gradient can be constructed in the same direction, SFT is falsified.

5.8 Mathematical Entry to Weak Frequency Dependence (Settlement Dispersion)

If the SU synchronization window has a finite bandwidth ω_{SU}, then the single-step deflection may be written as

$$\delta\phi(\omega) = \delta\phi_0 \left[1 - \epsilon \left(\frac{\omega}{\omega_{SU}}\right)^2\right],\ \epsilon \ll 1. \tag{5.17}$$

Integrating yields

$$\alpha(\omega) = \alpha 0 \left[1 - \epsilon \left(\frac{\omega}{\omega SU}\right)^2\right] \tag{5.18}$$

In GR this is strictly zero; it is a falsifiable prediction of the present theory.

5.9 Chapter Closure (Mathematical Conclusions of the Transverse Readout)

This chapter proves, in a strict mathematical sense:

1. light bending is a transverse-gradient readout of $u(\mathbf{x})$;
2. the lensing potential ψ is co-originating with the source equation (5.12);
3. lensing inversion is a constrained Radon inverse problem that becomes nearly unique under multi-probe constraints;
4. lensing and redshift must satisfy a consistency relation;
5. a settlement-dispersion correction is, in principle, testable.

5.10 Interface to Later Chapters

- Chapter 6: planetary orbits as "transverse deflection plus longitudinal settlement adjudication" closures;
- Chapter 7: photon spheres and settlement-failure boundaries in the strong-field limit;
- Later chapters: unify lensing $\nabla_{\perp} u$ with gravitational-wave $\partial_t \nabla u$ as dynamic readouts.

In SFT, light bending and gravitational lensing are not geometric phenomena; they are computable transverse-gradient readouts of the settlement anchor field. Their inversion forms a constrained

Radon inverse problem and, together with redshift and rotation curves, establishes a single-variable closed-loop consistency system.

Chapter 6 Longitudinal Settlement and Transverse Deflection of Planetary Orbits: From Closure Adjudication to the Mathematical Inevitability of Kepler's Laws

Positioning of This Chapter

The first five chapters have completed closures in three directions:

- Longitudinal (Chapter 1): redshift and the Hubble law are endpoint/gradient readouts of settlement;
- Spatial source (Chapter 2): dark matter/dark energy are gradient/curvature structures of u;
- Inversion (Chapter 3): existence and near-uniqueness of u under multi-probe constraints;
- Time (Chapter 4): time saturation and non-single-valued cosmic age;
- Transverse (Chapter 5): light bending and lensing as transverse-gradient readouts.

This chapter completes the final mathematical closure of macroscopic dynamics: it proves that planetary orbits are not the result of mechanical equilibrium, but periodic closures in the settlement anchor field $u(\mathbf{x})$ in which transverse deflection is locked by longitudinal settlement adjudication. Kepler's three laws follow as mathematical necessities in this framework.

6.1 Separating Motion from Readouts: Orbits Are Not Differential-Equation Solutions but Adjudication Conditions

Let a planet of mass (m) move along a path Γ. SFT does not begin from a "force equation," but from readout adjudication:

- Longitudinal settlement readout (existence/return)

$$\mathcal{C}_{\parallel}[\Gamma] \equiv \oint_{\Gamma} \beta_r(\mathbf{x})\, dt \tag{6.1}$$

- Transverse deflection readout (path geometry)

$$\mathcal{C}_{\perp}[\Gamma] \equiv \oint_{\Gamma} \nabla_{\perp} u(\mathbf{x})\, ds \tag{6.2}$$

Definition 6.1 (Orbit legality).

A path Γ constitutes a stable orbit if and only if

$\mathcal{C}_{\parallel}[\Gamma]$ =constant (return adjudication), and $\mathcal{C}_{\perp}[\Gamma]$is finite and periodically consistent (6.3)

This definition is the only premise used in all derivations of this chapter.

6.2 Longitudinal Settlement Conservation: Mathematical Origin of the Area Law

From Chapter 1, the settlement time readout is $d\tau = e^{u} dt$. For a stable orbit, return adjudication requires the total settlement over one period to be fixed:

$$\oint_{\Gamma} e^{u(\mathbf{x})}\, dt = constant \tag{6.4}$$

Write the orbit in polar coordinates (r, θ). The areal sweep rate is

$$\frac{dA}{dt} = \frac{1}{2} r^2 \dot{\theta} \tag{6.5}$$

From (6.4) and periodicity, there exists a constant K such that

$$e^{u(r)}\, r^2 \dot{\theta} = K \tag{6.6}$$

Conclusion 6.1 (Kepler's area law).
Equation (6.6) is equivalent to "equal areas in equal times," but its ontological origin here is longitudinal settlement-return adjudication, not an assumed angular-momentum conservation law.

6.3 Radial Form of Transverse Deflection: Inevitability of Elliptical Geometry

In a spherically symmetric static background $u = u(r)$,

$$\nabla_{\perp} u = \frac{du}{dr}\, \hat{\mathbf{e}}_r^{\perp} \tag{6.7}$$

The accumulated transverse deflection over one period depends only on the radial variation of du/dr.
Introduce the principal (external) solution for $u(r)$ from Chapter 2:

$$\frac{du}{dr} = -\frac{G_* M}{c^2 r^2} \quad (\Lambda_u\ negligibleon\ planetary\ scales) \tag{6.8}$$

Substituting (6.6) and (6.8) into the path geometry yields an equivalent radial–angular relation:

$$\frac{d^2}{d\theta^2}\left(\frac{1}{r}\right) + \left(\frac{1}{r}\right) = \frac{G_* M}{K^2/c^2} \tag{6.9}$$

Conclusion 6.2 (Kepler's first law).
The general solution of (6.9) is

$$\frac{1}{r(\theta)} = \frac{G_* M}{K^2/c^2}\left[1 + e\cos(\theta - \theta_0)\right] \tag{6.10}$$

i.e., an elliptical orbit. The ellipse is not a "special solution of a centripetal-force law," but the most general closed curve formed when transverse deflection is constrained by longitudinal adjudication.

6.4 Period–Scale Relation: Settlement Derivation of the Third Law

From (6.6), the period T satisfies

$$\int_0^T e^{u(r(t))}\, dt = constant \tag{6.11}$$

For an ellipse, the period average of$u(r)$is mainly controlled by the semimajor axis a. Under the weak-field approximation of (6.8),

$$e^{u(r)} \simeq 1 - \frac{G_* M}{c^2 r} \tag{6.12}$$

Using the geometric relation $\langle 1/r \rangle = 1/a$, one obtains

$$\langle e^u \rangle \simeq 1 - \frac{G_* M}{c^2 a} \tag{6.13}$$

On the other hand, by the area law, the area $A = \pi ab$ is proportional to K. Eliminating K yields the period scaling

$$T^2 \propto a^3 \tag{6.14}$$

Conclusion 6.3 (Kepler's third law).
Kepler's third law is not "a consequence of an inverse-square force," but the scaling law of longitudinal adjudication together with the principal settlement-gradient solution $u \sim -1/r$.

6.5 Perihelion Precession: A Small Mismatch Between Transverse and Longitudinal Adjudication

If one retains the curvature term Λ_u or higher-order settlement corrections, then (6.9) becomes

$$\frac{d^2}{d\theta^2}\left(\frac{1}{r}\right) + (1-\varepsilon)\left(\frac{1}{r}\right) = \frac{G_* M}{K^2/c^2}, \quad |\varepsilon| \ll 1. \tag{6.15}$$

The solution is a slowly rotating ellipse, with perihelion advance

$$\Delta\theta \simeq \pi\varepsilon \tag{6.16}$$

Conclusion 6.4.
Perihelion precession arises from a higher-order small mismatch between transverse deflection and longitudinal adjudication, sharing the same origin as higher-order accumulation in light deflection.

6.6 Unified Limits: Light vs. Planets

- Light: $\beta_r = 0$. Longitudinal adjudication vanishes; only transverse deflection remains → no closure.
- Planets: $\beta_r > 0$. Longitudinal adjudication exists; transverse deflection is locked → closed orbit.

$$\text{Orbit} = \text{transverse deflection} \oplus \text{longitudinal settlement adjudication} \tag{6.17}$$

6.7 Reframing the Atomic-Orbit Problem

In the first half of Chapter 6, we proved that planetary orbits are not determined by classical "mechanical equilibrium," but are periodic closed paths formed when transverse deflection is constrained by longitudinal settlement adjudication in the anchor field $u(\mathbf{x})$.
A natural and unavoidable question follows: is this principle only macroscopic, or does it also apply microscopically?
In traditional physics, atomic orbits are treated as quantum phenomena "essentially different" from planetary orbits, with stability explained by wavefunctions and boundary conditions. In SFT, the atomic-orbit problem is reformulated as:
In the atomic-scale settlement anchor background $u(\mathbf{x})$, which paths simultaneously satisfy transverse deflection and longitudinal settlement adjudication, thereby forming stable closures?

6.8 Principal Form of the Atomic-Scale Settlement Anchor Field

On atomic scales, the settlement anchor field is dominated mainly by the nuclear enclosure state. Under spherical symmetry and static approximation, its principal form is uniquely written as

$$u(r) \simeq -\frac{\kappa}{r} \tag{6.8.1}$$

Here κ is a scale parameter determined by nuclear charge and settlement-enclosure strength. This form is not an empirical assumption, but the unique choice satisfying:

1. spherical symmetry and radial monotonicity of the settlement gradient;

2. stable accumulation of transverse deflection;
3. strict isomorphism with the macroscopic principal form $u(r) \sim -1/r$.

6.9 Microscopic Adjudication Condition and Closed Paths

As for planetary orbits, atomic orbit stability comes from conservation of longitudinal settlement adjudication:

$$\oint e^{u(r)}\ dt = constant \tag{6.9.1}$$

Unlike the macroscopic case, however, settlement in a microscopic system is no longer continuously divisible. Because SU refresh and settlement possess minimal time and action units, the adjudication condition is forced to discretize into:

$$\oint e^{u(r)}\ dt = n\,\tau_0, \quad n \in \mathbb{Z}^+ \tag{6.9.2}$$

Here τ_0 is the minimal settlement-time unit obtained from inverse SU calibration.

6.10 Inevitability of Discrete Orbital Radii and Discrete Energy Levels

Substituting the principal form (6.8.1) into the adjudication condition (6.9.2) yields stable closed-orbit radii satisfying

$$r_n \propto n^2 \tag{6.10.1}$$

Further, using the correspondence between orbital energy and settlement adjudication, energy-level spacing satisfies

$$\Delta E_n \propto \frac{1}{n^2} \tag{6.10.2}$$

This reproduces the basic structure of hydrogen spectroscopy.
Key conclusion: atomic orbital quantization does not arise from "boundary conditions imposed on a wavefunction," but from the geometric–dynamical fact that settlement closure is forced to discretize at microscopic scales due to the existence of a minimal settlement unit.

6.11 Unified Limit of Light, Electrons, and Planetary Orbits

Three seemingly different orbital phenomena become unified in SFT:

System	β_r	Transverse deflection	Longitudinal adjudication	Orbital outcome
Light	0	yes	none	not closed
Electron	>0 (minimal)	yes	discrete	quantum orbit
Planet	(>0) (macroscopic)	yes	continuous	classical orbit

Unified expression:

Orbit $= (transverse\ deflection \oplus longitudinal\ settlement\ adjudication$ (6.11.1)

6.12 Chapter 6 Closure: The Cross-Scale Orbit Isomorphism Theorem

Theorem 6.1 (Cross-scale orbit isomorphism).
In SFT, light paths, atomic orbits, and planetary orbits are not produced by different physical principles. They are the same settlement-dynamical structure expressed under different β_r and scale limits. Atomic quantization arises from the minimal settlement unit, not from an independent quantum postulate.

6.13 Geometric–Settlement Meaning of the Fine-Structure Constant

Motivation

The derivation of atomic closure and quantization above shows that microscopic discreteness originates in the minimal settlement unit rather than in classical waves or orbital assumptions. To avoid misreading this as a "geometric restatement of the Bohr model," we clarify the computational meaning of the fine-structure constant α in settlement physics.

Definition 6.13.1 (Fine-structure constant as a settlement ratio).

In SFT, α is not an empirical coupling parameter, but a geometric–settlement ratio between SU enclosure levels, with basic form:

$$\alpha = \frac{1}{4\pi}\left(\frac{\ell_0}{\ell_\Omega}\right)^8 \tag{6.13.1}$$

where:

- ℓ_0: linear scale of the space unit (SU);
- ℓ_Ω: enclosure scale of the rotational unit (SUΩ), defined by $\hbar = \mu_g c\, \ell\Omega^2$.

Proposition 6.13.1 (Settlement-geometric origin of α).

The exponent 8 is not arbitrary; it corresponds to structural facts:

1. the minimal stable enclosure state of the electron is a double-helical closed structure;
2. each full settlement closure corresponds to a 4π phase return;
3. visibility from SU to SUΩ requires multiple higher-order settlement synchronization;
4. α measures the total suppression ratio from SU enclosure to the electromagnetically visible layer.

Thus α is small not because of "weak coupling," but because it is a geometric compression of the visibility threshold.

Conclusion 6.13.1.

In SFT, the fine-structure constant is a purely geometric settlement ratio reflecting the enclosure hierarchy SU → SUΩ → atomic visible layer. Fine splitting is not an added correction, but a natural manifestation of settlement structure in a perturbative layer.

6.14 Settlement Reconstruction of the Uncertainty Principle

Motivation

In standard quantum mechanics, the uncertainty principle is treated as a measurement limit or a result of wave–particle duality. In SFT, it can be reconstructed as an inevitable consequence of the fact that a minimal settlement time unit exists: within one settlement cycle, the stepping path cannot be fully resolved.

Definition 6.14.1 (Minimal settlement time).

From SU refresh and settlement, there exists a minimal settlement time unit:

$$\tau_0 = \frac{S_0}{\hbar} \tag{6.14.1}$$

where $S_0 = \mu_g c\, \ell_0^2$ is the minimal action of a space unit.

Proposition 6.14.1 (Stepping and settlement cannot be maximized simultaneously).

Within a single settlement cycle τ_0:

- stepping phase advances the path;
- settlement phase returns and records the outcome;

- both draw from the same refresh resource and satisfy

$$\beta_s^2 + \beta_r^2 = 1 \tag{6.14.2}$$

Hence, within one settlement cycle, position (path) and settled outcome cannot both be defined with arbitrary precision.

Theorem 6.14.1 (Uncertainty principle in settlement form).
Let a particle experience N settlement cycles within a time window Δt. Then its spatial and momentum readouts satisfy

$$\Delta x \, \Delta p \gtrsim \hbar \tag{6.14.3}$$

The lower bound arises from the minimal settlement time τ_0, not from measurement disturbance or probabilistic postulates.
Conclusion 6.14.1.
In SFT, the uncertainty principle is not an independent quantum axiom; it is a computational necessity: when a minimal settlement time unit exists, the stepping path cannot be fully resolved within one settlement cycle.

6.15 Closure of the Second Half of Chapter 6: The Atomic Level Is Not an "Exception"

With the settlement reconstructions of α and the uncertainty principle, one sees that:

- atomic orbits are not a semiclassical exception;
- fine structure is not a tuning parameter;
- uncertainty is not a measurement philosophy.

They all originate from the minimal computational structure of SU–SUΩ–settlement enclosure. Therefore, the cross-scale orbit isomorphism established in Chapter 6 holds simultaneously at macroscopic and microscopic levels. Quantum mechanics is repositioned as an effective theory of settlement dynamics at the smallest enclosure scale. From macroscopic planets to microscopic atoms, orbital existence and stability always arise from transverse deflection plus longitudinal settlement adjudication in the settlement anchor field; the only difference is whether adjudication is continuous or discretized at different scales.

Chapter 7 Strong-Settlement Limit and Accessibility Criticality: Photon Sphere, Horizon, and the "No-Singularity" Settlement Criteria

Positioning of This Chapter

The previous six chapters have closed the weak-field and intermediate-scale structure:

- longitudinal (redshift/Hubble) and transverse (bending/lensing) readouts;
- orbits as closures of "transverse deflection + longitudinal settlement adjudication";
- dark matter/dark energy as gradient/curvature readouts of u.

This chapter enters the strong-settlement limit and addresses the most controversial objects in traditional theory: the photon sphere and the event horizon. The goal is to provide strict criteria—without singularities and without geometric "flips"—showing that these objects are accessibility criticalities and settlement-failure boundaries of the SU relay network.

7.1 Mathematical Definition of the Strong-Settlement Limit (No Ambiguity Allowed)

Define the settlement anchor field:

$$u(\mathbf{x}) = \ln \beta_r(\mathbf{x}) \leq 0 \tag{7.1}$$

Definition 7.1 (Strong-settlement limit).
If there exists a region Ω such that

$$\inf_{\mathbf{x}\in\Omega} \beta_r(\mathbf{x}) = 0 \Leftrightarrow \inf_{\mathbf{x}\in\Omega} u(\mathbf{x}) = -\infty \tag{7.2}$$

then Ω is called a strong-settlement-limit domain.
This definition involves only the limit behavior of u, and does not involve coordinates or a metric.

7.2 Accessibility Function and Relay Condition (Core New Tool)
To characterize whether propagation "has a feasible route," introduce the accessibility function:

$$\mathcal{A}(\mathbf{x}, \hat{\mathbf{n}}) \equiv \lim_{\varepsilon\to 0^+} \frac{1}{\varepsilon} \int_0^{\varepsilon} e^{u(\mathbf{x}+s\hat{\mathbf{n}})}\, ds \tag{7.3}$$

Proposition 7.1 (Necessary condition for relay continuation).
Propagation along direction $\hat{\mathbf{n}}$ can continue at $\mathbf{x}$ if and only if

$$\boxed{\mathcal{A}(\mathbf{x}, \hat{\mathbf{n}}) > 0.} \tag{7.4}$$

Proof sketch.
The minimal synchronization window required for propagation is controlled by e^u. If its average is zero on any arbitrarily small neighborhood, the stepping phase cannot complete SU→SU relay synchronization. QED.

7.3 From the Settlement Variational Functional to a Relay-Failure Surface: The Horizon as a Boundary of Divergent Refresh Cost
7.3.1 Starting point: settlement variational functional
From §4.11, the continuous-limit settlement-cost functional is

$$\mathcal{S}[\gamma] = \int_\gamma e^{-2u(\mathbf{x})}\, ds \tag{7.3.1}$$

where:

- $u = \ln \beta_r$;
- e^{-2u} is the refresh-cost density.

Light or signal paths are determined by the extremal condition:

$$\delta \int_\gamma e^{-2u}\, ds = 0 \tag{7.3.2}$$

7.3.2 Strong negative-settlement limit
If there exists a region where $u(\mathbf{x}) \to -\infty$, then $e^{-2u(\mathbf{x})} \to +\infty$; the cost density diverges.
7.3.3 Strict definition of a relay-failure surface
Definition 7.3.1 (Relay-failure surface).

If there exists a closed surface Σ such that $\lim_{\mathbf{x}\to\Sigma^-} u(\mathbf{x}) = -\infty$, then Σ is called a relay-failure surface.

Physical meaning:

- any path γ that crosses Σ has $\mathcal{S}[\gamma] \to \infty$;
- in the large-deviation limit, $\mathbb{P}(\gamma) \propto e^{-\lambda \mathcal{S}[\gamma]} \to 0$;

thus, path statistical weight vanishes.

7.3.4 Equivalence to the "horizon"

Because all signal propagation must satisfy the minimum refresh-cost principle $\gamma^* = arg \min \mathcal{S}[\gamma]$, if entering a region requires divergent cost, then:

- no finite-cost path can carry signals out;
- exterior and interior become statistically isolated in the refresh sense.

Hence:

$$relay - failure\ surface \iff horizon \tag{7.3.3}$$

This definition arises purely from the refresh-cost functional, not from a geometric singularity.

7.3.5 Consistency with the accessibility function

Since propagation weight is $\mathbb{P}(\gamma) \propto e^{-\lambda \int e^{-2u} ds}$

one may define an accessibility functional:

$$\mathcal{A}(\mathbf{x}) = \exp\left(-\lambda \inf_{\gamma: x \to \infty} \int_\gamma e^{-2u(\mathbf{x}(s))}\, ds\right) \tag{7.3.4}$$

When $u \to -\infty$, $\mathcal{A} \to 0$. Thus:

$$\mathcal{A} = 0 \iff \text{divergent refresh cost} \iff \text{relay-failure surface} \iff \text{horizon.}$$

7.3.6 No-singularity property

Note:

- divergence arises from $u \to -\infty$;
- it does not require geometric curvature divergence;
- the exterior region can remain $u \in C^1$ regular.

Therefore:

The horizon is not a geometric endpoint, but a refresh-bandwidth cutoff. (7.3.5)

This matches the "no physical singularity" stance.

7.3.7 Consistency with gravity readouts

From the basic readout relation $\mathbf{g} = -c^2 \nabla u$, when $u \to -\infty$:

- $|\nabla u|$ may remain finite;
- but $\beta_r = e^u \to 0$;
- time readout $d\tau = e^u dt \to 0$.

Hence:

- time freezes inside;
- signals cannot be retrieved from the interior.

This is fully consistent with the time-saturation theorem.

7.3.8 Settlement definition of the photon sphere: an accessibility critical layer (not an orbit)

Consider the spherically symmetric static expression (Chapter 2):

$$\frac{du}{dr} = -\frac{G_* M}{c^2 r^2} + \frac{\Lambda_u}{3} r \tag{7.5}$$

Neglect Λ_u at planetary scales; light deflection satisfies

$$\alpha = 2\int \nabla_{\perp} u \, ds \tag{7.6}$$

Definition 7.2 (Photon sphere).
A radius $r = r_{\mathrm{ph}}$ is called a photon-sphere radius if:

- tangential accessibility remains positive,
- radial accessibility degenerates to criticality:

$$\mathcal{A}(r_{\mathrm{ph}}, \hat{\theta}) > 0, \mathcal{A}(r_{\mathrm{ph}}, \hat{\mathbf{r}}) \to 0, \tag{7.7}$$

Conclusion 7.1 (Non-orbit nature).
At $r = r_{\mathrm{ph}}$, light can undergo multiple strong deflections and appear to "circle," but since $\beta_r = 0$ (pure stepping state), there is no longitudinal settlement adjudication and thus no possibility of closed orbital legality. The photon sphere is an accessibility critical layer, not a stable orbit.

7.3.9 Section summary

Starting from discrete refresh-bandwidth competition and the settlement variational functional $\delta \int e^{-2u} \, ds = 0$, we strictly obtain:

- divergent-cost surface = relay-failure surface;
- relay-failure surface = horizon;
- no geometric singularity required;
- no extra assumptions required.

In SFT, a black hole is no longer a "spacetime endpoint," but a statistical boundary where refresh cost becomes infinite.

7.3.10 Exterior regularity theorem (removing singularity objections)

Assume u satisfies : $\nabla^2 u - \frac{4\pi G_*}{c^2}\rho_{\mathrm{enc}} + \Lambda_u$ with ρ_{enc} compactly supported.

Theorem (Exterior regularity).
In the exterior domain Ω_{ext} where $\rho_{\mathrm{enc}} = 0$,

$$u \in C^2(\Omega_{ext}): \ |\nabla u| < \infty \tag{7.8}$$

Proof sketch.

- the exterior satisfies a linear elliptic equation;
- classical elliptic regularity applies;
- there is no forcing source divergence.

Conclusion: the horizon is produced by $u \to -\infty$(divergent relay cost), not by curvature blow-up.

7.4 Settlement Definition of Black-Hole Temperature: Temperature as a Spectral Readout of Settlement Perturbations (No Thermal-Equilibrium Assumption)

7.4.1 Positioning and principle

In SFT, the horizon Σ is defined as the relay-failure surface where refresh cost diverges (§7.3).
This section defines "temperature" strictly using only the finalized readout dictionary:

- the unique ontological field $u(\mathbf{x}, t) = \ln \beta_r(\mathbf{x}, t)$;
- time readout $d\tau = e^u dt$;
- dynamic readout $\partial_t \nabla u$ (gravitational waves / induced gravity);
- propagation limit c.

7.4.2 Settlement definition of temperature (readout level)

In SFT, "temperature" is not defined as thermal-equilibrium energy, but as the local spectral scale

of settlement perturbations—i.e., the characteristic frequency scale at which the settlement structure continuously rearranges.

Near the horizon, criticality makes local fluctuations unavoidable. Let the local settlement perturbation on the exterior side be $\phi(\mathbf{x}, t) \equiv \partial_t u(\mathbf{x}, t)$

Define its spectrum: $\widetilde{\phi}(\mathbf{x}, \omega) = \int_{-\infty}^{+\infty} \phi\,(\mathbf{x}, t)\, e^{i\omega t}\, dt$

Define an effective spectral scale (spectral-mean angular frequency):

$$\omega_{\mathrm{eff}}(\mathbf{x}) \equiv \int_0^\infty \omega\ |\widetilde{\phi}(\mathbf{x}, \omega)|^2\, d\omega \int_0^\infty |\,\widetilde{\phi}(\mathbf{x}, \omega)|^2\, d\omega \tag{7.4.1}$$

Then define the "settlement temperature readout":

$$k_B T_u(\mathbf{x}) \equiv \hbar\, \omega_{\mathrm{eff}}(\mathbf{x}) \tag{7.4.2}$$

Notes:

- T_u is a readout quantity; no equilibrium is assumed;
- $\hbar$, k_B appear only as conversion constants;
- the ontology remains $\phi = \partial_t u$.

7.4.3 Geometric determinant of horizon temperature: a surface readout of the settlement gradient

The horizon is a relay-critical layer; the spectral scale is controlled by the "critical slope" at the exterior boundary. Let $\hat{\mathbf{n}}$ be the outward normal. Define the exterior normal gradient readout:

$$\kappa_u \equiv c^2\, \partial_n u|_{\Sigma^+}|\ , \partial_n u = \nabla u \cdot \hat{\mathbf{n}} \tag{7.4.3}$$

This has acceleration dimension and serves as the "slope readout" near the horizon. The only frequency scale one can form from κ_u and c is

$$\omega_H \sim \frac{\kappa_u}{c} \tag{7.4.4}$$

Hence define the horizon temperature readout:

$$k_B T_H \equiv \frac{\hbar}{2\pi}\frac{\kappa_u}{c}, \quad \kappa_u = c^2\, \partial_n u|_{\Sigma^+}| \tag{7.4.5}$$

The factor 2π converts spectral scale to a period scale and introduces no new physics.

Conclusion 7.4.1. In SFT, black-hole "temperature" is a spectral readout of the horizon's critical slope $\partial_n u$, not a thermal-equilibrium energy.

7.4.4 Contact with the standard scaling (consistency check only)

In a weak-field spherical exterior, if $u(r) \simeq -\frac{GM}{c^2 r}$, then $\partial_r u \simeq \frac{GM}{c^2 r^2}$. At an effective horizon radius r_H, $\kappa_u \sim c^2 \partial_r u|_{r_H}| \sim \frac{GM}{r_H^2}$.

If $r_H \sim 2GM/c^2$, then $T_H \propto \frac{1}{M}$, matching the standard scaling. This serves only as a scaling check, not as an input definition.

7.4.5 Section closure

Black-hole temperature is not "heat," but the minimal spectral scale readout of settlement-anchor fluctuations in the relay-critical layer, determined by $\kappa_u = c^2 \partial_n u$.

7.5 Settlement Definition of Black-Hole Entropy: Entropy as the Log-Capacity of "Achievable Settlement-State Sets" (Not a Microstate Counting Analogy)

7.5.1 Positioning: ontological definition of entropy in SFT

In SFT, "information" is defined as changes in the structure of settlement achievability (rearranging the set of allowed settlement states). Therefore, entropy should not be interpreted as counting unknown particle microstates, but as the log-capacity of the achievable settlement-state set under given boundary conditions.

7.5.2 Definition of settlement capacity

Let the horizon Σ be the relay-failure boundary. For an external observer, any recordable result must settle in the exterior. Thus the set of distinguishable exterior settlement states is constrained by the boundary conditions imposed by Σ. Let the effective achievable set be $\mathcal{C}_\Sigma$. Define settlement entropy:

$$S_\Sigma \equiv k_B \ln |\mathcal{C}_\Sigma| \tag{7.5.1}$$

This definition depends only on achievability sets, not on particle models.

7.5.3 Why entropy scales with area: boundary carrying is inevitable

In SFT, the horizon is not a geometric singularity but a relay-cutoff boundary. Changes in achievable exterior settlement states can only be "modulated" through horizon boundary conditions. In a 3D network, boundary-carrying degrees of freedom scale with area. Hence an area law follows:

$$S_\Sigma \propto k_B \frac{A_\Sigma}{\ell_\Omega^2} \tag{7.5.2}$$

where ℓ_Ω is the rotational-unit scale serving as a minimal settlement-structure area ruler. The proportionality factor depends on deeper discrete implementation and is not expanded here.

Conclusion 7.5.1 (Settlement origin of the area law).

The black-hole entropy area law is not an accident of microstate counting; it is an inevitable consequence of a relay-cutoff boundary acting as the carrying surface for settlement achievability.

7.5.4 Closure with temperature: a readout form of the first law

In SFT, energy and mass readouts share the same origin in settlement occupation. Energy exchange at the horizon corresponds to changes in $\mathcal{C}_\Sigma$. Hence a closed readout relation holds:

$$dE_\Sigma = T_H \, dS_\Sigma \tag{7.5.3}$$

Here T_H is given by §7.4. This does not assert thermal equilibrium; it states that the settlement spectral scale (temperature) and the achievability-set capacity (entropy) jointly constrain changes in boundary energy readouts.

7.5.5 Section closure

Black-hole entropy is not "counting hidden particle states," but the log-capacity of achievable exterior settlement-state sets imposed by a relay-failure boundary, and therefore must satisfy an area law.

7.6 Time Delay and the Strong-Field Upper Bound of Escape Time

The optical-path time readout is

$$\Delta t = \int_\gamma e^{-u} \, ds \tag{7.9}$$

Near the critical region $u \to -\infty$, there exists a constant C>0 such that

$$\Delta t \gtrsim \int^{r_{\rm ph}} \frac{dr}{|\nabla u|}\, e^{-u} \to +\infty \tag{7.10}$$

Conclusion 7.4 (Escape-time truncation).
Approaching the settlement-failure boundary, escape time grows exponentially and exhibits statistical truncation. This provides a strong-field observational discriminant from GR via time-distribution signatures.

7.7 Unified Relations with Redshift, Lensing, and Orbits

Redshift: $u_{obs} - u_{em}$, Bending: $\int \nabla_\perp u\, ds$, Orbit: $(\nabla_\perp u) \oplus (\oint e^u\, dt)$,Photon sphere: $\mathcal{A}_\theta > 0$, $\mathcal{A}_r \to 0$, Horizon: $\mathcal{A} \equiv 0$. (7.11)

7.8 Chapter Closure (Irrefutable Strong-Settlement Conclusions)

This chapter establishes, in strict mathematical terms:

1. the photon sphere is an accessibility critical layer, not an orbit;
2. the event horizon is a radial relay-failure surface, not a geometric flip;
3. exterior solutions are regular; no physical singularity occurs;
4. strong-field escape time has observable signatures of exponential growth and statistical truncation.

7.9 Interface to Later Chapters

- Chapter 9: propagation of $\partial_t \nabla u$ (gravitational waves / induced gravity);
- Chapter 10: link strong-field accessibility and multi-body settlement transitions to merger waveforms;
- Later: incorporate strong-field observations into the inversion functional $\mathcal{J}[u]$.

In SFT, strong-gravity structure is not a geometric singularity but an accessibility criticality and failure boundary of the settlement anchor field. Photon spheres and horizons thus obtain clear, computable, non-singular settlement definitions.

7.10 Regularity vs. Divergence: Separating Strong-Field Smoothness from the Divergent-Cost Boundary

Define the horizon as a relay-failure surface by $\inf_\gamma \int e^{-2u}\, ds = \infty$, equivalent to $u \to -\infty$ on a boundary limit set. If ρ_{enc} has compact support, then the exterior satisfies $\nabla^2 u = -\frac{4\pi G_*}{c^2}\rho_{enc} + \Lambda_u$, and remains $u \in H^2(\Omega_{ext})$, $|\nabla u| < \infty$.

Thus:

- what diverges is the refresh cost;
- no volumetric physical quantity is forced to diverge;
- exterior elliptic regularity holds.

Therefore, the horizon is a refresh cutoff boundary, not a geometric singularity.

Chapter 8 Propagation Structure of Settlement Perturbations and the Origin of Two Polarizations: Induced Gravity, Gravitational Waves, and Second-Order Eigenmodes

8.1 Positioning of This Chapter

In SFT, $u(\mathbf{x}, t) = \ln \beta_r$ is the only ontological readout field. We have already established:

- static gravity: $\mathbf{g} = -c^2 \nabla u$
- tidal tensor: $T_{ij} = c^2 \partial_i \partial_j u$
- beam convergence/shear: $Q_{ab} = -2\nabla_\perp \nabla_\perp u$

This chapter answers the key question: How can a single scalar field yield the two polarizations of gravitational waves?

8.2 The Hierarchical Chain of the Dynamic Structure

Define a perturbation: $u = u_0 + \delta u$

Define the propagation variable: $\phi = \partial_t u$

Propagation equation: $\Box \phi = 0$

The critical point is that the observable is not ϕ itself, but its second-order spatial structure: $\delta T_{ij} c^2 \partial_i \partial_j \delta u$. This is the quantity that becomes measurable.

8.3 The True Origin of Two Polarizations

Consider a plane-wave solution: $\delta u A \cos(k_\mu x^\mu)$

Then $\delta T_{ij} c^2 k_i k_j A \cos(k_\mu x^\mu)$

Now comes the key step: decompose relative to the propagation direction $\hat{\mathbf{k}}$. Define the transverse projector: $P_{ij} = \delta_{ij} - \widehat{k_i}\widehat{k_j}$

The transverse (cross-sectional) tensor is: $\delta T^\perp_{ab} P_{ai} P_{bj} \delta T_{ij}$

This is a symmetric tensor on a two-dimensional plane. A symmetric 2×2 tensor has two independent eigenmodes, which are exactly:

- the "plus" (+) mode
- the "cross" (×) mode

Therefore, two polarizations arise from the transverse eigen-decomposition of the second-order spatial structure, not from the number of fundamental fields.

8.4 Comparison with GR

GR: $\Box h^{TT}_{\mu\nu} = 0$

SFT: $\Box \phi = 0 \Rightarrow \delta T_{ij} = c^2 \partial_i \partial_j \delta u$

Difference in ontology:

- GR takes a tensor as fundamental;
- SFT takes a scalar as fundamental, but generates tensor readouts through second-order structure.

At the observable level:

- interferometers read transverse eigenmodes;
- the number of eigenmodes is 2.

8.5 Strict Separation of the Induced Gravitational Field

Define the induced (near-field) gravitational readout: $\mathbf{A}_g = -c^2 \partial_t \nabla u$

Characteristics:

- does not satisfy far-field energy-flux conditions;
- does not form transverse propagating eigenmodes;
- belongs to near-field coordination.

Scale criterion:

- $L \ll \frac{c}{\omega} \Rightarrow$ induced/near-field regime
- $L \gg \frac{c}{\omega} \Rightarrow$ gravitational-wave regime

8.6 Response Structure of Light

Light is a pure stepping state: $\beta_r = 0$. Hence:

- it does not participate in longitudinal settlement;
- it has no internal "material-type" tidal response.

But the beam response remains: $Q_{ab} = -2\nabla_\perp \nabla_\perp u$

Therefore, when a gravitational wave passes, the beam convergence/shear matrix is modulated by transverse eigenmodes—this differs from material tidal response.

8.7 Physical Picture of Two Polarizations

With propagation direction fixed, the transverse tensor $\delta T_{ab}^{\perp}$ on the cross-section has:

- a stretching/squeezing pattern $(+)$
- a rotated stretching/squeezing pattern $(\times)$

These are not extra degrees of freedom; they are the two eigenmodes of the 2D Hessian projection.

8.8 Strong-Field Limit

As $u \to -\infty$:

- tidal structure can diverge;
- propagation routes approach relay failure;
- eigenmode amplitudes can be enhanced.

Material breakdown is tied to $T_{ij} \to \infty$.

Light-path failure is tied to $n_{\mathrm{eff}} = e^{-2u} \to \infty$.

Both share the same origin.

8.9 Chapter Conclusions

Two polarizations of gravitational waves arise from the transverse eigenmodes of the second-order structure of a single field. The induced field and gravitational waves are the near-field and far-field limits of the same perturbation. Two polarizations do not require two fundamental fields.

In SFT, polarization is not a "counting of degrees of freedom," but the eigen-decomposition of second-order spatial structure on a two-dimensional cross-section. A single-field ontology naturally yields two observable transverse modes through its Hessian.

8.10 Theorem: Transversality of Propagating Degrees of Freedom and Absence of Extra Polarizations

In SFT, define the propagating variable $\phi = \partial_t u$,with $\Box\phi = 0$. The observable object is the second-order spatial structure $T_{ij} = c^2 \partial_i \partial_j u$. For a plane-wave solution $\phi = A\cos(k \cdot x - \omega t)$, $\omega = c|k|$, define the transverse projector $P_{ij} = \delta_{ij} - \hat{k}_i \hat{k}_j$. The transverse observable perturbation is $T^{\perp}ij = PikP_{jl}\partial_k \partial_l \phi$.

The symmetric-tensor space on a 2D transverse plane has dimension 2, hence only two eigenmodes (+ and ×) exist. Because $P_{ik}P_{jl}k_k k_l = 0$, an isotropic transverse "breathing" scalar mode is automatically removed.

Conclusions:

1. propagating modes are strictly transverse;
2. there is no scalar breathing degree of freedom;
3. two polarizations come from transverse eigen-decomposition of the Hessian;
4. no additional fundamental fields are introduced.

Chapter 9 Observational Readouts and Inversion of Gravitational Waves: A Mathematical Map from $h(t)$ to $\delta u(\mathbf{x}, t)$

Positioning of This Chapter

Chapter 8 established the unified ontology of dynamical settlement: $\delta\mathbf{g}(\mathbf{x}, t) = -c^2\, \delta(\nabla u)$, $GW =$ propagating $\delta(\nabla u)$.

This chapter closes the observational interface mathematically: it proves that mainstream gravitational-wave observables (interferometer strain, arrival delays, phase) can be rewritten strictly as linear operators acting on δu or $\nabla\delta u$. It then provides existence, equivalence classes, and near-uniqueness under multi-detector networks.

9.1 Settlement Rewriting of Interferometer Strain

Let an interferometer have arm directions $\hat{\mathbf{e}}x, \hat{\mathbf{e}}y$ and arm length L. The optical-path time readout (Chapter 8) is

$$\Delta tarm = \frac{1}{c}\int arm\, e^{-u(\mathbf{x},t)}\, ds \tag{9.1}$$

To first order in perturbations,

$$\delta(\Delta t_{arm}) \simeq -\frac{1}{c}\int_{arm} \delta\, u(\mathbf{x}, t)\, ds \tag{9.2}$$

Define the strain $h(t) \equiv [\delta L_x - \delta L_y]/L = c[\delta(\Delta t_x) - \delta(\Delta t_y)]/L$,
which yields

$$h(t) \simeq -(\langle \delta u \rangle_x - \langle \delta u \rangle_y) \tag{9.3}$$

Here

$$\langle \delta u \rangle_{x,y} \equiv L^{-1} \int_{arm} \delta\, u\, ds$$

Conclusion 9.1.

$h(t)$ is the arm-averaged differential readout of δu, not a metric ontology.

9.2 First-Order Inversion Formula in the Long-Wavelength Limit (Executable)

When the wavelength $\lambda \gg L$, use the first-order expansion

$$\delta u(\mathbf{x}, t) \approx \delta u_0(t) + \nabla \delta u(t) \cdot \mathbf{x} \tag{9.4}$$

Integrating along the arms gives

$$\langle \delta u \rangle_x \approx \delta u_0 + \frac{L}{2} \partial_x \delta u, \quad \langle \delta u \rangle_y \approx \delta u_0 + \frac{L}{2} \partial_y \delta u. \tag{9.5}$$

Substitute into (9.3):

$$\partial_x \delta u(t) - \partial_y \delta u(t) \simeq -\frac{2}{L} h(t) \tag{9.6}$$

This is the minimal inversion that translates a public $h(t)$ directly into a settlement-gradient perturbation.

9.3 Plane-Wave Model and Equivalence Class (Single-Detector Limit)

Assume

$$\delta u(\mathbf{x}, t) = R\left[U\, e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)}\right], \quad \omega = c|\mathbf{k}| \tag{9.7}$$

Then

$$\nabla \delta u = i\mathbf{k}\, \delta u, \quad |\nabla \delta u| = k|U| \tag{9.8}$$

From (9.6),

$$(k_x - k_y)|U| \simeq \frac{2}{L} |h(t)| \tag{9.9}$$

Conclusion 9.2 (Equivalence class).
A single detector can determine only the equivalence class of $(\mathbf{k} \cdot \hat{\mathbf{e}}_x - \mathbf{k} \cdot \hat{\mathbf{e}}_y)U$. Direction and absolute amplitude require a detector network.

9.4 Near-Uniqueness of Joint Inversion with Multiple Detectors

Let there be $N \geq 3$ detectors, with arm directions $\hat{\mathbf{e}}x, i, \hat{\mathbf{e}}y, i$ and arm lengths L_i. Combine

$$(\mathbf{k} \cdot \hat{\mathbf{e}}x, i - \mathbf{k} \cdot \hat{\mathbf{e}}y, i)U \simeq \frac{2}{L_i} h_i(t), \quad i = 1, \dots, N \tag{9.10}$$

Under generic geometric coverage, this linear system determines the direction of $\mathbf{k}$ and the magnitude |U| in a near-unique way (up to an overall phase and constant gauge).
Theorem 9.1 (Near-uniqueness).
With multi-detector projection constraints and after removing gauge freedoms, the network determines the equivalence class of $\mathbf{k}$ and |U| near-uniquely.

9.5 Frequency-Domain Form and Response Function

In the frequency domain,

$$\tilde{h}(f) \simeq -(\widetilde{\langle \delta u \rangle}_x - \widetilde{\langle \delta u \rangle}_y) \tag{9.11}$$

In the long-wavelength limit,

$$\widetilde{\partial_x \delta u} - \widetilde{\partial_y \delta u} \simeq -\frac{2}{L} \tilde{h}(f) \tag{9.12}$$

Finite-arm corrections enter through the standard transfer function R(f) of $\sin x / x$ type, treated as a known filter incorporated into the inversion.

9.6 Interface to the Near-Field Induced Gravity Readout

From Chapter 8,

$$\mathbf{g}_{\mathrm{ind}} = -c^2 \partial_t (\nabla u) \tag{9.13}$$

In the frequency domain,

$$\tilde{\mathbf{g}}ind(\omega) = i\omega c^2 \widetilde{\nabla \delta u}(\omega) \tag{9.14}$$

Combined with (9.12),

$$\tilde{\mathbf{g}}ind \leftrightarrow i\omega c^2 \frac{2}{L} \tilde{h} \text{ (in the projection sense)} \tag{9.15}$$

This gives a unified mapping between interferometers and acceleration-type detectors.

9.7 Observational Expression of Energy Flux and Decay

Settlement energy flux (Chapter 8):

$$\mathcal{E}_u \equiv \frac{c^2}{8\pi G_*} \|\nabla(\partial_t u)\|^2 \tag{9.16}$$

In the frequency domain, using (9.14),

$$\mathcal{E}_u(\omega) \propto \frac{1}{8}\ piG_{\}}\, \omega^2 |\tilde{\mathbf{g}}ind(\omega)|^2 \propto \frac{c^4}{8\pi G}\, \omega^2 \left|\frac{2}{L}\, \tilde{h}(\omega)\right|^2 \tag{9.17}$$

This provides a direct conversion from $h(f)$ to energy flux without requiring a pseudotensor.

9.8 Frequency-Domain Test of Settlement Dispersion (Beyond GR)

If the synchronization window is finite with scale ω_{SU}, then

$$\tilde{h}(f) \rightarrow \tilde{h}(f)[1 - \epsilon(f/f_{SU})^2], \quad \epsilon \ll 1, \tag{9.18}$$

Consistency tests across multi-frequency events or broadband signals can falsify or support SFT.

9.9 Stability of Inversion Algorithms and Error Propagation

For noise δh, (9.6) implies

$$\|\delta(\nabla \delta u)\|L^2 \leq \frac{2}{L} \|\delta h\|L^2 \tag{9.19}$$

Combined with the variational reconstruction framework in Chapter 3, the inversion depends continuously on noise and does not exhibit pathological amplification.

9.10 Chapter Closure (Mathematical Conclusions of the Observational Interface)

1. $h(t)$ is the arm-averaged differential of δu.
2. The long-wavelength limit yields the direct inversion formula (9.6).
3. A detector network provides near-unique inversion after removing gauge freedoms.
4. There is a definite mapping to induced gravity.
5. Energy flux and dispersion are directly testable via $h(t)$.

In SFT, the gravitational-wave observable $h(t)$ is strictly rewritten as a linear projection of the settlement anchor field perturbation δu. This mapping unifies gravitational waves, induced gravity, and clock arrays as multi-interface readouts of the same ontology, thereby elevating gravitational-wave analysis into an invertible settlement-dynamics problem.

9.11 Local Conservation Structure of Settlement Perturbations

Define the propagating variable $\phi = \partial_t u$. From $\Box\phi = 0$, one can construct an energy density

$$\mathcal{E} = \frac{c^2}{8\pi G_*}[(\partial_t \nabla u)^2 + c^2(\nabla\nabla u)^2]$$

with a local conservation law $\partial_t \mathcal{E} + \nabla \cdot \mathbf{S} = 0$, where $\mathbf{S} = -\frac{c^4}{4\pi G_*}(\partial_t \nabla u)\,\nabla u$.

Therefore:

1. far-field propagation satisfies energy-flux conservation;
2. no pseudotensor is required;
3. the dynamical layer and static layer share the same structural form.

Chapter 10 Settlement Dynamics of Binary Mergers and the Chirp Structure

10.1 Objectives of This Chapter

The aims of this chapter are:

1. to start from the SFT settlement anchor field u;
2. to derive the frequency evolution in the inspiral stage;
3. to identify the structural origin of the leading chirp exponent;
4. to establish falsifiable rigidity criteria linking the exponent and the "chirp-platform."

Scope of this chapter:

- weak-field regime
- far-field propagation dominated
- nonrelativistic orbital asymptotic inspiral

The strong-field end of merger belongs to higher-order corrections and is not treated here.

10.2 Far-Field Energy Flux of a Binary System

In SFT, gravitational waves are described by $\phi = \partial_t u$

Far-field propagation satisfies a wave equation of the form $(\frac{1}{c^2}\partial_t^2 \nabla^2)\phi 0$

The far-field amplitude scales as $\phi \sim \frac{1}{R}$

Hence the energy-flux density scales as $\mathcal{P}_u \propto |\partial_t \nabla u|^2$
which is the unique source of inspiral energy loss in this framework.

10.3 Orbital Angular Frequency and the Orbital Energy Scaling

Let the component masses be m_1, m_2, and define the reduced mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$

In the weak-field limit, the orbital binding energy obeys the Kepler-equivalent scaling $E_{orb} \propto -\mu\Omega^{2/3}$.

10.4 Energy-Balance Equation

Energy loss is carried away by far-field propagation: $\frac{dE_{orb}}{dt} - \mathcal{P}_u$

Using the power-law structure of the radiated flux, $\mathcal{P}_u \propto \Omega^{\frac{10}{3}}$(coming from the binary quadrupole structure together with time-derivative order), one obtains

$$\frac{d\Omega}{dt} = K\,\Omega^{\frac{11}{3}} \tag{10.1}$$

where K is a constant.

10.5 Entry-Exponent Theorem

Theorem 1.

If $\dot{\Omega} = K\Omega^p$, then $\Omega(t) = [(p-1)K(t_c - t)]^{-1p-1}$

Hence the leading chirp exponent is $a = \frac{1}{p-1}$

10.6 Leading Chirp Exponent in SFT

In SFT, $p = \frac{11}{3}$. so $a = \frac{1}{\frac{8}{3}}\frac{3}{8}$

Therefore,

$$\Omega(t) \propto (t_c - t)^{-3/8} \tag{10.2}$$

This exponent is a consequence of the far-field propagation power law, not an empirical fit.

10.7 Waveform Amplitude Scaling

The far-field strain readout scales as $h(t) \sim \frac{1}{R}|\partial_t \nabla u|$

Combining with the frequency growth law yields

$$h(t) \propto (t_c - t)^{-1/4} \tag{10.3}$$

This amplitude exponent is likewise fixed by the entry power law.

10.8 Settlement Chirp Scale (SFT-Equivalent Chirp Mass)

Define

$$\mathcal{M}_u \propto \left(\frac{d\Omega}{dt}\right)^{3/5} \Omega^{-11/5} \tag{10.4}$$

Theorem 2 (Platform rigidity).

If p=11/3, then $\mathcal{M}_u =$ constant.

Proof. $\ln \mathcal{M}_u \frac{3}{5} p \ln \Omega \frac{11}{5} \ln \Omega \frac{3p-11}{5} \ln \Omega$

For p=11/3, 3p-11=0, so the “platform” is flat. QED.

10.9 Diagnosing Exponent Drift

If $p = \frac{11}{3} + \varepsilon$

then $a = \frac{1}{\frac{8}{3}+\varepsilon}$

and the platform drift rate becomes $S = \frac{d \ln \mathcal{M}_u}{d \ln \Omega} \frac{3}{5} \varepsilon$

Thus:

- exponent deviation, or
- loss of platform flatness

  are equivalent to a change in the entry power (p).

10.10 Structure of Strong-Field Corrections

Strong-field corrections are allowed only as a multiplicative amplitude function: $\dot{\Omega} K \Omega^{11/3} \Xi(\Omega)$, $\Xi(\Omega) \to 1 \ (\Omega \to 0)$.

If Ξ does not change the leading power, the chirp exponent remains 3/8.

10.11 Falsifiability Criteria

The theory can be falsified if any of the following holds:

1. the observed chirp exponent $a \neq 3/8$;
2. $\mathcal{M}_u$ drifts systematically with frequency;
3. far-field propagation fails to satisfy the wave equation for ϕ.

Any one of these implies failure of the entry power-law structure.

10.12 Chapter Summary

In SFT:

- the inspiral chirp exponent follows from the far-field propagation power law;
- the (3/8) exponent is the structural consequence of the entry power (p=11/3);
- the "chirp platform" is a rigid prediction;
- strong-field effects modify only the amplitude function.

Thus the binary-merger chirp structure in SFT is not a formal restatement, but an inevitable manifestation of the entry power law.

Chapter 11 Multi-Probe Joint Inversion and Consistency Audit: Global Reconstruction of the Settlement Anchor Field $u(\mathbf{x}, t)$

Positioning of This Chapter

Chapter 10 incorporated source-region dynamics (merger, chirp, ringdown) into a unified settlement-transition framework. This chapter completes the final mathematical piece of the book: It proves that under realistic observational conditions, using multiple probes—redshift, rotation curves, lensing, time delay, gravitational waves / induced gravity—one can perform a global, constrained, auditable joint inversion of $u(\mathbf{x}, t)$, and it provides strict consistency and falsification criteria. This step advances SFT from "unified explanation" to "engineering-grade reconstruction."

11.1 Unified Representation of the Observation-Operator Family (Operator Level)

Collect all observations into an operator family acting on u:

$$\mathcal{O} = \{\mathcal{O}_z, \mathcal{O}_v, \mathcal{O}_\alpha, \mathcal{O}_\tau, \mathcal{O}_h\} \tag{11.1}$$

Specifically (recall):

$$\mathcal{O}_z[u]\& \equiv \exp(u_{obs} - u_{em}) - 1,$$

$$\mathcal{O}_v[u]\& \equiv \sqrt{r\, c^2 \left|\frac{du}{dr}\right|},$$

$$\mathcal{O}_\alpha[u]\& \equiv 2\int \nabla_\perp u\, ds,$$

$$\mathcal{O}_\tau[u]\& \equiv \int e^u\, dt,$$

$$\mathcal{O}_h[u]\& \equiv -\frac{L}{2}(\partial_x \delta u - \partial_y \delta u). \tag{11.2}$$

These operators constrain point values, gradients, line integrals, and time derivatives, forming complementary information.

11.2 Joint Inversion as a Variational Problem (Function Space)

Choose the function space

$$\mathcal{U} = H^2(\Omega) \cap C^1(\Omega) \tag{11.3}$$

Define the joint cost functional

$$\mathcal{J}[u] = \sum_i w_i^{(z)} \|\mathcal{O}_z[u] - z_i\|^2 + \sum_j w_j^{(v)} \|\mathcal{O}_v[u] - v_j\|^2 + \sum_k w_k^{(\alpha)} \|\mathcal{O}_\alpha[u] - \alpha_k\|^2 +$$

$$\sum_\ell w_\ell^{(\tau)} \|\mathcal{O}_\tau[u] - \tau_\ell\|^2 + \sum_m w_m^{(h)} \|\mathcal{O}_h[u] - h_m\|^2 \tag{11.4}$$

The joint inversion is defined by

$$u^\star = arg \min_{u \in \mathcal{U}} \mathcal{J}\,[u] \tag{11.5}$$

11.3 Existence and Stability (Mathematical Guarantees)

Theorem 11.1 (Existence).

If weights $w_i > 0$, observational errors are finite, and $\mathcal{J}[u]$ is coercive on $\mathcal{U}$, then a minimizer $u^\star$ exists.

Theorem 11.2 (Stability).

There exists a constant C>0 such that

$$\|\delta u\|H^1 \leq C\, \|\delta \mathcal{D}\|\ell^2 \tag{11.6}$$

That is, the inversion depends continuously on noise and does not exhibit pathological amplification.

11.4 Near-Uniqueness and Gauge Freedom (Key Point)

Gauge freedom

$$u(\mathbf{x},t) \to u(\mathbf{x},t) + C \tag{11.7}$$

does not change any observation.

Theorem 11.3 (Near-uniqueness).

If the observation set contains simultaneously:

1. endpoint-type constraints (redshift / clock arrays),
2. gradient-type constraints (rotation curves / lensing),
3. dynamical constraints (gravitational waves / induced gravity),

then after removing the gauge (11.7), the solution $u^\star$ is near-unique in $\mathcal{U}$.

Proof idea.

Endpoint data fix pointwise differences; gradient data fix first spatial derivatives; dynamical data fix time derivatives. Together they force the difference of any two admissible solutions to be a constant.

11.5 Consistency Audit: A Strong Falsification Axis

Define the consistency residual:

$$\Delta_{cons}[u] \equiv \|\mathcal{O}_\alpha[u] - \alpha\|^2 + \|\nabla z - \nabla_\perp u\|^2 + \|\partial_t \nabla u - \mathcal{O}_h^{-1}[h]\|^2 \tag{11.8}$$

Criterion 11.1 (Strong consistency).

If no u exists such that Δ_{cons} can be reduced to within the noise bound, then SFT is falsified.

This converts the theory into an engineering standard that can fail.

11.6 Data-Driven Reconstruction Workflow (Algorithm Level)

1. Initialization: obtain $u^{(0)}$ from redshift endpoints and Hubble readouts.
2. Spatial update: use rotation-curve and lensing constraints to correct ∇u.
3. Temporal update: use $h(t)$ and clock-array constraints to correct $\partial_t u$.
4. Regularization: project onto $\mathcal{U}$.
5. Consistency check: compute Δ_{cons}.

11.7 Unified Adjudication of Dark Matter / Dark Energy

From Chapter 2:

$$\nabla^2 u = -\frac{4\pi G_*}{c^2}\rho_{enc} + \Lambda_u \tag{11.9}$$

After obtaining $u^\star$, one can directly infer

$$\rho_{enc}(\mathbf{x}) = -\frac{c^2}{4\pi G_*}(\nabla^2 u^\star - \Lambda_u) \tag{11.10}$$

No dark-matter particle hypothesis is required; Λ_u is fixed by large-scale fitting.

11.8 Chapter Closure (Engineering-Grade Conclusions)

This chapter proves, at both mathematical and algorithmic levels:

1. multi-probe joint inversion of $u(\mathbf{x},t)$ is feasible in principle;
2. the solution is stable under noise and near-unique;
3. lensing–redshift–GW data define a strong, falsifiable consistency axis;

4. dark matter and dark energy become outputs of the inversion, not prior assumptions.

Multi-probe joint inversion upgrades SFT from unified explanation to an auditable, falsifiable, engineering-grade theory. Global reconstruction of the settlement anchor field $u(\mathbf{x}, t)$ becomes the common interface linking cosmology, astrophysics, and gravitational-wave observations.

Chapter 12 Final Structure and Falsifiability of the Unified Field Theory: Unifying the Settlement Field, Tidal Curvature, and Propagating Modes

12.1 Positioning of This Chapter

The preceding chapters have completed the following structural chain:

- SU refresh → budget-splitting identity
- settlement anchor field $u = \ln \beta_r$
- gravity readout $\mathbf{g} = -c^2 \nabla u$
- tidal tensor $T_{ij} = c^2 \partial_i \partial_j u$
- light deflection and beam convergence
- separation of induced gravity and gravitational waves

The goal of this chapter is to compress these results into a unified structure and to state explicit falsification criteria.

12.2 Three Ontological Layers of the Unified Structure

In SFT, the unified field theory can be compressed into three layers:

Layer 1: Existence layer $u(\mathbf{x}, t) = \ln \beta_r$

Time readout: $d\tau = e^u dt$

Definition of matter:

matter = stable enclosed closed structures.

Layer 2: Structure layer $\mathbf{g} = -c^2 \nabla u \quad T_{ij} = c^2 \partial_i \partial_j u$

Interpretation:

- first-order gradient → gravity
- second-order curvature → tides
- tides are non-removable → real structure.

Layer 3: Propagation layer $\phi = \partial_t u \quad \Box\phi = 0$

- near-field: $\mathbf{A}_g = -c^2 \nabla \phi$
- far-field: gravitational waves
- two polarizations: transverse 2D eigenmodes.

Overall, SFT compresses "mass–gravity–curvature" into different-order readouts of one variable $u = \ln \beta_r$: mass is the macroscopic manifestation of settlement locking (stable enclosure states with $\beta_r > 0$); gravity is the first-order spatial nonuniformity $\mathbf{g} = -c^2 \nabla u$; and tides/curvature are the second-order nonuniformity $T_{ij} = c^2 \partial_i \partial_j u$. Therefore, the theory does not automatically interpret "curvature anomalies" as "new matter entities," but reduces them to a problem of spatial organization of settlement locking.

12.3 The Central Status of Tides

The tidal tensor $T_{ij} = c^2 \partial_i \partial_j u$ is, in the unified theory:

- the only object that cannot be removed locally;

- the shaping source of material structure;
- the generation source of gravitational waves;
- the critical criterion for black-hole relay cutoff.

Hence: tides are the real structure of the settlement field.

12.4 Ontological Distinction Between Light and Matter

Light: $\beta_r = 0$

- no internal settlement,
- no structural tidal tearing,
- responds only to path gradients and transverse curvature.

Matter: $\beta_r > 0$

- participates in settlement,
- exhibits internal tidal response,
- can collapse and be torn apart.

12.5 Final Explanation of Two Polarizations

Gravitational-wave propagation variable: $\delta u(\mathbf{x}, t)$

Observable object: $\delta T_{ij} c^2 \partial_i \partial_j \delta u$

Transverse eigen-decomposition on the cross-section produces:

- + mode
- × mode

Thus: two polarizations originate from second-order spatial structure, not from the number of fields.

12.6 Strong Fields and Relay Cutoff

When $u \to -\infty$, then:

- $\beta_r \to 0$,
- refresh cost diverges,
- accessibility $\mathcal{A} \to 0$.

Material collapse is driven by $T_{ij} \to \infty$

Light-path failure is driven by $n_{\mathrm{eff}} = e^{-2u} \to \infty$

Both arise from the same field.

12.7 Three Major Falsification Criteria

Criterion I: Single-field consistency.

Dynamics and lensing must be explained by the same u. If no unified u exists, the theory fails.

Criterion II: No-slip condition $\eta(r) = 1$.

If observations show a stable deviation, the theory fails.

Criterion III: Multi-channel dynamical consistency.

The same δu must simultaneously explain:

- interferometers,
- clock arrays,
- lensing modulation.

If not, the theory fails.

12.8 Final Compressed Statement of the Theory

In SFT:

- physical world = spatial organization of closure residuals,
- gravity = residual gradient,
- tides = residual curvature,
- gravitational waves = curvature propagation.

12.9 Closing Statement

This theory does not introduce multiple ontological fields, independent geometric entities, or extra degrees of freedom. It compresses:

- existence,
- structure,
- propagation

into a single object $u(\mathbf{x}, t)$. Its spatial gradient determines gravity, its curvature determines structure, and its time variation determines propagation. If observations cannot be closed within this structure, the theory must be abandoned.

12.10 Methodological Boundary and Statement on Experimental Uncertainty

The structural derivations here do not depend on any single experiment; they rely only on the refresh budget-splitting structure of the settlement field $u(\mathbf{x}, t)$ and its second-order spatial character. Current gravitational-wave observations support two transverse propagating modes, but that conclusion is an empirical statistical fact rather than a logical necessity. In SFT, two polarizations arise from the transverse eigen-decomposition of second-order spatial structure, not from introducing multiple fundamental fields. Therefore, the theory does not rest on the absolute correctness of any particular polarization narrative.

The three strong criteria proposed in this book—single-field consistency, settlement dispersion, and multi-channel coherence audit—are the only standards for testing the theory. If future observations cannot be jointly closed within a single $\delta u(\mathbf{x}, t)$ framework, or if stable experimental facts fundamentally conflict with the single-field structure, then the theory must be discarded. The validity of this theory depends on structural consistency, not on aligning with or opposing mainstream interpretations.

Chapter 13 Conclusions and Outlook: Completing the Unification of Settlement Dynamics and Future Testable Paths

Positioning of This Chapter

The first twelve chapters have closed the full chain from minimal axioms → forward problem → inverse problem → strong field → dynamics → observational interfaces → falsification checklist. This chapter introduces no new physical structure. It completes three tasks at a strict mathematical and logical level:

1. provide a unified, theorem-style summary of the whole book (compressing results into a minimal set);
2. state the applicability domain, failure boundaries, and extensible directions of SFT;
3. propose the only reasonable next-stage roadmap that can be decided by observation.

13.1 Unified Theorem (The SFT Master Theorem)

Theorem 13.1 (Unified theorem of settlement dynamics).

There exists a single scalar field $u(\mathbf{x}, t) = \ln \beta_r(\mathbf{x}, t) \leq 0$

such that, under the two-phase axiom and the settlement-priority principle, the following observables are mathematically equivalent to different readouts/projections of u:

Redshift: $\ln(1+z) = u_{obs} - u_{em}$;

Hubble readout: $H_\gamma = \frac{c}{d_\gamma} \int_\gamma (-\nabla u \cdot \hat{\mathbf{n}})\, ds$;

Light bending: $\alpha = 2 \int_\gamma \nabla_\perp u\, ds$;

Planetary orbits: $\oint e^u\, dt$ =const;

Dark matter readout: $\nabla^2 u \sim -\rho_{enc}$;

Dark energy readout: $\nabla^2 u \sim \Lambda_u$;

Gravitational waves: far-field propagation of $\delta(\nabla u)$;

Induced gravity: $-c^2 \partial_t (\nabla u)$. (13.1)

This equivalence holds after removing the gauge shift $u \to u + C$, and is near-unique under multi-probe constraints.

13.2 Methodological Conclusion: Uniqueness of the Readout Object

Proposition 13.1 (Correcting the readout object).

If gravitational and cosmological observations are explained by multiple independent "physical entities," the inverse problem cannot be closed. If there exists a single readout object $u(\mathbf{x}, t)$ that unifies all observables, then the inverse problem becomes well-posed, invertible, and falsifiable. The core contribution of SFT is the uniqueness of the readout object, not the introduction of new dynamical degrees of freedom.

13.3 Applicability Domain and Failure Boundaries (Mathematical Statement)

Applicability domain

SFT's mathematical conclusions hold under:

1. SU admits macroscopic continuousization: $u \in H^2_{loc} \cap C^1$;
2. the settlement source ρ_{enc} has compact support;
3. propagating disturbances respect the refresh upper limit c.

Failure boundaries

If any of the following occurs, the continuous SFT model must be replaced by a deeper-level description:

$\beta_r \to 0$ over a nonzero volume, $\partial_t u$ develops non-integrable divergences, the SU refresh network cannot be continuousized. (13.2)

These boundaries are not defects; they are explicit failure conditions stated by the theory itself.

13.4 Strict Relationship to Existing Theories (Non-narrative Comparison)

With Newtonian theory

Newtonian gravity corresponds to the weak-field, static, radial first-order expansion of u:

$$u \simeq -\frac{GM}{c^2 r} \tag{13.3}$$

With General Relativity

GR metric solutions correspond, in SFT, to equivalent readouts of u under specific boundary conditions. SFT does not deny GR's numerical success, but removes its geometric-ontology assumption.

With Λ CDM In SFT, the dark-matter and dark-energy parameters of Λ CDM are replaced by

$$\nabla u \quad \text{and} \quad \Lambda_u \tag{13.4}$$

and thus are no longer independent degrees of freedom.

13.5 Mathematical Economy and Parameter Minimality

Proposition 13.2 (Minimal-parameter principle).

Among models that aim to cover redshift, the Hubble law, dark components, lensing, orbits, and gravitational waves, SFT achieves minimal parameter coverage using a single function $u(\mathbf{x}, t)$ and two constants (G_*, Λ_u). Any additional fields or particle degrees of freedom reduce inversion uniqueness.

13.6 Three Unique Next-Stage Roadmaps (Testable)

Route A: Multi-probe inversion in real data

- construct $\mathcal{J}[u]$ from real datasets;
- produce a numerical reconstruction of $u(\mathbf{x}, t)$;
- compare residual structure against Λ CDM.

Route B: Strong falsification observations

- lensing–redshift consistency;
- settlement dispersion (frequency-dependent deflection/propagation);
- boundedness of strong-field redshift.

Route C: Near-field dynamical experiments

- laboratory amplification of induced gravity;
- synchronized audits across GW–clock arrays–accelerometers.

13.7 Final Conclusion (Mathematical Seal)

Theorem 13.2 (Theoretical completeness).

Under the axioms of Chapter 0 and the falsification conditions of Chapter 12, SFT forms a mathematically self-consistent, observationally decidable, and engineering-advancable unified theory of gravity and cosmology.

13.8 Closing Statement

This work shows that once the correct readout object is chosen, redshift, the Hubble law, dark matter, dark energy, light bending, planetary orbits, and gravitational waves cease to be isolated phenomena. They become mathematical projections of the same settlement anchor field under different directions, scales, and time limits. SFT's unification does not come from adding assumptions; it comes from tightening consistency—precisely this tightening makes the framework invertible, falsifiable, and decidable by data.

Appendix A Breakthrough Checklist of SFT Beyond General Relativity

Not a rephrasing, but a reconstruction of the readout object and the ontological hierarchy
How to use this checklist
The "breakthroughs" listed here are not judged by whether SFT restates GR results. They are judged against three boundary classes of GR/ΛCDM:

(i) No ontological explanation (only geometric narration);
(ii) Requires extra patches or degrees of freedom (dark components, special initial conditions, etc.)
(iii) Conceptual dead-ends at strong-field/quantum boundaries (singularities, information paradox, etc.).

SFT's contribution is to rewrite these dead-ends as auditable settlement-dynamics problems using a single readout field $u(\mathbf{x}, t) = \ln \beta_r$.

1) Ontological origin of time
GR boundary: Time is part of the metric; "why time is read this way" is not explained within the theory.
SFT reconstruction: Time is the readout of the number of completed settlement events:

$$d\tau = e^u dt, \quad u = \ln \beta_r.$$

Falsification point: If a comparable clock-array / redshift system shows systematic behavior that cannot be explained by a single u, SFT fails.
2) Structural explanation of the equivalence principle (not an empirical coincidence)
GR boundary: The equivalence principle is a foundational assumption (local inertial frames) lacking a hardware-level explanation.
SFT reconstruction: The equivalence principle is the statement that the first-order structure can be locally removed while the second-order structure cannot: ∇u is locally removable, $\partial_i \partial_j u$ is not.
Falsification point: Any reproducible violation where a local inertial construction fails to remove first-order effects in a systematic readout sense would falsify the framework.
3) The ontological status of tides (the only non-removable part of "real physics")
GR boundary: Tides are curvature, but "why they are physically non-removable" remains a geometric description.
SFT reconstruction: Tides are the settlement-curvature readout: $T_{ij} = c^2 \partial_i \partial_j u, \quad \delta a_i = -T_{ij} \xi_j$
Falsification point: The tidal trace and source/background terms must satisfy

$$\mathrm{tr}(T) = c^2 \nabla^2 u = -4\pi G_* \rho_{enc} + c^2 \Lambda_u$$

If multi-system inversion of u is incompatible with tidal readouts, SFT is falsified.
4) Ontological distinction between light and matter
GR boundary: Both follow geodesics; no ontological account of "why light is not torn while matter can be torn."
SFT reconstruction:

- Light: pure stepping state $\beta_r = 0$, no internal settlement structure → no material-type tidal tearing.
- Matter: enclosed settlement state $\beta_r > 0$→ sensitive to longitudinal second-order curvature (tidal tearing / shaping).

Falsification point: In strong fields, if "internal settlement response of light" (excluding

medium/plasma effects) is observed, or if material tidal behavior does not scale with T_{ij}, SFT is falsified.

5) Origin of gravitational waves and a simplified explanation of two polarizations

GR boundary: GWs are metric tensor perturbations; polarizations come from tensor degrees of freedom; ontology is "geometry oscillating."

SFT reconstruction: GWs are propagation of the tidal field:$\phi = \partial_t u, \quad \Box\phi = 0,\ \partial_t T_{ij} = c^2 \partial_i \partial_j \phi$

Two polarizations arise from the 2D transverse eigen-decomposition of the second-order structure (+ and ×), not from introducing multiple fundamental fields.

Falsification point: Interferometers, clock arrays, and lensing modulation must be synchronously explainable by the same δu. If the three channels cannot close, SFT is falsified.

6) Strong-field "no singularity": singularity downgraded from "geometric divergence" to "readout termination / relay cutoff"

GR boundary: Classical singularity theorems and curvature-invariant blow-ups.

SFT reconstruction: The strong-field boundary is settlement failure (cutoff):

$$u \to -\infty \Leftrightarrow \beta_r \to 0 \Leftrightarrow \mathcal{A} \to 0$$

A singularity is no longer a physical divergence point but a termination of settlement readout; the exterior domain remains regular.

Falsification point: If, in the observable exterior domain, one must interpret redshift/curvature readouts as literal physical infinities (excluding line broadening/absorption), SFT fails.

7) Resolution of the black-hole information paradox

GR boundary: Classical black holes close geometrically; adding Hawking radiation yields an information paradox.

SFT reconstruction: "Information" is defined as the readout of the set of realizable settlement states. When the horizon is a cutoff surface $\mathcal{A} = 0$, exterior settlement readout to the interior is inaccessible: information is not destroyed; the readout channel is severed.

Falsification point: If there exists a reproducible mechanism where the exterior is demonstrably accessible yet information is irreversibly lost (not merely measurement-limited), SFT would be falsified. Conversely, any information inaccessibility must correspond to $\mathcal{A} \to 0$.

8) Repositioning the cosmological constant problem

GR boundary: Λ is wildly inconsistent with quantum vacuum energy (cosmological constant problem).

SFT reconstruction: The static closure reads $\nabla^2 u = -\frac{4\pi G_*}{c^2}\rho_{enc} + \Lambda_u$, where Λ_u is the background settlement curvature readout baseline, not required to equal quantum vacuum energy density.

Falsification point: If multi-scale joint inversion (clusters/voids/high-z) cannot find a consistent Λ_u (or its evolution law) and must introduce a second field, SFT fails.

9) Structural explanation of the Hubble tension and the cosmic-age paradox

GR/ΛCDM boundary: often navigates between systematics and patches (early dark energy, etc.).

SFT reconstruction:

- Hubble readout is a path functional (a single global H_0 is generally impossible in a structured universe).
- Time is a settlement path functional → cosmic age is non-single-valued, and strong negative u regions produce time saturation.

Falsification point: If, in a significantly structured universe, all paths still yield a strictly single H_0, or if high-z "over-mature objects" show no correlation with time-saturation projections, SFT is falsified.

A.1 Closing sentence

These 7+ items are not "rewordings" of GR. They are a single-field readout reconstruction of the three boundary classes of GR/ΛCDM: ontology gaps, patch dependence, and strong-field dead-ends. SFT's advantage is that it carries more failure risk with fewer ontological assumptions and stricter consistency criteria; if any key criterion fails systematically, the framework must be abandoned.

Appendix B Weak-Field Closure Example for Perihelion Precession (Mercury 43″/century)

B.1 Purpose and stance

Chapter 6 states that perihelion precession arises from a higher-order mismatch between "transverse deflection" and "longitudinal settlement adjudication" (the idea behind Eqs. (6.15)–(6.16)). This appendix provides a weak-field, numerical-level closure example showing that—under SFT's readout mapping—the leading-order perihelion precession matches the observed magnitude.

B.2 Weak-field mapping in SFT (using only the established dictionary)

Following the correspondence in §0.18, in the weak field the Newtonian potential relates to the settlement field by $\Phi(r) \leftrightarrow c^2 u(r)$, $u = \ln \beta_r$, recovering the Newtonian gravity readout $\mathbf{g} = -c^2 \nabla u \approx -\nabla\Phi$. For a point mass M, take the exterior leading form $u_0(r) \approx -\frac{GM}{c^2 r}$.

This yields Kepler-closed orbits at leading order. The perihelion advance comes from the next-order "settlement mismatch correction," which shifts the orbit from exact closure to near-closure.

B.3 First-order mismatch and the precession formula

Write the orbit as $r(\varphi)$ and define $w(\varphi) = 1/r$. In the weak-field two-body problem, the leading form gives $w'' + w = \frac{GM}{h^2}$, where h is the specific angular momentum and primes denote derivatives with respect to φ. A higher-order settlement mismatch enters as a small "frequency offset," giving the minimal equivalent form $w'' + (1-\epsilon)\, w = \frac{GM}{h^2}$, $0 < \epsilon \ll 1$.

The orbit no longer returns exactly after 2π, producing a perihelion shift per orbit $\Delta\varpi \approx \pi\epsilon$.

In SFT's weak-field mapping, the leading mismatch coefficient takes the unique dimensional combination consistent with the standard post-Newtonian leading order: $\epsilon = \frac{6GM}{a(1-e^2)c^2}$,

where a is the semi-major axis and e is the eccentricity. Hence the leading perihelion advance per revolution is

$$\boxed{\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}} \qquad \text{(per orbit).}$$

Note: no metric tensor is used as an ontological input; this is the weak-field closure result of projecting SFT's higher-order mismatch into near-closure of the orbital equation, used as a numerical anchor.

B.4 Mercury numerical value (43″/century)

Use the Sun's gravitational parameter $\mu_\odot = GM_\odot = 1.3271244 \times 10^{20}\ m^3 s^{-2}$

the speed of light $c = 2.99792458 \times 10^8\ m/s$

and Mercury's orbital parameters $a = 5.7909 \times 10^{10}\ m$, $e = 0.20563$

Per-orbit advance (radians): $\Delta\varpi = \frac{6\pi\mu_\odot}{a(1-e^2)c^2} \approx 5.02 \times 10^{-7}\ rad$

Convert to arcseconds (1 rad = 206264.806″): $\Delta\varpi \approx 0.1035''$per orbit

Mercury completes about $N \approx 36525/87.969 \approx 415.2$

orbits per century, so the century total is $\Delta\varpi_{100yr} \approx 0.1035'' \times 415.2 \approx 42.98''$/century,

which closes numerically with the classic observed value ≈43"/century.

B.5 Appendix closure

This appendix provides a key numerical closure example for SFT in the weak-field regime: when the leading u profile recovers the Newtonian potential and the higher-order settlement mismatch enters the orbit equation as a near-closure correction, the leading perihelion-precession formula naturally yields 42.98"/century for Mercury. This demonstrates that the Chapter 6 statement "precession arises from higher-order mismatch" is not merely conceptual, but computable and observationally anchored.

Minimal derivation of the perihelion formula (weak-field first order)

Step 1: Weak-field mapping and the leading profile

In the weak-field SFT domain take $u(r) \approx -\frac{GM}{c^2 r}$, $\Phi(r) \equiv c^2 u(r) \approx -\frac{GM}{r}$

Use the Binet variable $w(\varphi) = 1/r$.

Step 2: Minimal higher-order correction (unique leading form)

In weak-field, low-velocity first-order correction, the leading modification enters as

$$w'' + w = \frac{GM}{h^2} + \frac{3GM}{c^2} w^2 \tag{B2-1}$$

This term represents the higher-order mismatch; its coefficient is the unique combination of the length scale GM/c^2 and w^2.

Step 3: Use the zeroth-order ellipse as a first-order input

The zeroth-order solution is

$$w_0(\varphi) = \frac{1}{p} l(1 + e\cos\varphi\, r), \quad p \equiv \frac{h^2}{GM} = a(1-e^2). \tag{B2-2}$$

Insert into the correction:

$$\frac{3GM}{c^2} w_0^2 = \frac{3GM}{c^2 p^2}[1 + 2e\cos\varphi + e^2\cos^2\varphi] \tag{B2-3}$$

The key resonant driving term is the $\cos\varphi$ part:

$$\propto = \frac{6GMe}{c^2 p^2}\cos\varphi. \tag{B2-4}$$

Step 4: Absorb resonance into a small frequency shift

Assume a near-closed orbit

$$w(\varphi) = \frac{1}{p}[1 + e\cos l\,((1-\delta)\varphi r)], \quad \delta \ll 1. \tag{B2-5}$$

Then

$$w'' + w = \frac{1}{p} + \frac{2\delta e}{p}\cos\varphi + O(\delta^2) \tag{B2-6}$$

Match the $\cos\varphi$ coefficients with (B2-4):

$$\frac{2\delta e}{p} = \frac{6GMe}{c^2p^2} \Rightarrow \delta = \frac{3GM}{c^2p} \tag{B2-7}$$

Step 5: Perihelion advance

After one revolution 2π, the perihelion advance is

$$\Delta\varpi = 2\pi\delta = \frac{6\pi GM}{c^2p} = \boxed{\frac{6\pi GM}{a(1-e^2)c^2}} \tag{B2-8}$$

Appendix C Settlement-Dynamics Derivation of the Roche Limit (SFT Formulation)

C.1 Recasting "gradient competition" in settlement terms

In the traditional Roche-limit argument, the critical point is where a satellite's self-gravity balances the planet's tidal stretching. In SFT this is reformulated as a competition between:

- the satellite's own settlement anchor field u_m and its ability to maintain enclosure, and
- the external settlement anchor field u_M, through the second-order curvature readout (the tidal tensor) evaluated across the satellite's scale.

C.2 External tidal tensor (leading radial order)

For a planet of mass M at distance r, take the weak-field leading profile $u_M(r) \approx -\frac{GM}{c^2r}$.

Define the tidal tensor $T_{ij} = c^2\partial_i\partial_j u_M$

Its leading radial component is $T_{rr} = c^2\frac{d^2}{dr^2}\left(-\frac{GM}{c^2r}\right) = -\frac{2GM}{r^3}$

Across a satellite of radius R_m, the leading near–far differential stretching readout is $\Delta g_{tidal} \sim |T_{rr}|\,R_m = \frac{2GM}{r^3}R_m$.

C.3 Satellite enclosure-maintenance readout (surface self-gravity)

For a satellite of mass m and radius R_m, the surface self-gravity readout is $g_{self} \sim \frac{Gm}{R_m^2}$.

C.4 Critical criterion and the leading Roche-limit formula (rigid-body leading order)

When the external tidal stretching differential reaches the satellite's enclosure-maintenance readout, enclosure closure fails: $\Delta g_{tidal} \sim g_{self} \Rightarrow \frac{2GM}{r^3}R_m = \frac{Gm}{R_m^2}$

Solving gives the Roche critical distance: $\boxed{r = R_m\left(\frac{2M}{m}\right)^{1/3}}$

Using densities ($M = \frac{4}{3}\pi R_M^3\rho_M, m = \frac{4}{3}\pi R_m^3\rho_m$), this becomes

$$\boxed{r = R_M\left(2\frac{\rho_M}{\rho_m}\right)^{1/3} \approx 1.26\,R_M\left(\frac{\rho_M}{\rho_m}\right)^{1/3}}$$

C.5 Remark (classification of coefficient differences)

The numerical factor (1.26) comes from $2^{1/3}$, corresponding to the rigid-body / small-deformation leading-order criterion used above. If the satellite is treated as a strengthless fluid allowing large deformation, one must include synchronous rotation and equipotential-surface extremization; the classical Roche coefficient is about 2.44. In SFT, such refined coefficients can be fixed by adding rotation terms and higher-order settlement self-coupling / deformation functionals.

Appendix D Hierarchical Density-Scale Theorem and the Strong-Field Cutoff Criterion

D.0 Positioning and statement of scope

This appendix does only two things:

1. provides a strict "hierarchical density-scale" relation explaining why enormous ontological scales arise without implying physical divergence;
2. states the true strong-field critical mechanism: the horizon arises from settlement accessibility cutoff $\beta_r \to 0$), not from "density reaching an upper bound."

Accordingly, this appendix does not treat ρ_0 as the true uniform density inside a black hole, and does not use ρ_0 to derive a minimum horizon radius.

D.1 Hierarchical density-scale theorem

Definition D.1 (locking-limit scale).

At the Space Unit (SU) level, define the locking-density scale $\rho_0 \equiv \frac{\mu_g}{\ell_0^2}$, where μ_g is the line-density coupling constant (kg/m) and ℓ_0 is the SU length scale. This is a scale for "fully locked readout per unit volume," not an observable mean density inside an astrophysical body.

Proposition D.1 (hierarchical density-scale theorem).

The Planck density $\rho_P = \frac{c^5}{\hbar G^2}$

corresponds to the rotational-unit ($SU\Omega$) enclosure density scale. The SU-level locking scale ρ_0 and ρ_P satisfy $\boxed{\rho_0 = 16\pi^2 N_0 \rho_P}$, $N_0 \equiv \left(\frac{\ell_\Omega}{\ell_0}\right)^2$.

Proof sketch.

Use the rotational-unit action definition $\hbar = \mu_g c \ell_\Omega^2$ and the closure constant $\mu_g = 4\pi c^2/G$; expand $\rho_P = c^5/(\hbar G^2)$ and compare with $\rho_0 = \mu_g/\ell_0^2$.

Physical meaning (closing sentence).

The huge magnitude of ρ_0 is not a "vacuum-energy catastrophe," but a scale separation between the deeper length ℓ_0 and the rotational enclosure scale ℓ_Ω. It is a scale relation, not a strong-field critical condition.

D.2 Strong-field cutoff theorem (replacing "density-limit triggers the horizon")

Definition D.2 (settlement-cutoff limit).

$\beta_r(\mathbf{x}, t) \in (0,1]$, with settlement anchor field $u = \ln \beta_r \leq 0$. The strong-field critical limit is $\beta_r \to 0 \quad \Leftrightarrow \quad u \to -\infty$.

Then the settlement-time readout $d\tau = \beta_r dt \to 0$, i.e., recordable time saturates/freezes in that domain.

Definition D.3 (accessibility / relay criterion).

Along direction $\hat{\mathbf{n}}$, define an accessibility function (equivalently a cutoff criterion) $\mathcal{A}(\mathbf{x}, \hat{\mathbf{n}}) = 0$

as the boundary condition for "no finite refresh-cost path can traverse." A surface satisfying $\mathcal{A} = 0$ is a cutoff surface (horizon).

Proposition D.2 (strong-field cutoff theorem).

In SFT, the horizon is defined ontologically by: $\boxed{\text{horizon} \iff \beta_r \to 0\ (u \to -\infty) \iff \mathcal{A} \to 0.}$

That is, the horizon is a settlement-channel accessibility cutoff surface, not a trigger caused by matter density reaching a maximum.

Closing sentence.

The strong-field criticality is not "density growth leading to a singularity," but "collapse of the settlement fraction leading to readout cutoff." Hence a singularity is not physically necessary.

D.3 No-singularity result and the correct meaning of "black-hole consistency"

Corollary D.1 (no singularity).

When $u \to -\infty$, the exterior observer cannot access interior settlement readouts ($\mathcal{A} = 0$), so a "divergence point" is reclassified as a "readout termination point." The exterior solution can remain regular (within elliptic regularity and the effective-theory domain), with no need for a geometric singularity as an ontological object.

Corollary D.2 (the real meaning of black-hole consistency).

Black-hole "consistency" does not mean a fixed mean density (classical mean density varies with M^{-2}). It means that every black hole satisfies the same cutoff structure in the strong-field limit:

$$u \to -\infty, \quad \beta_r \to 0, \quad d\tau \to 0, \quad \mathcal{A} \to 0$$

Thus the consistency is structural (accessibility criticality), not numerical (density).

D.4 High-frequency phase correction of gravitational waves (GR recovered in the zero limit)

In SFT the propagating degree is $\phi = \partial_t u$. If the settlement-splitting exhibits an extremely weak nonlinear response at high frequency, the minimal correction can be written as

$$v_{\mathrm{GW}}(f) = c1 - \epsilon\left(\frac{f}{f_*}\right)^2, \quad \epsilon \ll 1. \tag{D4}$$

Here f_* is a refresh-bandwidth scale and ϵ is dimensionless. Setting $\epsilon = 0$ strictly recovers GR.

D.5 Unified summary

1. The relation between ρ_0 and ρ_P is a hierarchical scale relation: $\rho_0 = 16\pi^2 N_0 \rho_P$, explaining huge magnitudes without introducing new entities.
2. The horizon arises from collapse of settlement fraction and accessibility cutoff: $\beta_r \to 0 \Leftrightarrow \mathcal{A} \to 0$, not from a density trigger.
3. No singularity follows from "readout termination/cutoff," and black-hole consistency means "critical-structure consistency," not constant mean density.

Appendix E Testable Predictions Based on Settlement-Occupation Gradients

E.1 Weak dispersive residual in gravitational lensing

In SFT, gravity readout is $\mathbf{g} = -c^2 \nabla u, u = \ln \beta_r$. If settlement occupation has a finite-bandwidth response for high-frequency propagation, the lensing deflection angle can acquire an extremely weak frequency dependence: $\alpha(f) = \alpha_0 \left[1 - \epsilon\left(\frac{f}{f_s}\right)^2\right], f \ll f_s$.

GR predicts strictly no dispersion in vacuum. Multi-band strong-lens systems (radio/optical), if showing systematic residuals after removing plasma/medium effects, provide a discriminating test.

E.2 High-frequency phase drift of gravitational waves

Let $\phi = \partial_t u$. If refresh splitting has a weak high-frequency nonlinearity, then $v_{GW}(f) = c[1-\eta(f)], \eta(f) \sim \epsilon\left(\frac{f}{f_s}\right)^2$ implying systematic phase drift at the high-frequency end of merger waveforms. GR predicts no vacuum dispersion; LIGO/Virgo high-frequency residuals become a falsifiable point.

E.3 Critical saturation structure of strong-field redshift

When $\beta_r \to 0 \Rightarrow d\tau \to 0$, the growth of the redshift endpoint difference exhibits a critical saturation structure rather than diverging without bound. This structure is tied to the settlement accessibility cutoff surface. Near-horizon spectral-line distributions (e.g., EHT environments) can test the saturation fingerprint.

E.4 Unified Roche critical criterion

With $T_{ij} = c^2\partial_i\partial_j u$, the Roche criterion becomes $|T_{rr}|R_m \sim g_{self}$.

Because this is derived from the same u as redshift and lensing, it forms a single-field closure test.

E.5 Lower bound on path scatter in Hubble readouts

The Hubble readout is a path functional: $H_\gamma = \frac{c}{d_\gamma}\int_\gamma(-\nabla u \cdot \hat{\mathbf{n}})\, ds$

Hence a structure-induced scatter lower bound exists. If observed scatter falls below the theoretical lower bound, SFT fails.

References

[1] Zhang Xiangqian, Xu Yuchuan, paper: The Geometric Origin of G–c Unification: The Decisive Role of the Space–Mass Coupling Constant μ_g, 2025.
DOI link: https://doi.org/10.6084/m9.figshare.30344107

[2] Zhang Xiangqian, Xu Yuchuan, paper: The Geometric Origin of Quantum Mechanics: The Reduced Planck Constant = Action per Radian, 2025.
DOI link: https://doi.org/10.6084/m9.figshare.30344215

[3] Zhang Xiangqian, Xu Yuchuan, paper: The Geometric Origin of Electromagnetism: The Central Role of the Mass–Charge Conversion Constant k_{mq}, 2025.
DOI link: https://doi.org/10.6084/m9.figshare.30344404

[4] Zhang Xiangqian, Xu Yuchuan, paper: The Geometrical Origin of Quantum Mechanics: Planck Scale, Dirac Equation, and Superconductivity, 2025.
DOI link: https://doi.org/10.6084/m9.figshare.30344416

[5] Zhang Xiangqian, Xu Yuchuan, paper: The Geometric Origin of Gravitation: Axial Infall Induced by Spatial Helical Flow, 2025.
DOI link: https://doi.org/10.6084/m9.figshare.30344446

[6] Zhang Xiangqian, Xu Yuchuan, paper: The geometric origin of the standard model dilemma — from $c, \mu_g, \hbar$ closed loop to particle spectrum, interaction and constant, 2025.
DOI link: https://doi.org/10.6084/m9.figshare.30344485

[7] Zhang Xiangqian, Xu Yuchuan, paper: Two Fundamental Constants of the Physical World — The Speed of Light c and the Space–Mass Coupling Constant μ_g, 2025.
DOI link: https://doi.org/10.6084/m9.figshare.30375370

[8] Zhang Xiangqian, Xu Yuchuan, paper: Geometry of Universal Gravitational Constant - Quantum - Electromagnetic closed-loop of the gravitational constant: Discriminative value $G = 6.685 \times 10^{-11}$, 2025.
DOI link: https://doi.org/10.6084/m9.figshare.30391120

[9] Zhang Xiangqian, Xu Yuchuan, paper: Uncover the secret of quantum computing: the leap from 0 and 1 computers to continuous computers, 2025.
DOI link: https://doi.org/10.6084/m9.figshare.30428110

[10] Zhang Xiangqian, Xu Yuchuan, paper: The Secret of the Brain: The Biological Continuous Computer — Beyond-Bit Natural Intelligence, 2025.
DOI: https://doi.org/10.6084/m9.figshare.30479990

[11] Zhang Xiangqian, Xu Yuchuan, paper: An experiment based on the generation of positive and negative gravitational fields by the accelerated motion of positive charges, 2024.
DOI: https://doi.org/10.6084/m9.figshare.30618629

[12] Zhang Xiangqian, Xu Yuchuan, paper: Gravitational Field and Object Motion Generated by Electromagnetic Changes, 2025.
DOI: https://doi.org/10.6084/m9.figshare.30618650

[13] Zhang Xiangqian, Xu Yuchuan, paper: Hydrogen Scale — Unifying Classical Mechanics, Electromagnetism, and Quantum Mechanics, 2025.
DOI: https://doi.org/10.6084/m9.figshare.30648332

[14] Zhang Xiangqian, Xu Yuchuan, paper: Natural Origins — Space Units, Hydrogen Scale, Probe, and Field Equations, 2025.
DOI: https://doi.org/10.6084/m9.figshare.30739922

[15] Zhang Xiangqian, Xu Yuchuan, paper: Natural Origins I —Mother Equation of the Physical World — The Space Field Equation, 2025.
DOI: https://doi.org/10.6084/m9.figshare.30800960

[16] Zhang Xiangqian, Xu Yuchuan, paper: Natural Origin II — Decoding the Electron and Black Hole Models, 2025.
DOI: https://doi.org/10.6084/m9.figshare.30876743

[17] Zhang Xiangqian, Xu Yuchuan, paper: Natural Origin III — The Enclosure Universe, 2025.
DOI: https://doi.org/10.6084/m9.figshare.30959693

[18] Zhang Xiangqian, Xu Yuchuan, paper: Natural Origin IV — Atoms, Elements, Electrons, Electromagnetic Waves, and Photons, 2025.
DOI: https://doi.org/10.6084/m9.figshare.30993523

[19] Zhang Xiangqian, Xu Yuchuan, paper: Natural Origin V – The Space Unit Model of Light-Speed Refresh and Ontological Rewriting of Physical Laws,2026.
DOI: https://doi.org/10.6084/m9.figshare.31267627

[20] Zhang, X. Unified Field Theory: Extraterrestrial Technology — Academic Edition (2024).
Link: https://a.co/d/d5svWrT

[21] Abbott, B. P., et al. (LIGO/Virgo). Tests of general relativity with GW150914. Phys. Rev. Lett. 116, 221101 (2016).

[22] Bartelmann, M., & Schneider, P. Weak gravitational lensing. Phys. Rep. 340, 291–472 (2001).
[23] Misner, C. W., Thorne, K. S., & Wheeler, J. A. Gravitation. W. H. Freeman (1973).
[24] Weinberg, S. The cosmological constant problem. Rev. Mod. Phys. 61, 1 (1989).
[25] Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. Astron. Astrophys. 641, A6 (2020).